UNIVERSITÀ DEGLI STUDI ROMA TRE

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Universal Relations in Anisotropic Materials

Supervisor:
Prof. Giuseppe Tomassetti

Co-supervisor:
Prof. Giuseppe Zurlo

Master's Thesis by:
Luca Lombardo
Student ID: 558887

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"Un paese ci vuole, non fosse che per il gusto di andarsene via.
Un paese vuol dire non essere soli, sapere che nella gente,
nelle piante, nella terra c'è qualcosa di tuo, che anche quando
non ci sei resta ad aspettarti."

— Cesare Pavese, La luna e i falò

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Introduction

This thesis aims to explore some theoretical aspects related to materials exhibiting a deformational response characterized by a preferential direction distribution. In particular, we will focus on *transversely isotropic elastic materials*, with special attention to a subcategory of these: *isotropic magnetoelastic materials*. In the latter, the preferential direction is defined by internal magnetic fields, which obey *Maxwell*'s differential equations.

As will be illustrated, *universal relations* exist for these materials, linking measurable kinematic and stress quantities independently of the specific constitutive response functions of the material. The concept of *universal deformation* will also be introduced, namely a finite deformation that is controllable in any material belonging to the considered class (isotropic, transversely isotropic) solely through the application of realistically applicable loads, such as surface tractions. In this case, universal relations can be simplified into a series of specific relations for the universal deformation in question.

Since these relations are independent of the material's constitutive response functions, they can be used in the experimental phase to verify that the tested material actually belongs to the hypothesized class. Otherwise, this might suggest the need to adopt a more general material model to adequately characterize the specimen under examination.

Despite the effectiveness of these relations in the experimental phase, their underlying physical meaning is not yet completely clear. It is precisely in this direction that the present thesis fits, aiming to characterize these relations more thoroughly. We will see how they are inextricably linked to the *Eshelby tensor*, known to be responsible for the forces acting on material inhomogeneities, and how this connection can contribute to a better understanding of the phenomenon. Furthermore, since the mathematical procedure used so far in the literature is particularly laborious and involves numerous steps, it is believed that this has diverted attention from the fundamental hypothesis. In particular, we will clarify that universal relations are, in fact, a *direct* consequence of material symmetries.

Furthermore, numerical aspects related to magnetoelastic materials will be discussed. The coupling between the mechanical and magnetic problems requires a clear numerical procedure. Given the presence of large deformations, it will be necessary to introduce *Lagrangian* magnetic variables, which, although lacking

a direct physical meaning, are fundamental for solving Maxwell's equations in the reference configuration. Two distinct approaches to solving the problem will be explored: both will be used, but to treat different situations. Simulations of such materials, widely used as magnetic actuators, are essential for predicting their behavior as a function of boundary conditions. To obtain desired finite deformations, it will be necessary to resort to the construction of *controllable states*, a concept that will be treated first alongside the aforementioned controllable deformations and later deepened in the numerical part of the thesis, where we will explore the numerical reconstruction of such states for magnetoelastic materials.

In Chapter 1, we will introduce the basic theory of finite elasticity. This chapter will provide the theoretical foundations necessary to tackle the more complex topics discussed in subsequent chapters.

In Chapter 2, we will focus on material symmetries, defining isotropy and transverse isotropy. We will also provide an overview of the governing equations of magnetoelasticity, in both Eulerian and Lagrangian formulations. The concept of isotropic magnetoelastic materials will also be introduced.

In Chapter 3, we will explore universal relations in the classical context, providing expressions for isotropic, transversely isotropic, and isotropic magnetoelastic materials. Additionally, we will present a method to prove the existence of an exact number of these relations.

In Chapter 4, we will link universal relations to the Eshelby tensor. We will see how the skew-symmetric part of this tensor is involved when a configurational remodeling of the preferential direction is allowed. We will introduce a new method to derive universal relations, limiting ourselves to the case of transverse isotropy, and show an application example in which the preferential direction reorganizes, analyzing the physical implications of the phenomenon.

Starting from Chapter 5, we will discuss the numerical aspects related to solving the coupled nonlinear magnetoelastic problem. For educational purposes, we will write an algorithm to solve the problem using Newton's method. We will derive the corresponding weak forms starting from the governing partial differential equations in local form. We will apply Newton's method and proceed with the finite element approximation. We will conclude the chapter by presenting a monolithic weak formulation, which we will use within the finite element code.

In the final chapter, we will implement the problem using a Python code, making use of the libraries provided by `FEniCSx`. Initially, we will solve a classic magnetoelastic problem with vacuum, similar to others present in the literature,

and highlight all the precautions that were taken to obtain a correct resolution of the problem. At the end of the chapter, we will numerically reconstruct a controllable state belonging to the family of states introduced in Chapter 3 for magnetoelastic materials.

Notation

The following notations are used in this work:

Vector notation

Symbol	Description
Bold	for vectors and tensors in invariant representation
$[\mathbf{A}], \{\mathbf{b}\}$	matrix representation of tensors and vectors
a_i, B_{ij}	index notation of spatial vectors and tensors
a_A, B_{AB}	index notation of referential vectors and tensors

Algebraic operators

Symbol	Description
$\cdot$	scalar product
$\otimes$	tensor product
$\times$	vector product
$\mathbf{AB}$	product
$\mathrm{sym}[\mathbf{A}]$	symmetric part of tensor $\mathbf{A}$
$\mathrm{skw}[\mathbf{A}]$	skew-symmetric part of tensor $\mathbf{A}$
$\mathrm{tr}\,\mathbf{A}$	trace of tensor $\mathbf{A}$
$\det \mathbf{A}$	determinant of tensor $\mathbf{A}$
$[\![\cdot]\!]$	jump of a quantity across the interface between two different domains

Differential operators

Symbol	Description
grad, div, curl	gradient, divergence, and curl operators, respectively, with respect to spatial coordinates
Grad, Div, Curl	gradient, divergence, and curl operators, respectively, with respect to material coordinates
$\frac{d}{dt}(\cdot)$	material derivative operator
$\dot{(\cdot)}$	Eulerian time derivative operator
$\delta(\cdot)$	infinitesimal variation operator
$\frac{\partial}{\partial X_A}(\cdot) \equiv (\cdot)_{,A}$	derivative operator with respect to the material coordinate X_A
$\frac{\partial}{\partial x_i}(\cdot) \equiv (\cdot)_{,i}$	derivative operator with respect to the spatial coordinate x_i
$\frac{\partial^2}{\partial X_A \partial X_B}(\cdot) \equiv (\cdot)_{,AB}$	second derivative operator with respect to material coordinates X_A, X_B
$\frac{\partial^2}{\partial x_i \partial x_j}(\cdot) \equiv (\cdot)_{,ij}$	second derivative operator with respect to spatial coordinates x_i, x_j

Sets of tensors

Symbol	Description
Lin	set of all tensors
Lin^+	set of all tensors with a positive determinant
Orth	set of all orthogonal tensors
$\mathrm{Orth}^+ \equiv \mathrm{Rot}$	set of all rotations
Sym	set of all symmetric tensors
Skw	set of all skew-symmetric tensors
Psym	set of all positive definite symmetric tensors

Recurring quantities

Symbol	Description
$\mathbf{u}$	displacement
$\mathbf{F}$	deformation gradient
J	Jacobian of the transformation
$\mathbf{C}$	right Cauchy-Green tensor
$\mathbf{B}$	left Cauchy-Green tensor
σ	Cauchy stress tensor
$\mathbf{P}$	first Piola-Kirchhoff stress tensor
$\mathbf{S}$	second Piola-Kirchhoff stress tensor
$\mathbf{h}$	magnetic field
$\mathbf{b}$	magnetic induction
$\mathbf{j}_{\mathrm{f}}$	free current density
$\mathbf{k}_{\mathrm{f}}$	free surface current density
$\boldsymbol{\tau}$	total magnetic stress tensor
$\boldsymbol{\tau}_m$	Maxwell stress tensor
$\mathbf{h}_l$	Lagrangian magnetic field
$\mathbf{b}_l$	Lagrangian magnetic induction
φ_l	Lagrangian magnetic scalar potential
$\mathbf{a}_l$	Lagrangian magnetic vector potential
$\mathbf{j}_{\mathrm{F}}$	Lagrangian free current density
$\mathbf{k}_{\mathrm{F}}$	Lagrangian free surface current density
$\mathbf{T}$	nominal total magnetic stress tensor
$\mathbf{T}_m$	nominal Maxwell stress tensor
$\mathbf{E}$	Eshelby tensor

1 Review of Finite Elasticity

This chapter provides a brief overview of the main concepts related to finite elasticity, which are necessary for understanding the subsequent theoretical developments. The concepts presented in this chapter follow, with adaptations and simplifications, the exposition by Anand and Govindjee [1].

1.1 Kinematic relations

Let us consider a homogeneous body $\mathcal{B}$, identified by the region of space it occupies in a *reference configuration*, and let $\mathbf{X}$ denote a generic material point of $\mathcal{B}$. We define a *motion map*

$$\mathbf{x} = \chi(\mathbf{X}, t) \tag{1.1}$$

which associates each material point $\mathbf{X}$ with the position $\mathbf{x}$ occupied at time t in the *deformed configuration* $\mathcal{B}_t$.

We define the *displacement* $\mathbf{u}$ as

$$\mathbf{u}(\mathbf{X}, t) = \mathbf{x} - \mathbf{X} = \chi(\mathbf{X}, t) - \mathbf{X} \tag{1.2}$$

The *deformation gradient* associated with the map is defined as

$$\mathbf{F} = \operatorname{Grad} \chi \equiv \frac{\partial \chi(\mathbf{X}, t)}{\partial \mathbf{X}}. \tag{1.3}$$

The relationship between an infinitesimal volume element dv_R in the reference configuration $\mathcal{B}$ and the corresponding infinitesimal volume element dv in the deformed configuration $\mathcal{B}_t$ is given by

$$dv = J\, dv_R, \qquad \text{with} \qquad J = \det \mathbf{F} > 0, \tag{1.4}$$

where J represents the *Jacobian of the transformation*.

The relationship between an infinitesimal area element $\mathbf{N}\, da_R$ in the reference configuration $\mathcal{B}$ and its image $\mathbf{n}\, da$ in the deformed configuration $\mathcal{B}_t$ is provided by *Nanson's formula*:

$$\mathbf{n}\, da = J\, \mathbf{F}^{-\mathrm{T}} \mathbf{N}\, da_R. \tag{1.5}$$

Furthermore, we introduce the symmetric and positive definite tensors,

$$\mathbf{C} = \mathbf{F}^{\mathrm{T}}\mathbf{F}, \qquad \mathbf{B} = \mathbf{F}\mathbf{F}^{\mathrm{T}}, \tag{1.6}$$

called the *right Cauchy-Green tensor* and the *left Cauchy-Green tensor*, respectively.

1.2 Balance equations in Eulerian form

Below are the balance equations, in local Eulerian form, for linear and angular momentum, along with their respective boundary conditions, in the deformed configuration $\mathcal{B}_t$ (static case).

Linear momentum balance equation

$$\operatorname{div} \boldsymbol{\sigma} + \mathbf{b} = \mathbf{0} \qquad \text{in } \mathcal{B}_t \tag{1.7}$$

Angular momentum balance equation

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}^{\mathrm{T}} \qquad \text{in } \mathcal{B}_t \tag{1.8}$$

Neumann boundary condition

$$\boldsymbol{\sigma}\mathbf{n} = \mathbf{t} \qquad \text{on } \partial\mathcal{B}_t^t \tag{1.9}$$

Dirichlet boundary condition

$$\mathbf{u} = \tilde{\mathbf{u}} \qquad \text{on } \partial\mathcal{B}_t^u \tag{1.10}$$

where $\boldsymbol{\sigma}$ is the well-known Cauchy stress tensor, $\mathbf{b}$ denotes the body forces per unit *spatial* volume (i.e., in the deformed configuration), $\mathbf{t}$ represents the surface traction acting on the boundary $\partial\mathcal{B}_t^t$, and $\tilde{\mathbf{u}}$ is a prescribed displacement field on the boundary portion $\partial\mathcal{B}_t^u$.

1.3 Balance equations in Lagrangian form

At this point, we introduce the *first Piola-Kirchhoff stress tensor* $\mathbf{P}$, defined as

$$\mathbf{P} := J\,\sigma\mathbf{F}^{-\mathrm{T}} \tag{1.11}$$

The tensor $\mathbf{P}$ represents the stress measured per unit area in the reference configuration. In general, it is not symmetric.

For the sake of compactness, the intermediate steps leading to the balance equations are omitted, limiting ourselves to stating their final form.

Linear momentum balance equation

$$\mathrm{Div}\,\mathbf{P} + \mathbf{b}_R = \mathbf{0} \qquad \text{in } \mathcal{B} \tag{1.12}$$

Angular momentum balance equation

$$\mathbf{P}\mathbf{F}^{\mathrm{T}} = \mathbf{F}\mathbf{P}^{\mathrm{T}} \qquad \text{in } \mathcal{B} \tag{1.13}$$

Neumann boundary condition

$$\mathbf{P}\mathbf{N} = \mathbf{t}_R \qquad \text{on } \partial\mathcal{B}^t \tag{1.14}$$

Dirichlet boundary condition

$$\mathbf{u} = \tilde{\mathbf{u}} \qquad \text{on } \partial\mathcal{B}^u \tag{1.15}$$

where $\mathbf{b}_R = J\,\mathbf{b}$ is the body force vector per unit reference volume, while $\mathbf{t}_R$ represents the surface traction per unit area in the reference configuration.

1.4 Dissipation inequality under isothermal conditions

We want to write a local form of the *dissipation inequality* (second law of thermodynamics) as a function of the first Piola-Kirchhoff stress tensor $\mathbf{P}$. The principle states that the rate of increase of the internal energy of a generic part $\mathcal{P} \subset \mathcal{B}$ cannot be greater than the power expended on it. If we denote by ψ_R the energy density per unit reference volume, the *dissipation inequality under isothermal conditions* requires that, for every part $\mathcal{P}$,

$$\frac{d}{dt}\int_{\mathcal{P}} \psi_R \;\leq\; \int_{\partial\mathcal{P}} \mathbf{P}\mathbf{N}\cdot\dot{\chi} \;+\; \int_{\mathcal{P}} \mathbf{b}_R\cdot\dot{\chi} \tag{1.16}$$

Using the divergence theorem and integration by parts, we obtain

$$\frac{d}{dt}\int_{\mathcal{P}} \psi_R \;\leq\; \int_{\mathcal{P}} \mathbf{P}\cdot\dot{\mathbf{F}} \;+\; \int_{\mathcal{P}} \big(\operatorname{Div}\mathbf{P} + \mathbf{b}_R\big)\cdot\dot{\chi}. \tag{1.17}$$

Therefore, using the linear momentum balance equation (1.12) and, since the part $\mathcal{P}$ is arbitrary, applying the localization theorem, we obtain the local form

$$\dot{\psi}_R - \mathbf{P}\cdot\dot{\mathbf{F}} \;\leq\; 0. \tag{1.18}$$

1.5 Form-invariance of constitutive equations and material frame-indifference

Since constitutive equations describe the *internal interactions* between the parts of a body, they must not depend on the external reference frame used to represent them. As a general assumption in continuum mechanics, it is therefore required that

all constitutive equations are form-invariant under changes of observer.

Specifically, if a variable Φ is linked to the list of variables Λ via a constitutive response function $\hat{\Phi}(\cdot)$, then it must also hold that

$$\Phi^* = \hat{\Phi}(\Lambda^*) \qquad \text{for all changes of observer.} \tag{1.19}$$

In other words, all observers use the same constitutive response function $\hat{\Phi}(\cdot)$. This assumption is known as the *principle of material frame-indifference* and constitutes the foundation of the invariance of constitutive relations.

1.6 Transformation of kinematic quantities

Let $\chi(\mathbf{X}, t)$ be the motion map of a body. Following a change of observer, the motion map transforms as

$$\chi^*(\mathbf{X}, t) = \mathbf{y}(t) + \mathbf{Q}(t)\big(\chi(\mathbf{X}, t) - \mathbf{o}\big), \tag{1.20}$$

where $\mathbf{Q}(t) \in \mathrm{Rot}$ and $\mathbf{y}(t)$ is an arbitrary translation.

Differentiating with respect to $\mathbf{X}$, we obtain the transformation of the deformation gradient:

$$\mathbf{F}^* = \frac{\partial \chi^*}{\partial \mathbf{X}} = \mathbf{Q}(t)\,\mathbf{F}. \tag{1.21}$$

It follows that the right Cauchy-Green tensor $\mathbf{C}$ is invariant under a change of observer, since

$$\mathbf{C}^* = \mathbf{F}^{*\mathrm{T}}\mathbf{F}^* = \mathbf{F}^{\mathrm{T}}\mathbf{Q}^{\mathrm{T}}\mathbf{Q}\mathbf{F} = \mathbf{F}^{\mathrm{T}}\mathbf{F} = \mathbf{C}. \tag{1.22}$$

1.7 The Second Piola-Kirchhoff stress tensor

We now introduce the *second Piola-Kirchhoff stress tensor*:

$$\mathbf{S} = J\mathbf{F}^{-1}\boldsymbol{\sigma}\mathbf{F}^{-\mathrm{T}}. \tag{1.23}$$

Since $\boldsymbol{\sigma}$ is symmetric, $\mathbf{S}$ is also symmetric.

We then write the local form of the dissipation inequality in terms of the second Piola-Kirchhoff stress tensor. From the fact that $\mathbf{P} = \mathbf{F}\mathbf{S}$, it follows that

$$\begin{aligned} \mathbf{P}\cdot\dot{\mathbf{F}} &= \mathbf{S}\cdot\mathbf{F}^{\mathrm{T}}\dot{\mathbf{F}} \\ &= \frac{1}{2}\mathbf{S}\cdot\left(\dot{\mathbf{F}}^{\mathrm{T}}\mathbf{F} + \mathbf{F}^{\mathrm{T}}\dot{\mathbf{F}}\right) \\ &= \frac{1}{2}\mathbf{S}\cdot\dot{\mathbf{C}}, \end{aligned} \tag{1.24}$$

We can therefore rewrite (1.18) as

$$\dot{\psi}_R - \frac{1}{2}\mathbf{S}\cdot\dot{\mathbf{C}} \leq 0, \tag{1.25}$$

1.8 Assumption of form-invariance

It is reasonable to assume that two observers between whom the change of observer is described by the transformation (1.20) will measure the same energy density field over $\mathcal{B}$ for the same motion, that is

$$\psi_R^* = \psi_R, \tag{1.26}$$

therefore, the free energy is also invariant.

In accordance with the principle of material frame-indifference (1.19), we assume that the functional forms of the constitutive equations are the same for all observers. In other words, if, for example, the free energy depends on the deformation gradient $\mathbf{F}$, then the following must hold:

$$\psi_R = \hat{\psi}_R(\mathbf{F}), \qquad \psi_R^* = \hat{\psi}_R(\mathbf{F}^*), \tag{1.27}$$

where $\mathbf{F}^*$ is the deformation gradient transformed following a change of observer.

From relations (1.26) and (1.27), it immediately follows that

$$\hat{\psi}_R(\mathbf{QF}) = \hat{\psi}_R(\mathbf{F}) \qquad \forall\, \mathbf{Q} \in \mathrm{Rot}, \tag{1.28}$$

meaning that the constitutive function $\hat{\psi}_R$ is invariant under rigid rotations of the reference frame. This implies that the dependence of the energy function on $\mathbf{F}$ can be expressed solely in terms of the invariant combination $\mathbf{F}^{\mathrm{T}}\mathbf{F} \equiv \mathbf{C}$, namely

$$\psi_R = \hat{\psi}_R(\mathbf{F}) = \bar{\psi}_R(\mathbf{C}), \tag{1.29}$$

1.9 Constitutive theory

Since elastic materials are assumed to have a constitutive response that depends solely on the current kinematic state of the material, the following special set of constitutive equations for ψ_R and $\mathbf{S}$ is adopted:

$$\begin{cases} \psi_R = \hat{\psi}_R(\mathbf{F}) = \bar{\psi}_R(\mathbf{C}), \\ \mathbf{S} = \bar{\mathbf{S}}(\mathbf{C}), \end{cases} \tag{1.30}$$

Furthermore, since $\mathbf{C}$ and $\mathbf{S}$ are invariant under a change of observer[1], the constitutive equation $(1.30)_2$ defining the second Piola-Kirchhoff stress tensor is also invariant under a change of reference frame.

1.10 Thermodynamic constraints

Recalling that elastic materials do not dissipate energy, and given that

$$\dot{\psi}_R(\mathbf{C}) = \frac{\partial \bar{\psi}_R(\mathbf{C})}{\partial \mathbf{C}} \cdot \dot{\mathbf{C}}, \tag{1.31}$$

the local form of the dissipation inequality (1.25), in the non-dissipative case, requires that the constitutive equations (1.30) satisfy the following condition:

$$\left(\frac{1}{2}\mathbf{S}(\mathbf{C}) - \frac{\partial \bar{\psi}_R(\mathbf{C})}{\partial \mathbf{C}}\right) \cdot \dot{\mathbf{C}} = 0. \tag{1.32}$$

Since this expression must be satisfied for every $\mathbf{C}$ and for every $\dot{\mathbf{C}}$, it is deduced

[1]The invariance property of $\mathbf{S}$ is not proven here, but it is easily verifiable.

that

$$\mathbf{S} = \bar{\mathbf{S}}(\mathbf{C}) = 2\,\frac{\partial \bar{\psi}_R(\mathbf{C})}{\partial \mathbf{C}}. \tag{1.33}$$

Subsequently, since $\mathbf{P} = \mathbf{F}\mathbf{S}$, the first Piola-Kirchhoff stress tensor is given by a constitutive equation of the form

$$\mathbf{P} = 2\,\mathbf{F}\,\frac{\partial \bar{\psi}_R(\mathbf{C})}{\partial \mathbf{C}}. \tag{1.34}$$

Similarly, using (3.10) we obtain an expression for the Cauchy stress tensor:

$$\boldsymbol{\sigma} = 2J^{-1}\mathbf{F}\,\frac{\partial \bar{\psi}_R(\mathbf{C})}{\partial \mathbf{C}}\,\mathbf{F}^{\mathrm{T}}. \tag{1.35}$$

Materials consistent with relations (1.33), (1.34), and (1.35) are commonly called *hyperelastic*.

It will be useful for us, in the remainder of this work, to define the constitutive relations without making the direct dependence of the energy on $\mathbf{C}$ explicit. It is easily obtained that

$$\mathbf{P} = \frac{\partial \hat{\psi}_R(\mathbf{F})}{\partial \mathbf{F}}, \tag{1.36}$$

and, using relation (1.11) again, we have

$$\boldsymbol{\sigma} = J^{-1}\frac{\partial \hat{\psi}_R(\mathbf{F})}{\partial \mathbf{F}}\mathbf{F}^{\mathrm{T}}. \tag{1.37}$$

2 Material symmetries, isotropy, incompressibility. Magnetoelastic materials

This chapter summarizes the fundamental concepts related to material symmetries, isotropy, and incompressibility, largely following the approach of Anand and Govindjee [1]. The sections dedicated to transversely isotropic materials and magnetoelastic materials instead follow the treatment by Dorfmann and Ogden [10].

2.1 Material symmetry

A *material symmetry* is defined as a rotation of the reference configuration that leaves the material's response to deformation unchanged. Since, for an elastic material, the stress response is determined by the free energy function, it is sufficient to formulate the definition of symmetry directly in terms of the latter.

Let $\mathbf{Y}$ be an arbitrary point in $\mathcal{B}$, and let $\chi_{\mathbf{F}}$ be a *homogeneous deformation* with deformation gradient $\mathbf{F}$:

$$\chi_{\mathbf{F}}(\mathbf{X}) = \mathbf{Y} + \mathbf{F}(\mathbf{X} - \mathbf{Y}). \tag{2.1}$$

Then, given a rotation $\mathbf{Q}$, we consider the following ideal experiments:

- **Experiment 1.** Deform $\mathcal{B}$ with the homogeneous deformation $\chi_{\mathbf{F}}$. In this case, the deformation gradient is $\mathbf{F}$, and the free energy is

$$\psi_{R1} = \hat{\psi}_R(\mathbf{F}) = \bar{\psi}_R(\mathbf{C}). \tag{2.2}$$

- **Experiment 2.** First, transform $\mathcal{B}$ with a rotation $\mathbf{Q}$ through the homogeneous deformation

$$\chi_{\mathbf{Q}}(\mathbf{X}) = \mathbf{Y} + \mathbf{Q}(\mathbf{X} - \mathbf{Y}),$$

  and then deform the rotated body with the same homogeneous deformation $\chi_{\mathbf{F}}$ as in experiment 1. In this experiment, the composite deformation gradient relative to the original reference body is

$$\mathbf{F}_2 = \text{Grad}\ \chi_{\mathbf{F}}(\chi_{\mathbf{Q}}(\mathbf{X})) = \mathbf{F}\mathbf{Q}. \tag{2.3}$$

  Then

$$\mathbf{C}_2 = \mathbf{F}_2^{\mathrm{T}}\mathbf{F}_2 = \mathbf{Q}^{\mathrm{T}}\mathbf{F}^{\mathrm{T}}\mathbf{F}\mathbf{Q} = \mathbf{Q}^{\mathrm{T}}\mathbf{C}\mathbf{Q}. \tag{2.4}$$

The free energy in the second experiment is therefore given by

$$\psi_{R2} = \hat{\psi}_R(\mathbf{FQ}) = \bar{\psi}_R(\mathbf{Q}^{\mathrm{T}}\mathbf{CQ}). \tag{2.5}$$

In general, we would not expect ψ_{R1} to be equal to ψ_{R2}; however, if for a given rotation $\mathbf{Q}$, $\psi_{R1} = \psi_{R2}$ for every $\mathbf{F}$, then $\mathbf{Q}$ is a material symmetry or, simply, a symmetry.

Thus, by (2.2) and (2.5), a *symmetry* is a rotation $\mathbf{Q}$ such that

$$\hat{\psi}_R(\mathbf{F}) = \hat{\psi}_R(\mathbf{FQ}) \tag{2.6}$$

for every deformation gradient $\mathbf{F}$; or, equivalently

$$\bar{\psi}_R(\mathbf{C}) = \bar{\psi}_R(\mathbf{Q}^{\mathrm{T}}\mathbf{CQ}) \tag{2.7}$$

for every symmetric and positive definite tensor $\mathbf{C}$.

2.2 Isotropy

We say that a material is *isotropic* if *every rotation is a material symmetry*.

In other words, if the body $\mathcal{B}$ is composed of an isotropic material, then

$$\bar{\psi}_R(\mathbf{Q}^{\mathrm{T}}\mathbf{CQ}) = \bar{\psi}_R(\mathbf{C}) \tag{2.9}$$

for all rotations $\mathbf{Q}$ and for all symmetric and positive definite tensors $\mathbf{C}$.

Note that (2.9) trivially holds even when substituting $\mathbf{Q}$ with $-\mathbf{Q}$; consequently, the expression "for all rotations" can be replaced by "for all orthogonal tensors". A scalar or tensor function invariant with respect to the entire orthogonal group Orth is called an *isotropic function*. The properties and canonical forms of isotropic functions, along with their proofs, are reported in the Appendix.

Since relation (2.9) must hold for all rotations $\mathbf{Q}$, we can, without loss of generality, choose $\mathbf{Q} = \mathbf{R}^{\mathrm{T}}$, where $\mathbf{R}$ is the rotation tensor in the *polar decomposition*[1] of the deformation gradient,

$$\mathbf{F} = \mathbf{RU} = \mathbf{VR}. \tag{2.10}$$

[1]For a detailed definition and the properties of the polar decomposition, see for example [1].

Since it is easily verifiable that

$$\mathbf{B} = \mathbf{R}\mathbf{C}\mathbf{R}^{\mathrm{T}}, \tag{2.11}$$

we conclude that

$$\bar{\psi}_R(\mathbf{C}) = \bar{\psi}_R(\mathbf{B}), \tag{2.12}$$

and therefore, for isotropic materials, we can also express the free energy in terms of $\mathbf{B}$:

$$\psi_R = \bar{\psi}_R(\mathbf{B}). \tag{2.13}$$

We also know that

$$\dot{\psi}_R = \frac{1}{2}\mathbf{S} \cdot \dot{\mathbf{C}} = \mathbf{P} \cdot \dot{\mathbf{F}}, \tag{2.14}$$

and thus

$$\frac{\partial \bar{\psi}_R(\mathbf{B})}{\partial \mathbf{B}} \cdot \dot{\mathbf{B}} = \mathbf{P} \cdot \dot{\mathbf{F}}. \tag{2.15}$$

Furthermore,

$$\dot{\mathbf{B}} = \dot{\mathbf{F}}\mathbf{F}^{\mathrm{T}} + \mathbf{F}\dot{\mathbf{F}}^{\mathrm{T}} = 2\ \mathrm{sym}[\dot{\mathbf{F}}\mathbf{F}^{\mathrm{T}}], \tag{2.16}$$

and, since

$$\frac{\partial \bar{\psi}_R(\mathbf{B})}{\partial \mathbf{B}} \tag{2.17}$$

is symmetric,

$$\frac{\partial \bar{\psi}_R(\mathbf{B})}{\partial \mathbf{B}} \cdot \dot{\mathbf{B}} = 2\,\frac{\partial \bar{\psi}_R(\mathbf{B})}{\partial \mathbf{B}} \cdot (\dot{\mathbf{F}}\mathbf{F}^{\mathrm{T}}) = 2\left[\frac{\partial \bar{\psi}_R(\mathbf{B})}{\partial \mathbf{B}}\mathbf{F}\right] \cdot \dot{\mathbf{F}}. \tag{2.18}$$

Therefore,

$$\left[2\frac{\partial \bar{\psi}_R(\mathbf{B})}{\partial \mathbf{B}}\mathbf{F} - \mathbf{P}\right] \cdot \dot{\mathbf{F}} = 0, \tag{2.19}$$

and since $\dot{\mathbf{F}}$ is arbitrary, we arrive at the relation

$$\mathbf{P} = 2\frac{\partial \bar{\psi}_R(\mathbf{B})}{\partial \mathbf{B}}\mathbf{F}. \tag{2.20}$$

Finally, since $\boldsymbol{\sigma} = J^{-1}\mathbf{P}\mathbf{F}^{\mathrm{T}}$, we obtain an expression that provides the Cauchy stress as a function of $\mathbf{B}$ alone:

$$\boldsymbol{\sigma} = 2J^{-1}\frac{\partial\bar{\psi}_R(\mathbf{B})}{\partial\mathbf{B}}\mathbf{B}. \tag{2.21}$$

2.2.1 Free energy for isotropic materials expressed as a function of invariants

Using the representation theorem for isotropic scalar functions (see the Appendix), an isotropic scalar function of a symmetric tensor $\mathbf{B}$ can be expressed as a function of its principal invariants $I_1(\mathbf{B})$, $I_2(\mathbf{B})$, and $I_3(\mathbf{B})$. The constitutive relation for the free energy of an isotropic elastic solid can therefore be written in the form

$$\psi_R = \bar{\psi}_R(\mathcal{I}_{\mathbf{B}}), \tag{2.22}$$

where

$$\mathcal{I}_{\mathbf{B}} := \big(I_1(\mathbf{B}),\, I_2(\mathbf{B}),\, I_3(\mathbf{B})\big). \tag{2.23}$$

and

$$I_1(\mathbf{B}) = \mathrm{tr}(\mathbf{B}), \qquad I_2(\mathbf{B}) = \frac{1}{2}\big[(\mathrm{tr}\,\mathbf{B})^2 - \mathrm{tr}(\mathbf{B}^2)\big], \qquad I_3(\mathbf{B}) = \det(\mathbf{B}). \tag{2.24}$$

Then, from (2.21),

$$\boldsymbol{\sigma} = \frac{2}{\sqrt{I_3}}\left[\frac{\partial\bar{\psi}_R(\mathcal{I}_{\mathbf{B}})}{\partial\mathbf{B}}\right]\mathbf{B}. \tag{2.25}$$

The known expressions for the derivatives of the principal invariants of $\mathbf{B}$ are reported below, omitting the calculation steps for the sake of brevity:

$$\frac{\partial I_1(\mathbf{B})}{\partial\mathbf{B}} = \mathbf{I}, \quad \frac{\partial I_2(\mathbf{B})}{\partial\mathbf{B}} = I_1(\mathbf{B})\mathbf{I} - \mathbf{B}, \quad \frac{\partial I_3(\mathbf{B})}{\partial\mathbf{B}} = I_3(\mathbf{B})\mathbf{B}^{-1}. \tag{2.26}$$

Using the identities (2.26), the following expression for the Cauchy stress tensor in an isotropic material is obtained:

$$\boldsymbol{\sigma} = \frac{2}{\sqrt{I_3}}\left[\frac{\partial\bar{\psi}_R(\mathcal{I}_{\mathbf{B}})}{\partial I_1}\mathbf{I} + \frac{\partial\bar{\psi}_R(\mathcal{I}_{\mathbf{B}})}{\partial I_2}(I_1\mathbf{I} - \mathbf{B}) + I_3\frac{\partial\bar{\psi}_R(\mathcal{I}_{\mathbf{B}})}{\partial I_3}\mathbf{B}^{-1}\right]\mathbf{B}. \tag{2.27}$$

An alternative to (2.26) follows from the *Cayley-Hamilton equation*:

$$\mathbf{B}^3 - I_1(\mathbf{B})\mathbf{B}^2 + I_2(\mathbf{B})\mathbf{B} - I_3(\mathbf{B})\mathbf{I} = \mathbf{0}. \tag{2.28}$$

Multiplying (2.28) by $\mathbf{B}^{-1}$, we obtain

$$\mathbf{B}^2 = I_1\mathbf{B} - I_2\mathbf{I} + I_3\mathbf{B}^{-1}. \tag{2.29}$$

Using this relation to eliminate $\mathbf{B}^2$ from (2.27), it is found that

$$\boldsymbol{\sigma} = \beta_0(\mathcal{I}_\mathbf{B})\mathbf{I} + \beta_1(\mathcal{I}_\mathbf{B})\mathbf{B} + \beta_2(\mathcal{I}_\mathbf{B})\mathbf{B}^{-1}, \tag{2.30}$$

hence, by virtue of the representation theorem (see again the Appendix), $\boldsymbol{\sigma}$ turns out to be an isotropic function of $\mathbf{B}$.

The scalar quantities $\beta_i(\mathcal{I}_\mathbf{B})$ are defined by:

$$\begin{aligned}
\beta_0(\mathcal{I}_\mathbf{B}) &= \frac{2}{\sqrt{I_3}}\left(I_2\frac{\partial\bar{\psi}_R(\mathcal{I}_\mathbf{B})}{\partial I_2} + I_3\frac{\partial\bar{\psi}_R(\mathcal{I}_\mathbf{B})}{\partial I_3}\right),\\
\beta_1(\mathcal{I}_\mathbf{B}) &= \frac{2}{\sqrt{I_3}}\frac{\partial\bar{\psi}_R(\mathcal{I}_\mathbf{B})}{\partial I_1},\\
\beta_2(\mathcal{I}_\mathbf{B}) &= -2\sqrt{I_3}\frac{\partial\bar{\psi}_R(\mathcal{I}_\mathbf{B})}{\partial I_2}.
\end{aligned} \tag{2.31}$$

2.3 Transverse isotropy

Having discussed the case of isotropic materials, we now consider an important extension, which concerns materials endowed with a preferential direction: *transversely isotropic materials*.

A transversely isotropic material can be described by introducing a unit vector $\mathbf{M}$ in the reference configuration, which identifies the preferential direction. In the absence of this direction, the material would be isotropic with respect to $\mathcal{B}$. In general, $\mathbf{M}$ varies with position $\mathbf{X}$ and is a vector field that can be interpreted as a continuous representation of the fibers.

The material response is therefore indifferent to arbitrary rotations about the direction $\mathbf{M}$. Furthermore, it is customary to make no physical distinction between the directions $\mathbf{M}$ and $-\mathbf{M}$. Consequently, the response must not be affected by exchanging $\mathbf{M}$ with $-\mathbf{M}$ and can be represented in terms of $\mathbf{M}\otimes\mathbf{M}$.

The strain energy density $\hat{\psi}_R(\mathbf{F})$ must satisfy

$$\hat{\psi}_R(\mathbf{FQ}) = \hat{\psi}_R(\mathbf{F}) \tag{2.32}$$

for all rotations $\mathbf{Q}$ (in the reference configuration) such that $\mathbf{QM} = \pm\mathbf{M}$. Equiva-

lently, such a material can be characterized by a strain energy density that is an isotropic function of $\mathbf{F}$ and the tensor $\mathbf{M} \otimes \mathbf{M}$ jointly.

Since we know that $\hat{\psi}_R$ depends on $\mathbf{F}$ only through $\mathbf{C}$, this means that, writing the dependence as $\check{\psi}_R(\mathbf{C}, \mathbf{M} \otimes \mathbf{M})$, we must have

$$\check{\psi}_R(\mathbf{Q}\mathbf{C}\mathbf{Q}^{\mathrm{T}}, \mathbf{Q}\mathbf{M} \otimes \mathbf{Q}\mathbf{M}) = \check{\psi}_R(\mathbf{C}, \mathbf{M} \otimes \mathbf{M}) \quad \forall\, \mathbf{Q} \in \mathrm{Orth}. \tag{2.33}$$

Requirement (2.33) implies, by the representation theorem for isotropic scalar functions (see the Appendix), that $\check{\psi}_R$ depends on five invariants, namely the principal invariants I_1, I_2, I_3 of $\mathbf{C}$ together with two independent invariants that depend on $\mathbf{M}$. It is customary to take them as $I_4 = \mathbf{C}\mathbf{M} \cdot \mathbf{M}$ and $I_5 = \mathbf{C}^2\mathbf{M} \cdot \mathbf{M}$.

We use the notation

$$\check{\psi}_R(I_1, I_2, I_3, I_4, I_5) \tag{2.34}$$

to represent the strain energy when it is treated as a function of the aforementioned invariants.

To calculate the stresses, the derivatives are needed:

$$\frac{\partial I_4}{\partial \mathbf{F}} = 2\mathbf{M} \otimes \mathbf{F}\mathbf{M}, \quad \frac{\partial I_5}{\partial \mathbf{F}} = 2\big(\mathbf{M} \otimes \mathbf{F}\mathbf{C}\mathbf{M} + \mathbf{C}\mathbf{M} \otimes \mathbf{F}\mathbf{M}\big), \tag{2.35}$$

together with the derivatives of I_1, I_2, I_3 which can be quickly derived. The Piola tensor, recalling that $\mathbf{P} = \partial\psi_R/\partial\mathbf{F}$ and applying the chain rule, will be given by

$$\begin{aligned}\mathbf{P} = {} & 2\check{\psi}_1\mathbf{F} + 2\check{\psi}_2\mathbf{F}(I_1\mathbf{I} - \mathbf{C}) + 2I_3\check{\psi}_3\mathbf{F}^{-\mathrm{T}} + 2\check{\psi}_4\mathbf{F}\mathbf{M} \otimes \mathbf{M} \\ & + 2\check{\psi}_5\big(\mathbf{F}\mathbf{C}\mathbf{M} \otimes \mathbf{M} + \mathbf{F}\mathbf{M} \otimes \mathbf{C}\mathbf{M}\big),\end{aligned} \tag{2.36}$$

where $\check{\psi}_i = \partial\check{\psi}_R/\partial I_i$, $i = 1, \ldots, 5$. The result for an isotropic material is obtained by omitting the terms in $\check{\psi}_4$ and $\check{\psi}_5$.

Equation (2.36) provides the stresses in a fiber-reinforced material for which the fiber direction corresponds to $\mathbf{M}$ locally in the reference configuration. The Cauchy stress tensor is obtained from (2.36) using the general formula $\boldsymbol{\sigma} = J^{-1}\mathbf{P}\mathbf{F}^{\mathrm{T}}$, which yields

$$\begin{aligned}\boldsymbol{\sigma} = {} & 2J^{-1}\big[\check{\psi}_1\mathbf{B} + \check{\psi}_2(I_1\mathbf{B} - \mathbf{B}^2) + I_3\check{\psi}_3\mathbf{I} \\ & + \check{\psi}_4\mathbf{m} \otimes \mathbf{m} + \check{\psi}_5(\mathbf{B}\mathbf{m} \otimes \mathbf{m} + \mathbf{m} \otimes \mathbf{B}\mathbf{m})\big],\end{aligned} \tag{2.37}$$

where $\mathbf{m} = \mathbf{F}\mathbf{M}$ is the preferential direction in the deformed configuration.

2.4 Incompressibility

An *incompressible material* is a material for which only isochoric motions are possible. This constraint is expressed mathematically as

$$J = \det \mathbf{F} \equiv 1. \tag{2.38}$$

This constraint guarantees that every part $\mathcal{P} \subset \mathcal{B}$ deforms into a spatial region $\mathcal{P}_t$ with the same volume; that is, $\mathrm{vol}(\mathcal{P}_t) = \mathrm{vol}(\mathcal{P})$ for every $\mathcal{P} \subset \mathcal{B}$. An immediate consequence of (2.38) is that

$$\det \mathbf{C} \equiv 1 \quad \text{and} \quad \det \mathbf{B} \equiv 1. \tag{2.39}$$

In what follows, it will be useful to note that

$$0 = \dot{\overline{\det \mathbf{C}}} = (\det \mathbf{C})\mathbf{C}^{-1} \cdot \dot{\mathbf{C}} \tag{2.40}$$

for an incompressible material. The constraint (2.39) is therefore equivalent to

$$\mathbf{C}^{-1} \cdot \dot{\mathbf{C}} = 0. \tag{2.41}$$

2.4.1 Incompressible elastic materials

The constitutive hypothesis for elastic materials (1.33) together with the thermodynamic requirement that elastic materials do not dissipate energy leads to

$$\left(\frac{1}{2}\mathbf{S}(\mathbf{C}) - \frac{\partial \bar{\psi}_R(\mathbf{C})}{\partial \mathbf{C}} \right) \cdot \dot{\mathbf{C}} = 0. \tag{2.42}$$

Now, for incompressible materials, $\mathbf{C}$ is restricted to the set of positive definite tensors with a determinant identically equal to one, and $\mathbf{C}$ satisfies $\mathbf{C}^{-1} \cdot \dot{\mathbf{C}} = 0$. Therefore, unlike (1.32), we cannot conclude that the quantity in brackets in (2.42) must be zero. Instead, we can conclude that it must be a scalar multiple of $\mathbf{C}^{-1}$. Without loss of generality, we assume the scalar multiplier to be $-(1/2)p$, that is

$$\frac{1}{2}\mathbf{S}(\mathbf{C}) - \frac{\partial \bar{\psi}_R(\mathbf{C})}{\partial \mathbf{C}} = -\frac{1}{2} p\, \mathbf{C}^{-1}, \tag{2.43}$$

where p is an arbitrary scalar field.

Thus, the second Piola-Kirchhoff stress tensor is given by

$$\mathbf{S} = \mathbf{S}(\mathbf{C}) = -p\,\mathbf{C}^{-1} + 2\frac{\partial \bar{\psi}_R(\mathbf{C})}{\partial \mathbf{C}}. \tag{2.44}$$

Equivalently, recalling that $\mathbf{P} = \mathbf{F}\mathbf{S}$ and $\boldsymbol{\sigma} = J^{-1}\mathbf{P}\mathbf{F}^{\mathrm{T}}$,

$$\mathbf{P} = -p\mathbf{F}^{-\mathrm{T}} + 2\mathbf{F}\frac{\partial \bar{\psi}_R(\mathbf{C})}{\partial \mathbf{C}}, \quad \boldsymbol{\sigma} = -p\mathbf{I} + 2\mathbf{F}\frac{\partial \bar{\psi}_R(\mathbf{C})}{\partial \mathbf{C}}\mathbf{F}^{\mathrm{T}}, \tag{2.45}$$

where p remains an arbitrary scalar field; note that $J = 1$ in the incompressible case.

Following a similar procedure without making the dependence on $\mathbf{C}$ explicit, since for an incompressible material

$$\dot{\overline{\det \mathbf{F}}} = \mathbf{F}^{-\mathrm{T}} \cdot \dot{\mathbf{F}} \tag{2.46}$$

then it trivially follows that

$$\mathbf{P} = -p\mathbf{F}^{-\mathrm{T}} + \frac{\partial \hat{\psi}_R(\mathbf{F})}{\partial \mathbf{F}} \qquad \boldsymbol{\sigma} = -p\mathbf{I} + \frac{\partial \hat{\psi}_R(\mathbf{F})}{\partial \mathbf{F}}\mathbf{F}^{\mathrm{T}} \tag{2.47}$$

Free energy for incompressible isotropic and anisotropic elastic materials

Let us now assume that the body is incompressible and isotropic. In this case, keeping in mind the condition $J = 1$, the free energy function can be written as

$$\bar{\psi}_R(\mathbf{B}) = \tilde{\psi}_R(\mathcal{I}_{\mathbf{B}}), \tag{2.48}$$

where

$$\mathcal{I}_{\mathbf{B}} := (I_1(\mathbf{B}),\, I_2(\mathbf{B})) \tag{2.49}$$

represents the list of non-trivial principal invariants.

The Cauchy stress tensor then takes the form

$$\boldsymbol{\sigma} = -p\mathbf{I} + 2\,\frac{\partial \tilde{\psi}_R(\mathcal{I}_{\mathbf{B}})}{\partial \mathbf{B}}\mathbf{B}, \tag{2.50}$$

which can be equivalently expressed as

$$\boldsymbol{\sigma} = -p\mathbf{I} + (\beta_1(\mathcal{I}_{\mathbf{B}}) - I_1\beta_2(\mathcal{I}_{\mathbf{B}}))\mathbf{B} + \beta_2(\mathcal{I}_{\mathbf{B}})\mathbf{B}^2, \tag{2.51}$$

or, in an alternative form,

$$\boldsymbol{\sigma} = -p\mathbf{I} + \beta_1(\mathcal{I}_{\mathbf{B}})\mathbf{B} + \beta_2(\mathcal{I}_{\mathbf{B}})\mathbf{B}^{-1}, \tag{2.52}$$

where

$$\beta_1(\mathcal{I}_{\mathbf{B}}) = 2\,\frac{\partial\tilde{\psi}_R(\mathcal{I}_{\mathbf{B}})}{\partial I_1}, \qquad \beta_2(\mathcal{I}_{\mathbf{B}}) = -2\,\frac{\partial\tilde{\psi}_R(\mathcal{I}_{\mathbf{B}})}{\partial I_2}. \tag{2.53}$$

In the case of an incompressible transversely isotropic material, for which it is assumed $\psi_R = \check{\psi}_R(I_1, I_2, I_4, I_5)$, the corresponding relation for the Cauchy stress tensor becomes

$$\boldsymbol{\sigma} = 2\check{\psi}_1\mathbf{B} + 2\check{\psi}_2(I_1\mathbf{B} - \mathbf{B}^2) + 2\check{\psi}_4\mathbf{m}\otimes\mathbf{m} + 2\check{\psi}_5(\mathbf{B}\mathbf{m}\otimes\mathbf{m} + \mathbf{m}\otimes\mathbf{B}\mathbf{m}) - p\,\mathbf{I}, \tag{2.54}$$

where $\check{\psi}_i = \partial\check{\psi}_R/\partial I_i$ for $i = 1, \ldots, 5$.

2.5 Magnetoelastic materials

Having discussed the purely elastic cases, we now extend the treatment to *magnetoelastic materials*, in which mechanical and magnetic effects are coupled.

For a complete formulation of the theory, including balance equations and constitutive relations, the reader is referred to the text by Dorfmann and Ogden [10].

In this section, we will provide a summary of the fundamental equations of magnetoelasticity, both in the Eulerian and Lagrangian formulations, and illustrate the corresponding boundary conditions.

2.5.1 Eulerian formulation

Maxwell's equations for magnetostatics, considering a deformed magnetizable material occupying the region $\mathcal{B}_t$ at time t, are

$$\operatorname{curl}\mathbf{h} = \mathbf{j}_{\mathrm{f}}, \quad \operatorname{div}\mathbf{b} = 0 \quad \text{in } \mathcal{B}_t, \tag{2.55}$$

where $\mathbf{h}$ is the magnetic field, $\mathbf{j}_{\mathrm{f}}$ is the free current density, and $\mathbf{b}$ is the magnetic induction field.

The boundary conditions are

$$[\![\mathbf{h}]\!] \times \mathbf{n} = \mathbf{k}_{\mathrm{f}}, \quad [\![\mathbf{b}]\!] \cdot \mathbf{n} = 0 \quad \text{on } \partial\mathcal{B}_t, \tag{2.56}$$

where $\mathbf{k}_{\mathrm{f}}$ is the free surface current density and $[\![\cdot]\!]$ denotes the "jump" of the quantity between the inside and the outside of the material, that is

$$[\![\mathbf{h}]\!] = \mathbf{h} - \mathbf{h}^*, \quad [\![\mathbf{b}]\!] = \mathbf{b} - \mathbf{b}^*, \tag{2.57}$$

with $\mathbf{h}^*$ and $\mathbf{b}^* = \mu_0 \mathbf{h}^*$ representing the magnetic field and induction outside the material, respectively, and $\mathbf{n}$ is the unit normal vector to the surface $\partial\mathcal{B}_t$.

The linear momentum balance equation is given by

$$\operatorname{div} \boldsymbol{\tau} + \mathbf{f} = \mathbf{0} \quad \text{in } \mathcal{B}_t, \tag{2.58}$$

where $\boldsymbol{\tau}$ is the *Eulerian total magnetic stress tensor* (or *Cauchy stress tensor*), which includes both the mechanical and magnetic parts:

$$\boldsymbol{\tau} = \boldsymbol{\sigma} + \boldsymbol{\tau}_m, \tag{2.59}$$

with $\boldsymbol{\sigma}$ being the Cauchy stress tensor (mechanical stress) and $\boldsymbol{\tau}_m$ the *Maxwell stress tensor*, defined as

$$\boldsymbol{\tau}_m := \mathbf{b} \otimes \mathbf{h} - \frac{1}{2}\mu_0(\mathbf{h} \cdot \mathbf{h})\mathbf{I}. \tag{2.60}$$

The body forces $\mathbf{f}$ are of a mechanical nature.

The angular momentum balance equation is

$$\boldsymbol{\tau} = \boldsymbol{\tau}^{\mathrm{T}}, \tag{2.61}$$

which indicates the symmetry of the total stress tensor $\boldsymbol{\tau}$.

The boundary condition on the surface traction is

$$\boldsymbol{\tau}\mathbf{n} = \mathbf{t}_a + \mathbf{t}_m^* \quad \text{on } \partial\mathcal{B}_t^t, \tag{2.62}$$

where $\mathbf{t}_m^* = \boldsymbol{\tau}_m^* \mathbf{n}$ represents the magnetic traction, with $\boldsymbol{\tau}_m^*$ being the Maxwell stress tensor outside the body and $\mathbf{t}_a$ the mechanical surface traction.

Finally, the boundary condition on the displacement field $\mathbf{u}$ is

$$\mathbf{u} = \tilde{\mathbf{u}} \quad \text{on } \partial\mathcal{B}_t^u. \tag{2.63}$$

Let it be clear that, although it is possible to formulate all equations also in terms of the Cauchy stress tensor $\boldsymbol{\sigma}$, the choice to use $\boldsymbol{\tau}$ is more appropriate from a physical point of view, since the total stress tensor is the one that best describes the

overall interaction between mechanical and magnetic effects in the material.

2.5.2 Lagrangian formulation

We now turn to the Lagrangian formulation of the problem.

Writing equations (2.55) in integral form, we obtain the following conditions for the magnetic fields in the deformed configuration:

$$0 = \int_{\mathcal{B}_t} \operatorname{div} \mathbf{b} = \int_{\partial\mathcal{B}_t} \mathbf{b} \cdot \mathbf{n}, \tag{2.64}$$

$$\int_{\mathcal{S}_t} \mathbf{j}_{\mathrm{f}} \cdot \mathbf{n} = \int_{\mathcal{S}_t} \operatorname{curl} \mathbf{h} \cdot \mathbf{n} = \int_{\partial\mathcal{S}_t} \mathbf{h} \cdot d\mathbf{x}, \tag{2.65}$$

where $\mathcal{B}_t$ and $\partial\mathcal{B}_t$ represent a volume and its boundary in the deformed configuration, while $\mathcal{S}_t$ and $\partial\mathcal{S}_t$ are a surface and its bounding closed curve. Here, $\mathbf{n}$ is the unit normal vector to the surface in both equations. We have applied the divergence theorem and Stokes' theorem.

Given the reference volume $\mathcal{B}$ and the reference surface $\mathcal{S}$, and considering Nanson's formula (1.5) and the relation $d\mathbf{x} = \mathbf{F}d\mathbf{X}$ for line elements, we obtain the following expressions:

$$0 = \int_{\mathcal{B}} \left(J\mathbf{F}^{-1}\mathbf{b}\right) \cdot \mathbf{N} = \int_{\mathcal{B}} \operatorname{Div} \left(J\mathbf{F}^{-1}\mathbf{b}\right), \tag{2.66}$$

$$\int_{\mathcal{S}} (J\mathbf{F}^{-1}\mathbf{j}_{\mathrm{f}}) \cdot \mathbf{N} = \int_{\partial\mathcal{S}} \left(\mathbf{F}^{\mathrm{T}}\mathbf{h}\right) \cdot d\mathbf{X} = \int_{\mathcal{S}} \operatorname{Curl} \left(\mathbf{F}^{\mathrm{T}}\mathbf{h}\right) \cdot \mathbf{N}, \tag{2.67}$$

where we have again used the divergence theorem and Stokes' theorem.

Localizing the equations, we obtain:

$$\operatorname{Div} \left(J\mathbf{F}^{-1}\mathbf{b}\right) = 0, \quad \operatorname{Curl} \left(\mathbf{F}^{\mathrm{T}}\mathbf{h}\right) = J\mathbf{F}^{-1}\mathbf{j}_{\mathrm{f}} \quad \text{in } \mathcal{B}. \tag{2.68}$$

By definition, the Lagrangian fields are expressed as:

$$\mathbf{h}_l := \mathbf{F}^{\mathrm{T}}\mathbf{h}, \quad \mathbf{b}_l := J\mathbf{F}^{-1}\mathbf{b}, \quad \mathbf{j}_{\mathrm{F}} := J\mathbf{F}^{-1}\mathbf{j}_{\mathrm{f}} \tag{2.69}$$

The balance equations in the Lagrangian formulation therefore become:

$$\operatorname{Curl} \mathbf{h}_l = \mathbf{j}_{\mathrm{F}}, \quad \operatorname{Div} \mathbf{b}_l = 0 \quad \text{in } \mathcal{B}. \tag{2.70}$$

The boundary conditions are analogous to those of the Eulerian formulation:

$$[\![\mathbf{h}_l]\!] \times \mathbf{N} = \mathbf{k}_{\mathrm{F}}, \quad [\![\mathbf{b}_l]\!] \cdot \mathbf{N} = 0 \quad \text{on } \partial\mathcal{B}, \tag{2.71}$$

where $\mathbf{k}_{\mathrm{F}}$ is the free current density per unit referential area. Regarding the linear momentum balance, we introduce the *nominal* total magnetic stress tensor $\mathbf{T}$ as follows:

$$\mathbf{T} = J\boldsymbol{\tau}\mathbf{F}^{-\mathrm{T}}, \tag{2.72}$$

which leads to the following balance equation:

$$\mathrm{Div}\,\mathbf{T} + \mathbf{f}_R = \mathbf{0} \quad \text{in } \mathcal{B}, \tag{2.73}$$

where $\mathbf{f}_R$ represents the mechanical body forces per unit reference volume.

The angular momentum balance equation is given by:

$$\mathbf{T}\mathbf{F}^{\mathrm{T}} = \mathbf{F}\mathbf{T}^{\mathrm{T}} \quad \text{in } \mathcal{B}. \tag{2.74}$$

The boundary condition on the surface traction is:

$$\mathbf{T}\mathbf{N} = \mathbf{t}_A + \mathbf{t}^*_M \quad \text{on } \partial\mathcal{B}^t, \tag{2.75}$$

where $\mathbf{t}_A$ is the mechanical surface traction per unit reference area and $\mathbf{t}^*_M$ is the magnetic traction, defined as $\mathbf{t}^*_M = \mathbf{T}^*_M\mathbf{N}$, with $\mathbf{T}^*_M = J\boldsymbol{\tau}^*_m\mathbf{F}^{-\mathrm{T}}$.

Finally, the boundary condition on the displacement field:

$$\mathbf{u} = \tilde{\mathbf{u}} \quad \text{on } \partial\mathcal{B}^u. \tag{2.76}$$

2.5.3 Constitutive relations

For the sake of compactness and functionality, we will report only two constitutive formulations: the one in $\Omega^*(\mathbf{F}, \mathbf{b}_l)$ and $\Omega(\mathbf{F}, \mathbf{h}_l)$, depending on which magnetic state variable we choose to use. The energy functions Ω^* and Ω are expressed per unit reference volume. Without retracing all the steps leading to the constitutive formulation, the formulation in $\Omega^*(\mathbf{F}, \mathbf{b}_l)$ is given by:

$$\mathbf{T} = \frac{\partial\Omega^*}{\partial\mathbf{F}}, \quad \mathbf{h}_l = \frac{\partial\Omega^*}{\partial\mathbf{b}_l} \tag{2.77}$$

or, expressing the Eulerian variables:

$$\boldsymbol{\tau} = J^{-1}\frac{\partial\Omega^*}{\partial\mathbf{F}}\mathbf{F}^{\mathrm{T}}, \quad \mathbf{h} = \mathbf{F}^{-\mathrm{T}}\frac{\partial\Omega^*}{\partial\mathbf{b}_l} \tag{2.78}$$

One can pass from Ω^* to Ω via the *Legendre transformation*:

$$\Omega = \Omega^* - \mathbf{b}_l \cdot \mathbf{h}_l \tag{2.79}$$

In terms of Ω, we obtain the counterparts of (2.77):

$$\mathbf{T} = \frac{\partial\Omega}{\partial\mathbf{F}}, \quad \mathbf{b}_l = -\frac{\partial\Omega}{\partial\mathbf{h}_l} \tag{2.80}$$

2.5.4 Incompressibility

We are well aware that for an incompressible material, the incompressibility constraint is given by:

$$\det\mathbf{F} = 1. \tag{2.81}$$

By adopting a procedure similar to that for isotropic elastic materials, we obtain the following expressions for the total stress tensor and the nominal stress tensor, respectively:

$$\boldsymbol{\tau} = \frac{\partial\Omega^*}{\partial\mathbf{F}}\mathbf{F}^{\mathrm{T}} - p^*\mathbf{I}, \quad \mathbf{T} = \frac{\partial\Omega^*}{\partial\mathbf{F}} - p^*\mathbf{F}^{-\mathrm{T}} \tag{2.82}$$

where p^* is a scalar associated with the incompressibility constraint. Expressions $(2.77)_2$ and $(2.78)_2$ remain unchanged, except for the constraint $\det\mathbf{F} = 1$.

Passing to the formulation in terms of Ω, we obtain:

$$\boldsymbol{\tau} = \frac{\partial\Omega}{\partial\mathbf{F}}\mathbf{F}^{\mathrm{T}} - p\mathbf{I}, \quad \mathbf{T} = \frac{\partial\Omega}{\partial\mathbf{F}} - p\mathbf{F}^{-\mathrm{T}} \tag{2.83}$$

where, in general, the incompressibility multiplier p is different from p^*.

2.5.5 Isotropic magnetoelastic materials

A magnetoelastic material is said to be *isotropic* if the energy function Ω^* is an isotropic function of the two tensors $\mathbf{C}$ and $\mathbf{b}_l \otimes \mathbf{b}_l$ (or, equivalently, if the energy function Ω is an isotropic function of $\mathbf{C}$ and $\mathbf{h}_l \otimes \mathbf{h}_l$).

We note that the definition of an isotropic magnetoelastic material is completely analogous to that of a transversely isotropic material, where the preferential

direction is represented by the magnetic induction vector (or the magnetic field).

The form of Ω^* thus reduces to depending on the principal invariants I_1, I_2, I_3 of $\mathbf{C}$, together with three invariants depending on $\mathbf{b}_l$:

$$I_4 = \mathbf{b}_l \cdot \mathbf{b}_l, \quad I_5 = \mathbf{C}\mathbf{b}_l \cdot \mathbf{b}_l, \quad I_6 = \mathbf{C}^2\mathbf{b}_l \cdot \mathbf{b}_l. \tag{2.84}$$

It is important to specify that in the case of transverse isotropy, the corresponding invariant I_4 is unity, since the material's preferential direction is identified by the unit vector $\mathbf{M}$ (see Section 2.3).

In what follows, the subscripts $1, 2, \ldots, 6$ on Ω^* denote the derivative with respect to $I_1, I_2, I_3, I_4, I_5, I_6$. Then, expanding the expressions of (2.78), we obtain:

$$\begin{aligned} \boldsymbol{\tau} = J^{-1} \big[& 2\Omega_1^*\mathbf{B} + 2\Omega_2^*(I_1\mathbf{I} - \mathbf{B}^2) + 2I_3\Omega_3^*\mathbf{I} + 2\Omega_5^*\mathbf{b} \otimes \mathbf{b} \\ & + 2\Omega_6^*(\mathbf{b} \otimes \mathbf{B}\mathbf{b} + \mathbf{B}\mathbf{b} \otimes \mathbf{b}) \big] , \end{aligned} \tag{2.85}$$

where $\Omega_i^* = \frac{\partial \Omega^*}{\partial I_i}$, and

$$\mathbf{h} = \Omega_4^*\mathbf{B}^{-1}\mathbf{b} + \Omega_5^*\mathbf{b} + \Omega_6^*\mathbf{B}\mathbf{b}. \tag{2.86}$$

where, again, $\mathbf{B} = \mathbf{F}\mathbf{F}^{\mathrm{T}}$ is the left Cauchy-Green strain tensor. The corresponding forms in terms of Lagrangian variables are obtained from (2.72) and $(2.69)_1$.

For an incompressible material, $I_3 = 1$ and (2.85) becomes:

$$\begin{aligned} \boldsymbol{\tau} = {} & 2\Omega_1^*\mathbf{B} + 2\Omega_2^*(I_1\mathbf{I} - \mathbf{B}^2) - p^*\mathbf{I} + 2\Omega_5^*\mathbf{b} \otimes \mathbf{b} \\ & + 2\Omega_6^*(\mathbf{b} \otimes \mathbf{B}\mathbf{b} + \mathbf{B}\mathbf{b} \otimes \mathbf{b}), \end{aligned} \tag{2.87}$$

while (2.86) remains unchanged.

The analogy between (2.87) and (2.37) is evident, where the role of the fiber in the current configuration $\mathbf{m}$ is played by the Eulerian magnetic induction.

If, instead, we work with Ω rather than Ω^*, the invariants are now based on $\mathbf{h}_l$. The invariants I_1, I_2, I_3 are always the principal invariants of $\mathbf{C}$, while the invariants I_4, I_5, I_6, depending on $\mathbf{h}_l$, are now:

$$I_4 = \mathbf{h}_l \cdot \mathbf{h}_l, \quad I_5 = \mathbf{C}\mathbf{h}_l \cdot \mathbf{h}_l, \quad I_6 = \mathbf{C}^2\mathbf{h}_l \cdot \mathbf{h}_l. \tag{2.88}$$

The associated formulas for $\boldsymbol{\tau}$ are similar to those based on Ω^*, and we give only

those for an incompressible material, with $\Omega = \Omega(I_1, I_2, I_4, I_5, I_6)$. In particular, we have:

$$\boldsymbol{\tau} = 2\Omega_1 \mathbf{B} + 2\Omega_2 (I_1 \mathbf{I} - \mathbf{B}) - p\mathbf{I} + 2\Omega_5 \mathbf{Bh} \otimes \mathbf{Bh}$$
$$+2\Omega_6 \left(\mathbf{Bh} \otimes \mathbf{B}^2\mathbf{h} + \mathbf{B}^2\mathbf{h} \otimes \mathbf{Bh} \right). \tag{2.89}$$

The magnetic induction takes the form:

$$\mathbf{b} = -2 \left(\Omega_4 \mathbf{Bh} + \Omega_5 \mathbf{B}^2\mathbf{h} + \Omega_6 \mathbf{B}^3\mathbf{h} \right). \tag{2.90}$$

In the equations above, Ω_i is defined as $\frac{\partial \Omega}{\partial I_i}$, for $i = 1, 2, 4, 5, 6$.

When $\mathbf{b}_l$ is used as the independent magnetic variable, it must satisfy $\operatorname{Div} \mathbf{b}_l = 0$, and it is therefore convenient to use a vector potential $\mathbf{a}_l$, so that $\mathbf{b}_l = \operatorname{Curl} \mathbf{a}_l$. The resulting field $\mathbf{h}_l$, calculated from $(2.77)_2$, must satisfy the vector equation $\operatorname{Curl} \mathbf{h}_l = \mathbf{0}$.

On the other hand, if we start from $\mathbf{h}_l$ as the independent magnetic variable, it must satisfy $\operatorname{Curl} \mathbf{h}_l = \mathbf{0}$, hence we use a scalar potential function φ_l such that $\mathbf{h}_l = -\operatorname{Grad} \varphi_l$, and then the resulting field $\mathbf{b}_l$, calculated from $(2.80)_2$, must satisfy the scalar equation $\operatorname{Div} \mathbf{b}_l = 0$.

3 Universal Relations

In this chapter we will examine *universal relations* in the following contexts:

- *Isotropic elastic* materials,
- *Transversely isotropic elastic* materials,
- *Isotropic magnetoelastic* materials.

3.1 Universal relations in isotropic elastic materials

This section follows the treatment by Beatty [3].

Universal relations are *equations that hold for every material belonging to a given class*. Their importance is underscored by the fact that if the data obtained in a laboratory test properly designed for a given material do not satisfy said relation, that material cannot be considered a candidate for inclusion in that specific class.

Recalling the characteristic constitutive equation of isotropic elastic materials introduced in Section 2.2:

$$\boldsymbol{\sigma} = \beta_0 \mathbf{I} + \beta_1 \mathbf{B} + \beta_2 \mathbf{B}^{-1} \tag{3.1}$$

The three *response functions* β_i $(i = 0, 1, 2)$ depend, as we well know, on the three principal invariants of $\mathbf{B}$. We also recall that when the material is incompressible, β_0 must be replaced by an arbitrary pressure $-p$ and $\mathbf{B}$ must satisfy the incompressibility constraint $I_3 \equiv \det \mathbf{B} = 1$; the other two functions β_1 and β_2 depend on the remaining I_1 and I_2.

It is an immediate consequence of (3.1) that the following equation holds *for any isotropic elastic material, compressible or incompressible, whatever the form of the response functions* β_i:

$$\boldsymbol{\sigma}\mathbf{B} = \mathbf{B}\boldsymbol{\sigma}. \tag{3.2}$$

This simple rule states that the tensor $\boldsymbol{\sigma}\mathbf{B}$ is symmetric. Therefore, having defined a Cartesian basis defined by the unit vectors $\{\mathbf{e}_i\}$, $i = 1, 2, 3$, the general universal relation (3.2) translates into the following three scalar equations expressed in terms of the physical components $\sigma_{ij} = \boldsymbol{\sigma} \cdot \mathbf{e}_{ij}$ and $B_{ij} = \mathbf{B} \cdot \mathbf{e}_{ij}$, where $\mathbf{e}_{ij} := \mathbf{e}_i \otimes \mathbf{e}_j$:

$$
\begin{aligned}
B_{12}(\sigma_{11} - \sigma_{22}) &= (B_{11} - B_{22})\sigma_{12} + B_{13}\sigma_{32} - \sigma_{13}B_{32}, \\
B_{23}(\sigma_{22} - \sigma_{33}) &= (B_{22} - B_{33})\sigma_{23} + B_{21}\sigma_{13} - \sigma_{21}B_{13}, \\
B_{31}(\sigma_{33} - \sigma_{11}) &= (B_{33} - B_{11})\sigma_{31} + B_{32}\sigma_{21} - \sigma_{32}B_{21}.
\end{aligned}
\tag{3.3}
$$

It can be shown that property (3.2) is necessary and sufficient for the coincidence of the principal directions of the symmetric tensors $\boldsymbol{\sigma}$ and $\mathbf{B}$ (see Appendix).

However, (3.2) does not imply, in general, that $\boldsymbol{\sigma}$ and $\mathbf{B}$ have equal corresponding principal values. The result that, for an isotropic elastic material, the principal directions of $\boldsymbol{\sigma}$ coincide with those of $\mathbf{B}$, and that there exists a one-to-one correspondence between the eigenvalues of $\boldsymbol{\sigma}$ and the eigenvalues of $\mathbf{B}$, are consequences of (3.2) alone. But the most interesting result for the case of coincident principal values of $\boldsymbol{\sigma}$ requires more. The scalar relations (3.3) were applied by Batra [2] to show, conversely, that for an isotropic material (3.1) the eigenvectors of $\mathbf{B}$ coincide with those of $\boldsymbol{\sigma}$; and if two or three of the principal values of $\boldsymbol{\sigma}$ are equal, $\mathbf{B}$ must have corresponding equal principal values, provided the following empirical inequalities hold:

$$
\beta_1 > 0, \quad \beta_2 \leq 0. \tag{3.4}
$$

Note that this theorem itself expresses a property valid for any isotropic elastic material, compressible or incompressible, whose behavior is reasonably well-defined, in the sense of the inequalities (3.4), but otherwise unspecified; it is practically universal.

3.1.1 Controllable deformations, Ericksen's theorem and universal deformations

A deformation that can be *produced solely by the application of surface tractions* is called a *controllable deformation*.

Before proceeding, it is appropriate to state a classical result of finite elasticity, known as *Ericksen's Theorem*:

—

Ericksen's Theorem. *If a deformation can be maintained by surface tractions alone in every member of the set $\mathscr{I}$ of homogeneous, isotropic, elastic materials, then such deformation must be homogeneous.*

—

The proof of this theorem is provided in the Appendix. Ericksen has thus shown that the only controllable deformations in every homogeneous, isotropic elastic material, whether compressible or incompressible, are homogeneous ones. These deformations are defined as *universal deformations*.

For several varieties of deformations examined below, it turns out that, with a suitable numbering of the components, $\mathbf{B}$ has the representation

$$\mathbf{B} = B_{11}\mathbf{e}_{11} + B_{22}\mathbf{e}_{22} + B_{33}\mathbf{e}_{33} + B_{12}\left(\mathbf{e}_{12} + \mathbf{e}_{21}\right). \tag{3.5}$$

It is evident that $\mathbf{B}^{-1}$ will have a similar representation. Thus, (3.1) shows that $\boldsymbol{\sigma}$ has the corresponding form in physical components

$$\boldsymbol{\sigma} = \sigma_{11}\mathbf{e}_{11} + \sigma_{22}\mathbf{e}_{22} + \sigma_{33}\mathbf{e}_{33} + \sigma_{12}\left(\mathbf{e}_{12} + \mathbf{e}_{21}\right). \tag{3.6}$$

Hence, in every case where $\mathbf{B}$ has the representation (3.5), it follows that (3.3) reduces to the single universal relation

$$\frac{\sigma_{11} - \sigma_{22}}{\sigma_{12}} = \frac{B_{11} - B_{22}}{B_{12}}. \tag{3.7}$$

Several examples of universal relations generated by (3.2) will be presented below. We begin with a *universal* deformation of homogeneous shear.

3.1.2 A universal relation for controlled shear

Let us define x_1, x_2, x_3 and X_1, X_2, X_3 as the components of $\mathbf{x}$ and $\mathbf{X}$ in the basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$. The uniform shear and extension of a block are defined by the equations

$$x_1 = DX_1 + EX_2, \quad x_2 = FX_2, \quad x_3 = GX_3, \tag{3.8}$$

where D, E, F, G are constants. Using (3.8) in $\mathbf{B} = \mathbf{F}\mathbf{F}^{\mathrm{T}}$ yields

$$\mathbf{B} = (D^2 + E^2)\mathbf{e}_{11} + F^2\mathbf{e}_{22} + G^2\mathbf{e}_{33} + EF(\mathbf{e}_{12} + \mathbf{e}_{21}). \tag{3.9}$$

This universal deformation is possible in an incompressible material only when $DFG = 1$ ($\det \mathbf{F} = 1$). Substituting (3.9) into (3.7) provides, for any compressible or incompressible isotropic material, the following universal relation for a uniform shear with extension:

$$\sigma_{11} - \sigma_{22} = M\sigma_{12} \quad \text{with} \quad M \equiv \frac{D^2 + E^2 - F^2}{EF}. \tag{3.10}$$

This family includes the case of simple shear ($D = F = G = 1$), for which the universal relation reduces to

$$\sigma_{11} - \sigma_{22} = E\,\sigma_{12}. \tag{3.11}$$

Some examples of non-homogeneous controllable deformations will be presented below.

3.1.3 Universal relations for four families of deformations

In the relations that follow, (r, θ, x_3) and (r, θ, ϕ) are, respectively, the cylindrical and spherical coordinates of a point at position $\mathbf{x}$ in the deformed configuration; (R, Θ, X_3) and (R, Θ, Φ) have the same meaning for $\mathbf{X}$ in the reference configuration.

The parameters A, B, C, D, E, F are finite constants that cannot be zero when they appear as divisors; otherwise, except for signs dictated by physical conditions, they can be considered arbitrary. Families 1, 2, and 4 are isochoric, so such deformations are potential candidates for both compressible and incompressible materials. Families 3 and 5 are possible in an incompressible material only when they respect the constraint $\det \mathbf{F} = 1$.

The following results, based solely on the use of (3.2), provide *possible* universal relations for homogeneous isotropic elastic materials, both compressible and incompressible.

Family 1: Bending, stretching, and shearing of a rectangular block are described by

$$r = \sqrt{2AX_1}, \quad \theta = BX_2, \quad x_3 = \frac{X_3}{AB} - BCX_2, \tag{3.12}$$

from which we derive an expression for $\mathbf{B}$:

$$\mathbf{B} = r^2B^2\mathbf{e}_{11} + \left(B^2C^2 + \frac{1}{A^2B^2}\right)\mathbf{e}_{22} + \frac{A^2}{r^2}\mathbf{e}_{33} - rB^2C(\mathbf{e}_{12} + \mathbf{e}_{21}). \tag{3.13}$$

For this isochoric family, the universal relation obtained from (5.7) is given by

$$\sigma_{11} - \sigma_{22} = \sigma_{12}\frac{1 + A^2B^4(C^2 - r^2)}{A^2B^4Cr}. \tag{3.14}$$

Family 2: Straightening, stretching, and shearing of a hollow cylinder sector are characterized by the isochoric map

$$x_1 = \frac{1}{2}AB^2R^2, \quad x_2 = \frac{\Theta}{AB}, \quad x_3 = \frac{X_3}{B} + \frac{C\Theta}{AB}. \tag{3.15}$$

A similar use of (3.7) yields the following universal relation

$$\sigma_{11} - \sigma_{22} = \frac{\sigma_{12}}{B^2C}\Big[B^2(1 - C^2) - 2AX_1\Big]. \tag{3.16}$$

Family 3: Inflation, bending, torsion, extension, and shearing of an annular sector are described by

$$r = \sqrt{AR^2 + B}, \quad \theta = C\Theta + DX_3, \quad x_3 = E\Theta + FX_3. \tag{3.17}$$

If the material is incompressible, the parameters must satisfy the constraint $A(CF - DE) = 1$. In both cases, the universal formula derived from (3.7) is

$$\sigma_{11} - \sigma_{22} = \sigma_{12}\frac{r^2(C^2 + D^2R^2) - (E^2 + F^2R^2)}{r(CE + DFR^2)}. \tag{3.18}$$

Family 4: The inflation of a spherical shell sector is described by

$$r = \big[\pm R^3 + A\big]^{1/3}, \quad \theta = \pm\Theta, \quad \phi = \Phi. \tag{3.19}$$

Since the matrix **B**, in the chosen reference frame, is diagonal, this isochoric family has no universal relation of the type (3.7).

Family 5: Inflation, bending, extension, and azimuthal shearing of an annular sector are characterized by

$$r = AR, \quad \theta = B\ln\frac{R}{D} + C\Theta, \quad x_3 = FX_3. \tag{3.20}$$

When the material is incompressible, the constraint $A^2CF = 1$ must be respected. In both cases, (3.7) provides the universal relation

$$\sigma_{11} - \sigma_{22} = \sigma_{12}\,\frac{1 - B^2 - C^2}{B}. \tag{3.21}$$

The universal relations (3.14), (3.16), and (3.18) are valid at every material point. However, they assume particular relevance on the boundary, where the quantities

involved can actually be measured.

It is thus evident that, starting from the general universal relation $\boldsymbol{\sigma}\mathbf{B} = \mathbf{B}\boldsymbol{\sigma}$, it is possible to deduce, for each family of deformations, a corresponding *local* universal relation, expressed as a function of the characteristic parameters of the deformation itself.

3.1.4 An example of universal deformation for a magnetoelastic material

The concepts of *controllable deformation* and, consequently, of *universal deformation* can also be extended to magnetoelastic materials. Reinterpreting the definition provided by Singh and Pipkin [22] for dielectric elastic materials, a *controllable state* is a state that can exist in the absence of body forces and/or free current density. A controllable state is said to be *universal* if it is realizable in every isotropic magnetoelastic material.

We will limit ourselves here to considering the simplest case of a universal state, corresponding to a homogeneous deformation in the presence of a uniform magnetic field, treated in [22] and [4].

Let us consider an infinite plate in the X_1 and X_2 directions, bounded by the surfaces $X_3 = \pm h$ in the undeformed configuration. The deformation is described by

$$x_1 = \lambda_1 X_1 + \kappa_1 \lambda_3 X_3, \qquad x_2 = \lambda_2 X_2 + \kappa_2 \lambda_3 X_3, \qquad x_3 = \lambda_3 X_3. \tag{3.22}$$

Let us assume that a uniform magnetic induction $\{\mathbf{b}\} = (b_1, b_2, b_3)^{\mathrm{T}}$ acts within the plate. Since the deformation gradient $\mathbf{F}$ is also constant, it follows that the invariants introduced in (2.84) are also constant. From the constitutive relations (2.85) and (2.86), it then follows that $\mathbf{h}$ and $\boldsymbol{\tau}$ are also constant. In this way, the equilibrium equations

$$\operatorname{curl} \mathbf{h} = \mathbf{0}, \qquad \operatorname{div} \boldsymbol{\tau} = \mathbf{0}, \tag{3.23}$$

are automatically satisfied. A similar result is obtained in the case where a uniform magnetic field $\mathbf{h}$ is prescribed instead of the induction $\mathbf{b}$.

From the boundary conditions (2.56), the components h_1^* and h_2^* of the external magnetic field, and the component b_3^* of the external magnetic induction, relative to the surfaces $x_3 = \pm\lambda_3 h$ of the deformed plate, are derived. From the relation

$$\mathbf{b}^* = \mu_0 \mathbf{h}^*, \tag{3.24}$$

it is possible to determine the missing components. Uniform external fields with

such values satisfy Maxwell's equations (2.55).

Finally, from the boundary condition for the traction (2.62), one can determine the mechanical traction $\mathbf{t}_a$ to be applied on the surfaces $x_3 = \pm\lambda_3 h$ to obtain the desired state. Indeed, the tensors $\boldsymbol{\tau}$ and $\boldsymbol{\tau}^*_m$ are obtained from (2.87) and from the vectors $\mathbf{h}^*$ and $\mathbf{b}^*$, respectively. For example, on the face $x_3 = \lambda_3 h$, having the normal $\{\mathbf{n}\} = (0, 0, 1)^{\mathrm{T}}$, we have

$$t_{a,i} = \tau_{i3} - \tau^*_{m,i3}, \qquad i = 1, 2, 3. \tag{3.25}$$

We conclude by observing that, in the final chapter, we will explicitly test this universal state by prescribing a simple shear deformation and a uniform magnetic induction. We will then derive the surface tractions and external fields necessary to sustain such a configuration. Subsequently, we will verify the results by solving the direct problem: through a FEM analysis, we will impose the obtained tractions and external fields and check that the resulting deformation and induction match the prescribed ones.

3.2 The universal manifold method

In this section we present the *universal manifold method*, introduced by Pucci and Saccomandi [17], to verify the existence of universal relations.

We will limit ourselves to considering all those materials for which the constitutive equations for the Cauchy stress tensor can be represented as follows

$$\boldsymbol{\sigma} = \sum_{k=1}^{m} \mu_k \mathbf{D}^{(k)}, \tag{3.26}$$

where the tensors $\mathbf{D}^{(k)} \in \mathrm{Sym}$ are *kinematic* quantities and the *response coefficients* μ_k are certain scalar functions of the variables $\mathbf{x}$, t and $\mathbf{D}^{(k)}$.

We will consider a universal deformation (or *solution*) and calculate the corresponding tensors $\mathbf{D}^{(k)}$ and the response coefficients μ_k. Once a time $\bar{t}$ and a position $\bar{\mathbf{X}}$ in the reference configuration are fixed, the components of the tensors $\bar{\mathbf{D}}^{(k)} = \mathbf{D}^{(k)}(\bar{\mathbf{X}}, \bar{t})$ are determined, while the coefficients $\bar{\mu}_k = \mu_k(\bar{\mathbf{X}}, \bar{t})$ can still vary in subsets of $\mathbb{R}$, given that these coefficients change depending on the material.

Let us denote by σ_{ij} the components of $\boldsymbol{\sigma}$ with respect to an arbitrary basis $\{\mathbf{e}_i\}$

and introduce the space of dimension $(6+m)$

$$\mathscr{S}^{6+m} \quad \text{with coordinates} \quad \boldsymbol{\tau} \equiv (\tau_1, \ldots, \tau_6) \quad \text{and} \quad \bar{\boldsymbol{\mu}} \equiv (\bar{\mu}_1, \ldots, \bar{\mu}_m) \tag{3.27}$$

where

$$\tau_1 = \sigma_{11}, \quad \tau_2 = \sigma_{12}, \quad \tau_3 = \sigma_{13}, \quad \tau_4 = \sigma_{22}, \quad \tau_5 = \sigma_{23}, \quad \tau_6 = \sigma_{33}. \tag{3.28}$$

The relations

$$\boldsymbol{\sigma} = \sum_{k=1}^{m} \bar{\mu}_k \bar{\mathbf{D}}^{(k)}, \tag{3.29}$$

can be interpreted as a representation of an m-dimensional linear manifold, $\mathscr{V}_m$, parameterized by the coefficients $\bar{\mu}_k$.

Our interest is focused on the projection, $\Pi(\mathscr{V}_m)$, of $\mathscr{V}_m$ onto the subspace $\mathscr{S}^6(\boldsymbol{\tau})$.

In deriving such a projection, we consider the following matrix

$$\begin{pmatrix} \bar{D}^{(1)}_{11} & \bar{D}^{(1)}_{12} & \bar{D}^{(1)}_{13} & \bar{D}^{(1)}_{22} & \bar{D}^{(1)}_{23} & \bar{D}^{(1)}_{33} \\ \bar{D}^{(2)}_{11} & \bar{D}^{(2)}_{12} & \bar{D}^{(2)}_{13} & \bar{D}^{(2)}_{22} & \bar{D}^{(2)}_{23} & \bar{D}^{(2)}_{33} \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ \bar{D}^{(m)}_{11} & \bar{D}^{(m)}_{12} & \bar{D}^{(m)}_{13} & \bar{D}^{(m)}_{22} & \bar{D}^{(m)}_{23} & \bar{D}^{(m)}_{33} \end{pmatrix} \tag{3.30}$$

where the entries are the components of the tensors $\bar{\mathbf{D}}^{(k)}$ with respect to the basis $\{\mathbf{e}_i\}$. If the rank r of the matrix (3.30) is less than 6, then the manifold $\Pi(\mathscr{V}_m)$ is not empty and its dimension is r; in what follows we will denote it as $\Pi(\mathscr{V}_m)_r$. Such a situation certainly occurs when $m < 6$.

The linear manifold $\Pi(\mathscr{V}_m)_r$ is the *universal* manifold represented by $6-r$ linear, homogeneous and independent equations in the variables $\boldsymbol{\tau}$, that is, by a set of equations of the form

$$\mathscr{H}_q\left(\boldsymbol{\tau}, \bar{\mathbf{D}}^{(1)}, \ldots, \bar{\mathbf{D}}^{(m)}\right) \equiv \sum_{p=1}^{6} \tau_p \, \phi_{pq}\left(\bar{\mathbf{D}}^{(1)}, \ldots, \bar{\mathbf{D}}^{(m)}\right) = 0, \quad q = 1, \ldots, 6-r, \tag{3.31}$$

Then the equations

$$\mathscr{H}_q\left(\boldsymbol{\sigma}, \mathbf{D}^{(1)}, \ldots, \mathbf{D}^{(m)}\right) = 0, \quad q = 1, \ldots, 6-r, \tag{3.32}$$

constitute a set of linear universal relations that we were looking for. Clearly, these

are not the only possible linear universal relations (given that there are many ways to represent $\Pi(\mathscr{V}_m)_r$), but every linear universal relation can be obtained as a linear combination of these.

3.3 Universal relations in transversely isotropic elastic materials

In this section we analyze the approach used in [19] and [20] through which the (only) universal relation is derived in a compact form, in the case of transversely isotropic elastic materials. We simplify the constitutive equation for a transversely isotropic elastic material (2.37), with the axis of transverse isotropy along the unit vector $\mathbf{m}$ in the current configuration, in the following way:

$$\sigma = \gamma_1 \mathbf{I} + \gamma_2 \mathbf{B} + \gamma_3 \mathbf{B}^2 + \gamma_4\, \mathbf{m} \otimes \mathbf{m} + \gamma_5\, (\mathbf{m} \otimes \mathbf{Bm} + \mathbf{Bm} \otimes \mathbf{m}), \tag{3.33}$$

where the response coefficients γ_i $(i = 1, \ldots, 5)$ are functions of the invariants I_1, I_2, I_3, I_4, I_5 introduced in Section 2.3.

From (3.33),

$$\begin{aligned} \sigma\mathbf{B} - \mathbf{B}\sigma = \gamma_4 \big[(\mathbf{m} \otimes \mathbf{m})\mathbf{B} - \mathbf{B}(\mathbf{m} \otimes \mathbf{m})\big] \\ +\gamma_5 \big[(\mathbf{m} \otimes \mathbf{Bm} + \mathbf{Bm} \otimes \mathbf{m})\mathbf{B} - \mathbf{B}(\mathbf{m} \otimes \mathbf{Bm} + \mathbf{Bm} \otimes \mathbf{m})\big]. \end{aligned} \tag{3.34}$$

Since $\mathbf{B} = \mathbf{B}^{\mathrm{T}}$, we have $(\mathbf{m} \otimes \mathbf{m})\mathbf{B} = \mathbf{m} \otimes \mathbf{Bm}$. Let us therefore write (3.34) as

$$\sigma\mathbf{B} - \mathbf{B}\sigma = \gamma_4 \big[\mathbf{m} \otimes \mathbf{Bm} - \mathbf{Bm} \otimes \mathbf{m}\big] + \gamma_5 \big[\mathbf{m} \otimes \mathbf{B}^2\mathbf{m} - \mathbf{B}^2\mathbf{m} \otimes \mathbf{m}\big]. \tag{3.35}$$

Let us recall that to every skew-symmetric tensor $\mathbf{W}$ corresponds an axial vector $\mathbf{W}_\times \equiv \mathbf{u}$, such that for every vector $\mathbf{v}$ one has $\mathbf{Wv} = \mathbf{u} \times \mathbf{v}$. Furthermore, for two generic vectors $\mathbf{a}$ and $\mathbf{b}$ the following identity holds

$$(\mathbf{a} \otimes \mathbf{b} - \mathbf{b} \otimes \mathbf{a})_\times = -\mathbf{a} \times \mathbf{b}. \tag{3.36}$$

Using these identities, equation (5.30) simplifies to

$$(\sigma\mathbf{B} - \mathbf{B}\sigma)_\times = -\mathbf{m} \times \big(\gamma_4 \mathbf{Bm} + \gamma_5 \mathbf{B}^2\mathbf{m}\big). \tag{3.37}$$

Therefore, the universal relation we are looking for is

$$(\boldsymbol{\sigma}\mathbf{B} - \mathbf{B}\boldsymbol{\sigma})_{\times} \cdot \mathbf{m} = 0. \tag{3.38}$$

Although the following consideration is not reported in [19] and [20], it is possible to derive, following a similar procedure, an *alternative* universal relation expressed as a function of the first Piola–Kirchhoff stress tensor $\mathbf{P}$, starting from expression (2.36). Omitting the intermediate steps, the final expression turns out to be:

$$\left(\mathbf{P}^{\mathrm{T}}\mathbf{F} - \mathbf{F}^{\mathrm{T}}\mathbf{P}\right) \cdot \mathbf{M} = 0 \tag{3.39}$$

where $\mathbf{M}$ represents the preferential direction in the undeformed configuration, such that $\mathbf{M} = \mathbf{F}^{-1}\mathbf{m}$. Relation (3.39) will be useful to us later.

For an isotropic material, the physical interpretation of the three universal relations is based on the coaxiality between the Cauchy stress tensor and the Cauchy-Green strain tensor. In the case of transversely isotropic materials, the meaning of the universal relation (3.38) lies instead in the orthogonality between the axial vector associated with $\boldsymbol{\sigma}\mathbf{B} - \mathbf{B}\boldsymbol{\sigma}$ and the axis of transverse isotropy $\mathbf{m}$.

To confirm that for these materials there is only one universal relation, we apply the universal manifold method described in section 3.2. We consider a universal solution and the corresponding tensors $\mathbf{B}$ and fix a point $\mathbf{X}$ in the reference configuration. We introduce in the 11-dimensional space $\mathscr{S}$ the coordinates

$$\boldsymbol{\tau} = (\tau_1, \tau_2, \tau_3, \tau_4, \tau_5, \tau_6), \quad \boldsymbol{\gamma} = (\gamma_1, \gamma_2, \gamma_3, \gamma_4, \gamma_5), \tag{3.40}$$

where

$$\tau_1 = \sigma_{11},\ \tau_2 = \sigma_{12},\ \tau_3 = \sigma_{13},\ \tau_4 = \sigma_{22},\ \tau_5 = \sigma_{23},\ \tau_6 = \sigma_{33}. \tag{3.41}$$

In this space (3.33) represents a five-dimensional homogeneous linear manifold $\mathscr{V}$ parameterized by $\boldsymbol{\gamma}$. This manifold can be projected in the subspace $\mathscr{S}^*(\boldsymbol{\tau})$ of $\mathscr{S}$ onto the manifold $\Pi(\mathscr{V})$ whose dimensions r equal the rank of the matrix

$$\begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ B_{11} & B_{12} & B_{13} & B_{22} & B_{23} & B_{33} \\ B^2_{11} & B^2_{12} & B^2_{13} & B^2_{22} & B^2_{23} & B^2_{33} \\ m_1^2 & m_1 m_2 & m_1 m_3 & m_2^2 & m_2 m_3 & m_3^2 \\ \mathbb{B}_{11} & \mathbb{B}_{12} & \mathbb{B}_{13} & \mathbb{B}_{22} & \mathbb{B}_{23} & \mathbb{B}_{33} \end{pmatrix} \tag{3.42}$$

where $\mathbb{B} := \mathbf{Bm} \otimes \mathbf{m} + \mathbf{m} \otimes \mathbf{Bm}$ and B^2_{ij} are components of the tensor $\mathbf{B}^2$. The universal manifold $\Pi(\mathscr{V})$ is linear and homogeneous and can be represented by

a set of homogeneous linear equations (the universal relations). The number of these equations is $6 - r$. The rank of the matrix (3.42) turns out to be equal to 5[1], therefore we have *only one universal relation*.

Universal relations can be a powerful tool for quantitatively understanding the meaning of a constitutive parameter. For example, let us consider a transversely isotropic elastic material with $\gamma_5 = 0$. For this class of materials a new universal relation holds, namely

$$(\boldsymbol{\sigma}\mathbf{B} - \mathbf{B}\boldsymbol{\sigma})_\times \cdot \mathbf{B}\mathbf{n} = 0. \tag{3.43}$$

If the experimental data for a transversely isotropic elastic material do not satisfy (3.42), then clearly $\gamma_5 \neq 0$.

3.4 Universal relations in magnetoelastic materials

In this section we will see the method used in [9], [4] and [14] to derive universal relations for an *incompressible* magnetoelastic material. First, we reorganize (2.87) into a more convenient form:

$$\boldsymbol{\tau} = -p\mathbf{I} + \zeta_1\mathbf{B} + \zeta_2\mathbf{B}^2 + \zeta_5\mathbf{b}\otimes\mathbf{b} + \zeta_6(\mathbf{b}\otimes\mathbf{B}\mathbf{b} + \mathbf{B}\mathbf{b}\otimes\mathbf{b}), \tag{3.44}$$

where we have introduced the coefficients $\zeta_1, \zeta_2, \zeta_5, \zeta_6$.

Following a procedure completely analogous to the case of transversely isotropic materials, from (3.44) we obtain:

$$\boldsymbol{\tau}\mathbf{B} - \mathbf{B}\boldsymbol{\tau} = \zeta_5\big(\mathbf{b}\otimes\mathbf{B}\mathbf{b} - \mathbf{B}\mathbf{b}\otimes\mathbf{b}\big) + \zeta_6\big(\mathbf{b}\otimes\mathbf{B}^2\mathbf{b} - \mathbf{B}^2\mathbf{b}\otimes\mathbf{b}\big). \tag{3.45}$$

Applying property (3.36), we will have that

$$\big(\boldsymbol{\tau}\mathbf{B} - \mathbf{B}\boldsymbol{\tau}\big)_\times = (\zeta_5\mathbf{B}\mathbf{b} + \zeta_6\mathbf{B}^2\mathbf{b}) \times \mathbf{b}, \tag{3.46}$$

from which the universal relation

$$\big(\boldsymbol{\tau}\mathbf{B} - \mathbf{B}\boldsymbol{\tau}\big)_\times \cdot \mathbf{b} = 0. \tag{3.47}$$

Here too we apply the universal manifold method to verify that the derived universal relation is unique. For this purpose we define an 11-dimensional space

[1]The rank of the matrix was verified symbolically using the `sympy` package in Python.

$\mathscr{S}$, with coordinates

$$\boldsymbol{\tau} = \{\tau_1, \tau_2, \tau_3, \tau_4, \tau_5, \tau_6\}, \quad \boldsymbol{\zeta} = \{-p, \zeta_1, \zeta_2, \zeta_5, \zeta_6\}, \tag{3.48}$$

where

$$\tau_1 = \tau_{11}, \quad \tau_2 = \tau_{12}, \quad \tau_3 = \tau_{13}, \quad \tau_4 = \tau_{22}, \quad \tau_5 = \tau_{23}, \quad \tau_6 = \tau_{33}. \tag{3.49}$$

In this space equation (3.44) represents a homogeneous linear manifold which we denote by $\mathscr{V}_\tau$, of dimension six and parameterized by the variable $\boldsymbol{\zeta}$. This manifold can be projected in the subspace $\mathscr{S}^*_\tau(\boldsymbol{\tau})$ of $\mathscr{S}$ onto the manifold, denoted by $\Pi_\tau(\mathscr{V}_\tau)$, whose dimension r is given by the rank of the matrix

$$\begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ B_{11} & B_{12} & B_{13} & B_{22} & B_{23} & B_{33} \\ B^2_{11} & B^2_{12} & B^2_{13} & B^2_{22} & B^2_{23} & B^2_{33} \\ b^2_1 & b_1 b_2 & b_1 b_3 & b^2_2 & b_2 b_3 & b^2_3 \\ \mathbb{B}_{11} & \mathbb{B}_{12} & \mathbb{B}_{13} & \mathbb{B}_{22} & \mathbb{B}_{23} & \mathbb{B}_{33} \end{pmatrix}, \tag{3.50}$$

where $\mathbb{B} := \mathbf{Bb} \otimes \mathbf{b} + \mathbf{b} \otimes \mathbf{Bb}$, $\mathbb{B}_{ij}$ are the components of $\mathbb{B}$, B_{ij} and B^2_{ij} are the components of $\mathbf{B}$ and $\mathbf{B}^2$ and b_i are those of $\mathbf{b}$.

Here too the universal manifold $\Pi_\tau(\mathscr{V}_\tau)$ is linear and homogeneous and can be represented by $6 - r$ homogeneous linear equations, that is the universal relations. The rank of the matrix turns out to be equal to 5, so the universal relation (3.47) is also the only one.

If the energy function is independent of I_5 ($\zeta_5 = 0$), then from (3.46) I have that:

$$\left(\boldsymbol{\tau}\mathbf{B} - \mathbf{B}\boldsymbol{\tau}\right)_\times = -\zeta_5 \mathbf{b} \times \mathbf{Bb}, \tag{3.51}$$

so that, added to (3.47), we obtain the new universal relation

$$\left(\boldsymbol{\tau}\mathbf{B} - \mathbf{B}\boldsymbol{\tau}\right)_\times \cdot \mathbf{Bb} = 0. \tag{3.52}$$

If $\mathbf{b}$ is an eigenvector of $\mathbf{B}$ then (3.52) does not provide any further information with respect to (3.47). Conversely, if this is not the case, then the combination of (3.47) and (3.52) shows that the vector $\left(\boldsymbol{\tau}\mathbf{B} - \mathbf{B}\boldsymbol{\tau}\right)_\times$ is orthogonal to the plane defined by the vectors $\mathbf{b}$ and $\mathbf{Bb}$.

4 A physical interpretation of universal relations

4.1 The Eshelby tensor

In this paragraph we follow the procedure presented by Eshelby [11] to systematically introduce the so-called *elastic energy-momentum tensor* (or *Eshelby tensor*, in his honor).

Let us suppose we have a *non-homogeneous* material, for which the energy density per unit reference volume depends on the deformation gradient $\mathbf{F}$ and the position $\mathbf{X}$ in the reference configuration, that is

$$W = W(\mathbf{F}, \mathbf{X}) \tag{4.1}$$

By imposing the stationarity of the potential energy (in the absence of body forces) of a body $\mathcal{B}$ in the undeformed configuration, we obtain the following Euler-Lagrange equations (linear momentum balance):

$$\frac{\partial}{\partial X_A} \frac{\partial W}{\partial x_{i,A}} = 0 \tag{4.2}$$

where

$$x_{i,A} \equiv \frac{\partial x_i}{\partial X_A} \equiv F_{iA} \tag{4.3}$$

and index notation has been adopted.

Distinguishing between the ordinary gradient

$$\frac{\partial W}{\partial X_L} \equiv (\mathrm{Grad}\, W)_L \tag{4.4}$$

and the derivative

$$\left(\frac{\partial W}{\partial X_L}\right)_{\mathrm{exp}} \tag{4.5}$$

which represents the explicit dependence on X_L, by means of the chain rule one obtains

$$\frac{\partial W}{\partial X_L} = \frac{\partial W}{\partial x_{i,A}} x_{i,AL} + \left(\frac{\partial W}{\partial X_L}\right)_{\text{exp}} \tag{4.6}$$

where

$$x_{i,AL} \equiv \frac{\partial^2 x_i}{\partial X_A \partial X_L} \tag{4.7}$$

Given (4.2), the following equation will also be valid:

$$\frac{\partial}{\partial X_A} \frac{\partial W}{\partial x_{i,A}} x_{i,L} = 0, \tag{4.8}$$

so we can rewrite (4.6) as follows:

$$\frac{\partial W}{\partial X_L} = \frac{\partial}{\partial X_A} \frac{\partial W}{\partial x_{i,A}} x_{i,L} + \frac{\partial W}{\partial x_{i,A}} x_{i,AL} + \left(\frac{\partial W}{\partial X_L}\right)_{\text{exp}}. \tag{4.9}$$

This expression is equivalent to

$$\frac{\partial W}{\partial X_L} = \frac{\partial}{\partial X_A} \left(\frac{\partial W}{\partial x_{i,A}} x_{i,L}\right) + \left(\frac{\partial W}{\partial X_L}\right)_{\text{exp}}, \tag{4.10}$$

from which one derives

$$\left(\frac{\partial W}{\partial X_L}\right)_{\text{exp}} = \frac{\partial}{\partial X_A} \left(W \delta_{LA} - \frac{\partial W}{\partial x_{i,A}} x_{i,L}\right). \tag{4.11}$$

The *Eshelby tensor* **E** is then defined as

$$E_{LA} = W \delta_{LA} - \frac{\partial W}{\partial x_{i,A}} x_{i,L} \tag{4.12}$$

which, in compact tensor form, is

$$\mathbf{E} = W\mathbf{I} - \mathbf{F}^{\mathrm{T}}\mathbf{P}. \tag{4.13}$$

where we have introduced the first Piola-Kirchhoff stress tensor $\mathbf{P} = \partial W / \partial \mathbf{F}$. Equation (4.11) therefore becomes

$$\frac{\partial E_{LA}}{\partial X_A} = \left(\frac{\partial W}{\partial X_L}\right)_{\text{exp}}. \tag{4.14}$$

In the case of a homogeneous portion of material, one has, locally, the conservation equation

$$\frac{\partial E_{LA}}{\partial X_A} = 0, \tag{4.15}$$

or, in tensor form:

$$\mathrm{Div}\,\mathbf{E} = \mathbf{0}. \tag{4.16}$$

The Eshelby tensor is thus the source of all those *configurational* forces acting on *inhomogeneities* (defects, dislocations) of the material. For the actual calculation of these forces, the reader is referred to Eshelby's original treatment [11].

4.2 Invariance of internal energy with respect to rotations of the reference configuration

The objective of this section is to derive a universal relation, in the case of transverse isotropy, by imposing the invariance of the (integral) internal energy with respect to rotations of the reference configuration.

Suppose we have a body $\mathcal{B}$ in the reference configuration. The internal energy Π of $\mathcal{B}$ is the functional

$$\Pi\big[\mathbf{F}, \mathbf{A}\big] = \int_{\mathcal{B}} W(\mathbf{F}, \mathbf{A})\,, \tag{4.17}$$

where $W = W(\mathbf{F}, \mathbf{A})$ is the energy density per unit reference volume, $\mathbf{F}$ is the deformation gradient, and $\mathbf{A} = \mathbf{a} \otimes \mathbf{a}$ is the fiber orientation tensor, identified by the direction $\mathbf{a}$.

We impose an infinitesimal rotation $\mathbf{Q}(\varepsilon)$ around the axis defined by $\mathbf{a}$, such that

$$\mathbf{Q}(\varepsilon) = \mathbf{I} + \varepsilon\mathbf{W} + o(\varepsilon^2), \quad \varepsilon > 0, \quad \mathbf{W} \in \mathrm{Skw}, \quad \mathbf{Wa} = \mathbf{0}. \tag{4.18}$$

We have seen, in section 2.1, that following a rotation of the reference configuration, the deformation gradient transforms according to the law

$$\mathbf{F}^* = \mathbf{FQ}. \tag{4.19}$$

We then calculate the variation $\delta\mathbf{F}$ following the rotation $\mathbf{Q}(\varepsilon)$:

$$\begin{aligned}\delta\mathbf{F} &= \frac{d}{d\varepsilon}\mathbf{F}(\varepsilon)\bigg|_{\varepsilon=0} \\ &= \frac{d}{d\varepsilon}\big[\mathbf{F}\mathbf{Q}(\varepsilon)\big]\bigg|_{\varepsilon=0} \\ &= \frac{d}{d\varepsilon}\big[\mathbf{F}+\varepsilon\mathbf{F}\mathbf{W}\big]\bigg|_{\varepsilon=0} \\ &= \mathbf{F}\mathbf{W}.\end{aligned} \tag{4.20}$$

The variation of the total energy functional is given by

$$\delta\Pi[\mathbf{F},\mathbf{A}] = \int_{\mathcal{B}} \frac{\partial W}{\partial \mathbf{F}}\cdot\delta\mathbf{F} + \frac{\partial W}{\partial \mathbf{A}}\cdot\delta\mathbf{A}. \tag{4.21}$$

Recalling the definition of the first Piola-Kirchhoff stress tensor $\mathbf{P} = \partial W/\partial\mathbf{F}$ and noting that for infinitesimal rotations around $\mathbf{a}$ one has $\delta\mathbf{A} = \mathbf{0}$, one obtains

$$\delta\Pi[\mathbf{F}] = \int_{\mathcal{B}} \mathbf{F}^{\mathrm{T}}\mathbf{P}\cdot\mathbf{W}. \tag{4.22}$$

Imposing stationarity, we obtain

$$\delta\Pi[\mathbf{F}] = \int_{\mathcal{B}} \mathbf{F}^{\mathrm{T}}\mathbf{P}\cdot\mathbf{W} = 0, \qquad \forall\,\mathbf{W}\in\{\mathbf{W}\in\mathrm{Skw}\mid\mathbf{W}\mathbf{a}=0\}, \tag{4.23}$$

which is equivalent to

$$\delta\Pi[\mathbf{F}] = \int_{\mathcal{B}} \mathrm{skw}\big[\mathbf{F}^{\mathrm{T}}\mathbf{P}\big]\cdot\mathbf{W} = 0, \qquad \forall\,\mathbf{W}\in\{\mathbf{W}\in\mathrm{Skw}\mid\mathbf{W}\mathbf{a}=0\}. \tag{4.24}$$

from which, if I define $\boldsymbol{\omega} := (\mathbf{W})_\times$, I can write

$$\int_{\mathcal{B}} \big(\mathbf{P}^{\mathrm{T}}\mathbf{F}-\mathbf{F}^{\mathrm{T}}\mathbf{P}\big)_\times\cdot\boldsymbol{\omega} = 0, \qquad \forall\,\boldsymbol{\omega}\in\{\boldsymbol{\omega}\in\mathbb{R}^3\mid\boldsymbol{\omega}\times\mathbf{a}=\mathbf{0}\}. \tag{4.25}$$

If the quantity in (4.25) must be zero for every vector $\boldsymbol{\omega}$ parallel to $\mathbf{a}$, then, for this to be true, the following must hold:

$$\big(\mathbf{P}^{\mathrm{T}}\mathbf{F}-\mathbf{F}^{\mathrm{T}}\mathbf{P}\big)_\times\cdot\mathbf{a} = 0. \tag{4.26}$$

This is, to all intents and purposes, the universal relation (3.39), but obtained via a different procedure.

The result obtained is very important for two reasons:

1. We obtained the universal relation not by mathematically manipulating the constitutive relations, but by applying the invariance to rotation of the reference configuration to the *total energy*, i.e., by imposing, to all intents and purposes, an integral (and not a pointwise, as in Chapter 2) condition of isotropy.

2. Given expression (4.13) for the Eshelby tensor, we immediately notice that

$$(\mathbf{P}^{\mathrm{T}}\mathbf{F} - \mathbf{F}^{\mathrm{T}}\mathbf{P}) = -\operatorname{skw}[\mathbf{E}], \tag{4.27}$$

so we could rewrite (4.26) as

$$\big(\operatorname{skw}[\mathbf{E}]\big)_{\times} \cdot \mathbf{a} = 0. \tag{4.28}$$

We can summarize the meaning of relation (4.28) as follows: a direct consequence of transverse isotropy is the orthogonality, at every material point, between the axial vector associated with the skew-symmetric part of the Eshelby tensor (to which we will later give the name of *Eshelbian couple*) and the fiber axis in the reference configuration. A deeper physical interpretation will be provided in the following paragraph.

Equation (4.26) can also be written as

$$\big(\mathbf{P}^{\mathrm{T}}\mathbf{F} - \mathbf{F}^{\mathrm{T}}\mathbf{P}\big) \cdot \tilde{\mathbf{a}} = 0, \tag{4.29}$$

where $\tilde{\mathbf{a}}$ is the skew-symmetric tensor associated with $\mathbf{a}$.

Since

$$\mathbf{P}^{\mathrm{T}}\mathbf{F} - \mathbf{F}^{\mathrm{T}}\mathbf{P} = J\,\mathbf{F}^{-1}\big(\sigma\mathbf{B} - \mathbf{B}\sigma\big)\mathbf{F}^{-\mathrm{T}}, \tag{4.30}$$

and, if we denote by $\boldsymbol{\alpha} = \mathbf{F}\mathbf{a}$ the vector that identifies the fiber direction in the current configuration and by $\tilde{\boldsymbol{\alpha}}$ its associated skew-symmetric tensor, it can be shown that

$$\tilde{\mathbf{a}} = J^{-1}\mathbf{F}^{\mathrm{T}}\tilde{\boldsymbol{\alpha}}\mathbf{F}, \tag{4.31}$$

it follows that

$$\begin{aligned}\left(\mathbf{P}^{\mathrm{T}}\mathbf{F}-\mathbf{F}^{\mathrm{T}}\mathbf{P}\right)\cdot\tilde{\mathbf{a}} &= \mathbf{F}^{-1}\left(\sigma\mathbf{B}-\mathbf{B}\sigma\right)\mathbf{F}^{-\mathrm{T}}\cdot\mathbf{F}^{\mathrm{T}}\tilde{\boldsymbol{\alpha}}\mathbf{F}\\ &= \left(\sigma\mathbf{B}-\mathbf{B}\sigma\right)\cdot\tilde{\boldsymbol{\alpha}} = 0.\end{aligned} \tag{4.32}$$

or, equivalently,

$$\left(\sigma\mathbf{B}-\mathbf{B}\sigma\right)_{\times}\cdot\boldsymbol{\alpha} = 0. \tag{4.33}$$

We recognize in this expression the universal relation (4.33), already known for transversely isotropic materials.

4.3 Internal remodeling of fibers

Let us now assume that the orientation of the fibers in the reference configuration, which we will identify here with the unit vector $\mathbf{a}_0$, can modify over time under the action of an external stimulus of any nature. The following considerations are based on the work of Ciambella and Nardinocchi [8].

The orientation tensor associated with $\mathbf{a}_0$ is defined as

$$\mathbf{A}_0 = \mathbf{a}_0\otimes\mathbf{a}_0. \tag{4.34}$$

The remodeling process is described by a time-dependent rotation:

$$\mathbf{A}(t) = \mathbf{R}(t)\,\mathbf{A}_0\,\mathbf{R}(t)^{\mathrm{T}}, \tag{4.35}$$

where $\mathbf{A} = \mathbf{a}\otimes\mathbf{a}$ represents the remodeled orientation tensor and

$$\mathbf{a}(t) = \mathbf{R}(t)\,\mathbf{a}_0, \tag{4.36}$$

the orientation of the fiber in the remodeled configuration.

The energy density per unit reference volume W is a function of the deformation gradient $\mathbf{F}$ and the *remodeled* orientation tensor $\mathbf{A}$, i.e. $W = W(\mathbf{F},\mathbf{A})$. By the chain rule one has

$$\dot{W} = \frac{\partial W}{\partial\mathbf{F}}\cdot\dot{\mathbf{F}} + \frac{\partial W}{\partial\mathbf{A}}\cdot\dot{\mathbf{A}}. \tag{4.37}$$

Let us then consider the local form of the dissipation inequality:

$$\dot{W} \le \mathbf{P}\cdot\dot{\mathbf{F}} + \boldsymbol{\Sigma}\cdot\dot{\mathbf{R}}\mathbf{R}^{\mathrm{T}}, \tag{4.38}$$

where $\boldsymbol{\Sigma}$ is the internal remodeling couple, which expends power on the rotation

of the fibers (see [8]).

Furthermore, one has

$$\dot{\mathbf{A}} = \dot{\mathbf{R}}\,\mathbf{A}_0\,\mathbf{R}^{\mathrm{T}} + \mathbf{R}\,\mathbf{A}_0\,\dot{\mathbf{R}}^{\mathrm{T}}, \tag{4.39}$$

and, since

$$\mathbf{R}\mathbf{R}^{\mathrm{T}} = \mathbf{I}, \tag{4.40}$$

differentiating with respect to time yields

$$\dot{\mathbf{R}}\mathbf{R}^{\mathrm{T}} + \mathbf{R}\dot{\mathbf{R}}^{\mathrm{T}} = \mathbf{0}. \tag{4.41}$$

It follows that

$$\begin{aligned} \dot{\mathbf{R}} &= -\,\mathbf{R}\dot{\mathbf{R}}^{\mathrm{T}}\mathbf{R}, \\ \dot{\mathbf{R}}^{\mathrm{T}} &= -\,\mathbf{R}^{\mathrm{T}}\dot{\mathbf{R}}\mathbf{R}^{\mathrm{T}}, \end{aligned} \tag{4.42}$$

and therefore

$$\begin{aligned} \dot{\mathbf{A}} &= -\mathbf{R}\dot{\mathbf{R}}^{\mathrm{T}}\mathbf{R}\,\mathbf{A}_0\,\mathbf{R}^{\mathrm{T}} - \mathbf{R}\,\mathbf{A}_0\,\mathbf{R}^{\mathrm{T}}\dot{\mathbf{R}}\mathbf{R}^{\mathrm{T}} \\ &= \dot{\mathbf{R}}\mathbf{R}^{\mathrm{T}}\mathbf{A} - \mathbf{A}\,\dot{\mathbf{R}}\mathbf{R}^{\mathrm{T}}. \end{aligned} \tag{4.43}$$

Moreover,

$$\frac{\partial W}{\partial \mathbf{A}} \cdot \dot{\mathbf{A}} = \frac{\partial W}{\partial \mathbf{A}} \cdot \left(\dot{\mathbf{R}}\mathbf{R}^{\mathrm{T}}\mathbf{A} - \mathbf{A}\,\dot{\mathbf{R}}\mathbf{R}^{\mathrm{T}}\right) = \left(\frac{\partial W}{\partial \mathbf{A}}\mathbf{A} - \mathbf{A}\frac{\partial W}{\partial \mathbf{A}}\right) \cdot \dot{\mathbf{R}}\mathbf{R}^{\mathrm{T}}. \tag{4.44}$$

Substituting into the dissipation inequality, one obtains:

$$\left(\frac{\partial W}{\partial \mathbf{F}} - \mathbf{P}\right) \cdot \dot{\mathbf{F}} + \left[\left(\frac{\partial W}{\partial \mathbf{A}}\mathbf{A} - \mathbf{A}\frac{\partial W}{\partial \mathbf{A}}\right) - \mathbf{\Sigma}\right] \cdot \dot{\mathbf{R}}\mathbf{R}^{\mathrm{T}} \leq 0. \tag{4.45}$$

If the dissipation is solely due to the remodeling of the fibers, the consequent constitutive relations are:

$$\begin{aligned} \mathbf{P} &= \frac{\partial W}{\partial \mathbf{F}} = 2\mathbf{F}\frac{\partial W}{\partial \mathbf{C}}, \\ \mathbf{\Sigma} &= \left(\frac{\partial W}{\partial \mathbf{A}}\mathbf{A} - \mathbf{A}\frac{\partial W}{\partial \mathbf{A}}\right) + \mathbf{\Sigma}_d, \end{aligned} \tag{4.46}$$

where an elastic part has been distinguished in $\mathbf{\Sigma}$,

$$\mathbf{\Sigma}_e = \frac{\partial W}{\partial \mathbf{A}}\mathbf{A} - \mathbf{A}\frac{\partial W}{\partial \mathbf{A}}, \tag{4.47}$$

and a dissipative part $\mathbf{\Sigma}_d$.

If the material is transversely isotropic, W is an isotropic function of $\mathbf{C}$ and $\mathbf{A}$ (see Section 2.3). In this case, W depends on the invariants I_1, I_2, I_3 of $\mathbf{C}$ and on the two additional invariants:

$$I_4 = \mathbf{C} \cdot \mathbf{A}, \qquad I_5 = \mathbf{C}^2 \cdot \mathbf{A}. \tag{4.48}$$

The constitutive relation for the second Piola-Kirchhoff stress tensor $\mathbf{S} = \mathbf{F}^{-1}\mathbf{P}$ then follows:

$$\mathbf{S} = 2(W_1 + I_1 W_2)\mathbf{I} - 2W_2\mathbf{C} + 2I_3 W_3 \mathbf{C}^{-1} + 2W_4\mathbf{A} + 2W_5\big(\mathbf{AC} + \mathbf{CA}\big), \tag{4.49}$$

where $W_i = \partial W / \partial I_i,\ i = 1, \dots, 5$.

Furthermore,

$$\frac{\partial I_4}{\partial \mathbf{A}} = \mathbf{C}, \qquad \frac{\partial I_5}{\partial \mathbf{A}} = \mathbf{C}^2, \tag{4.50}$$

whence

$$\frac{\partial W}{\partial \mathbf{A}} = W_4\mathbf{C} + W_5\mathbf{C}^2, \tag{4.51}$$

and therefore

$$\begin{aligned} \frac{\partial W}{\partial \mathbf{A}}\mathbf{A} - \mathbf{A}\frac{\partial W}{\partial \mathbf{A}} &= \big(W_4\mathbf{C} + W_5\mathbf{C}^2\big)\mathbf{A} - \mathbf{A}\big(W_4\mathbf{C} + W_5\mathbf{C}^2\big) \\ &= \mathbf{C}\left[W_4\mathbf{A} + W_5\big(\mathbf{AC} + \mathbf{CA}\big)\right] - \left[W_4\mathbf{A} + W_5\big(\mathbf{AC} + \mathbf{CA}\big)\right]\mathbf{C} \\ &= \frac{1}{2}\big(\mathbf{CS} - \mathbf{SC}\big). \end{aligned} \tag{4.52}$$

Therefore, we have demonstrated that

$$\mathbf{\Sigma}_e = \frac{1}{2}\big(\mathbf{CS} - \mathbf{SC}\big). \tag{4.53}$$

Since it is easily verifiable that the Eshelby tensor, starting from expression (4.13), can be rewritten as

$$\mathbf{E} = W\mathbf{I} - \mathbf{CS}, \tag{4.54}$$

it then turns out that

$$\mathrm{skw}[\mathbf{E}] = \frac{1}{2}\big(\mathbf{CS} - \mathbf{SC}\big) = \mathbf{\Sigma}_e. \tag{4.55}$$

At this point we can summarize the important result obtained: relation (4.28) indicates to us that, in transversely isotropic materials, the Eshelbian couple is

always orthogonal to the fiber axis. This implies that it performs no work on rotations around the fiber axis. However, in the case of *configurational remodeling* of the fibers, the work performed by the Eshelbian couple will, in general, be *non-zero*.

4.4 Application example

With the following example we wish to verify the validity of the universal relations in the case of a body endowed with an axis of anisotropy. Subsequently, we will admit a remodeling of the fibers and evaluate the work performed by the Eshelbian couple. The Python `sympy` package was used to perform the calculations.

Non-reorientable fibers. We introduce a Cartesian basis defined by the unit vectors $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$. Let us then consider a body of transversely isotropic elastic material, whose fiber orientation in the reference configuration is defined by the unit vector $\mathbf{M}$, parallel to the axis defined by the unit vector $\mathbf{e}_3$. In the basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ it therefore admits the following representation by components:

$$\{\mathbf{M}\} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \tag{4.56}$$

The solid is subjected to the following simple shear deformation:

$$x_1 = X_1, \quad x_2 = X_2 + KX_3, \quad x_3 = X_3 \tag{4.57}$$

Consequently, the deformation gradient $\mathbf{F}$ and the left Cauchy–Green tensor $\mathbf{B}$ assume the form:

$$[\mathbf{F}] = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & K \\ 0 & 0 & 1 \end{pmatrix}, \qquad [\mathbf{B}] = [\mathbf{F}]\,[\mathbf{F}]^{\mathrm{T}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & K^2+1 & K \\ 0 & K & 1 \end{pmatrix} \tag{4.58}$$

while the orientation of the fibers in the current configuration will be defined by $\mathbf{m} = \mathbf{FM}$:

$$\{\mathbf{m}\} = \begin{pmatrix} 0 \\ K \\ 1 \end{pmatrix} \tag{4.59}$$

Recalling the constitutive relation in the case of transverse isotropy (2.37), we

report below the components of $\boldsymbol{\sigma}$. The component notation is adopted solely for reasons of typographical space, and not for other considerations:

$$
\begin{aligned}
\sigma_{11} &= 2\check{\psi}_1 + 2\check{\psi}_2(K^2+2) + 2\check{\psi}_3, \\
\sigma_{22} &= 2K^2\check{\psi}_4 + 4K^2\check{\psi}_5(K^2+2) + 2\check{\psi}_1 - 2\check{\psi}_2(K^4+2K^2-2) + 2\check{\psi}_3, \\
\sigma_{33} &= 2\check{\psi}_1 + 4\check{\psi}_2 + 2\check{\psi}_3(K^2+1) + 2\check{\psi}_4 + 4\check{\psi}_5(K^2+1), \\
\sigma_{23} &= \sigma_{32} = -2K\left[\check{\psi}_2(K^2+2) + \check{\psi}_3 - \check{\psi}_4 - \check{\psi}_5(2K^2+3)\right], \\
\sigma_{12} &= \sigma_{13} = 0.
\end{aligned}
\tag{4.60}
$$

At this point we calculate the tensor $\big([\boldsymbol{\sigma}][\mathbf{B}] - [\mathbf{B}][\boldsymbol{\sigma}]\big)$:

$$
[\boldsymbol{\sigma}][\mathbf{B}] - [\mathbf{B}][\boldsymbol{\sigma}] = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -2K\left[\check{\psi}_4 + (K^2+2)\check{\psi}_5\right] \\ 0 & 2K\left[\check{\psi}_4 + (K^2+2)\check{\psi}_5\right] & 0 \end{pmatrix}. \tag{4.61}
$$

As can be observed, this tensor is not zero, unlike the case of complete isotropy, and is skew-symmetric. Therefore, there exists an axial vector associated with it, with component representation:

$$
\big([\boldsymbol{\sigma}][\mathbf{B}] - [\mathbf{B}][\boldsymbol{\sigma}]\big)_\times = \begin{pmatrix} 2K\left[\check{\psi}_4 + (K^2+2)\check{\psi}_5\right] \\ 0 \\ 0 \end{pmatrix} \tag{4.62}
$$

The universal relation is therefore verified

$$
(\boldsymbol{\sigma}\mathbf{B} - \mathbf{B}\boldsymbol{\sigma})_\times \cdot \mathbf{m} = 0, \tag{4.63}
$$

i.e. the axial vector associated with the tensor $(\boldsymbol{\sigma}\mathbf{B} - \mathbf{B}\boldsymbol{\sigma})$ is orthogonal to the direction of the fiber in the current configuration.

Let us now analyze the specific case for which $\check{\psi}_5 = 0$. We will have that

$$
\big([\boldsymbol{\sigma}][\mathbf{B}] - [\mathbf{B}][\boldsymbol{\sigma}]\big)_\times = \begin{pmatrix} 2K\check{\psi}_4 \\ 0 \\ 0 \end{pmatrix} \tag{4.64}
$$

Let us also calculate the vector $[\mathbf{B}]\{\mathbf{m}\}$:

$$[\mathbf{B}]\{\mathbf{m}\} = \begin{pmatrix} 0 \\ K^3 + 2K \\ K^2 + 1 \end{pmatrix} \tag{4.65}$$

Therefore, the second universal relation (3.43) will also be verified

$$(\boldsymbol{\sigma}\mathbf{B} - \mathbf{B}\boldsymbol{\sigma})_\times \cdot \mathbf{B}\mathbf{m} = 0 \tag{4.66}$$

Hence the axial vector associated with the tensor $\boldsymbol{\sigma}\mathbf{B} - \mathbf{B}\boldsymbol{\sigma}$ turns out to be perpendicular to the plane defined by the vectors $\mathbf{m}$ and $\mathbf{B}\mathbf{m}$, that is to the plane $(\mathbf{e}_2, \mathbf{e}_3)$. Note that, in the particular case of simple shear considered, this orthogonality holds valid even without imposing $\check{\psi}_5 = 0$.

We also report (returning to the general case, i.e. without imposing $\check{\psi}_5 = 0$) the calculation of the tensor $\left([\mathbf{P}]^{\mathrm{T}}[\mathbf{F}] - [\mathbf{F}]^{\mathrm{T}}[\mathbf{P}]\right)$:

$$[\mathbf{P}]^{\mathrm{T}}[\mathbf{F}] - [\mathbf{F}]^{\mathrm{T}}[\mathbf{P}] = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -2K\left[\check{\psi}_4 + (K^2+2)\check{\psi}_5\right] \\ 0 & 2K\left[\check{\psi}_4 + (K^2+2)\check{\psi}_5\right] & 0 \end{pmatrix} \tag{4.67}$$

so the axial vector associated with it will be

$$\left([\mathbf{P}]^{\mathrm{T}}[\mathbf{F}] - [\mathbf{F}]^{\mathrm{T}}[\mathbf{P}]\right)_\times = \begin{pmatrix} 2K\left[\check{\psi}_4 + (K^2+2)\check{\psi}_5\right] \\ 0 \\ 0 \end{pmatrix} \tag{4.68}$$

wherefore it is verified that

$$\left(\mathbf{P}^{\mathrm{T}}\mathbf{F} - \mathbf{F}^{\mathrm{T}}\mathbf{P}\right)_\times \cdot \mathbf{M} = 0. \tag{4.69}$$

Reorientable fibers. At this point, we impose an internal remodeling of the fibers, described by the 2D rotation $\mathbf{R}(\theta(t))$ around the axis defined by the unit vector $\mathbf{e}_2$, that is

$$[\mathbf{R}(\theta(t))] = \begin{pmatrix} \cos\theta(t) & 0 & \sin\theta(t) \\ 0 & 1 & 0 \\ -\sin\theta(t) & 0 & \cos\theta(t) \end{pmatrix}. \tag{4.70}$$

Hence

$$\mathbf{M}(t) = \mathbf{R}(t)\,\mathbf{M}_0, \quad \mathbf{M}_0 \equiv \mathbf{e}_3. \tag{4.71}$$

Writing $\mathbf{M}(t)$ in components,

$$\{\mathbf{M}(t)\} = \begin{pmatrix} \sin\theta(t) \\ 0 \\ \cos\theta(t) \end{pmatrix}. \tag{4.72}$$

Introducing the *angular velocity* vector $\boldsymbol{\omega} := (\dot{\mathbf{R}}\mathbf{R}^{\mathrm{T}})_{\times}$, which will be directed along the axis defined by $\mathbf{e}_2$,

$$\{\boldsymbol{\omega}\} = \begin{pmatrix} 0 \\ \dot{\theta} \\ 0 \end{pmatrix} \tag{4.73}$$

Omitting the calculations, we directly report the power $\mathscr{P}$ per unit reference volume expended by the Eshelbian couple:

$$\mathscr{P} = \left(\mathbf{P}^{\mathrm{T}}\mathbf{F} - \mathbf{F}^{\mathrm{T}}\mathbf{P}\right)_{\times} \cdot \boldsymbol{\omega} = K^2\left[\check{\psi}_4 + \check{\psi}_5\left(K^2+3\right)\right]\sin(2\theta)\;\dot{\theta}. \tag{4.74}$$

As can be observed, this power is, in general, non-zero.

5 Numerical solution of the magnetoelastic problem

5.1 Weak formulation of the problem starting from the strong form

The main reference articles dealing with numerical methods for the resolution of magnetoelastic problems ([21], [5], [6]) present the field equations, keeping however implicit the weak formulation directly implementable in a finite element solver.

It will therefore be our care to derive a weak form of the problem starting from the corresponding strong form. Alternatively, this formulation can also be obtained by means of variational principles [10], but this path will not be the subject of further study in the present work.

The reference theory for the construction of weak forms and the subsequent finite element approximation starting from the strong forms of partial differential equations is taken from the text by Larson and Bengzon [16].

In the following treatment we will use the (Lagrangian) formulation in $\Omega = \Omega(\mathbf{F}, \mathbf{h}_l)$:

$$\mathbf{T} = \frac{\partial \Omega}{\partial \mathbf{F}}, \qquad \mathbf{b}_l = -\frac{\partial \Omega}{\partial \mathbf{h}_l}. \tag{5.1}$$

We define a domain $\mathcal{B} \cup \mathcal{B}'$, with $\mathcal{B} \cap \mathcal{B}' = \emptyset$, in which $\mathcal{B}$ is occupied by a magnetoelastic material and $\mathcal{B}'$ by vacuum. The boundary $\partial\mathcal{B}$ separates the material from the vacuum, while $\partial\mathcal{B}' = \partial\mathcal{B} \cup \partial\mathcal{B}'_\infty$, where $\partial\mathcal{B}'_\infty$ represents the boundary "at infinity" of the domain occupied by the vacuum.

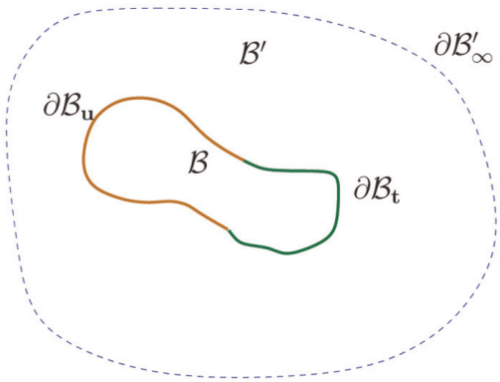

Figure 5.1: Problem domain [21]

Let us assume that there exists a *magnetic scalar potential* φ_l such that

$$\mathbf{h}_l = -\operatorname{Grad}\varphi_l, \tag{5.2}$$

whence

$$\operatorname{Curl}\mathbf{h}_l = \mathbf{0}, \tag{5.3}$$

is automatically satisfied.

The equation to be solved regarding the magnetic part is therefore

$$\operatorname{Div}\mathbf{b}_l = 0. \quad \text{in } \mathcal{B}\cup\mathcal{B}' \tag{5.4}$$

Regarding the mechanical part, in the absence of body forces, the following holds instead

$$\operatorname{Div}\mathbf{T} = \mathbf{0}. \quad \text{in } \mathcal{B} \tag{5.5}$$

We therefore have two coupled, non-linear partial differential equations valid in the domain $\mathcal{B}\cup\mathcal{B}'$ (reference configuration), in the unknowns $\mathbf{u}$ (displacement field) and φ_l.

Concerning the boundary or continuity conditions of the magnetic variables, we have:

$$[\![\mathbf{h}_l]\!]\times\mathbf{N} = \mathbf{0}, \qquad [\![\mathbf{b}_l]\!]\cdot\mathbf{N} = 0, \qquad \text{on } \partial\mathcal{B}. \tag{5.6}$$

Using the definition of magnetic scalar potential, these conditions become:

$$[\![\varphi_l]\!] = 0, \qquad [\![\mathbf{b}_l]\!]\cdot\mathbf{N} = 0, \qquad \text{on } \partial\mathcal{B}. \tag{5.7}$$

For the distant surface $\partial\mathcal{B}'_\infty$ additional boundary conditions are necessary. In this work it is assumed that:

$$\varphi_l = \tilde{\varphi}_l \quad \text{on } \partial\mathcal{B}'^{\varphi}_\infty, \qquad \mathbf{b}_l\cdot\mathbf{N} = 0 \quad \text{on } \partial\mathcal{B}'^{\mathbf{b}}_\infty, \tag{5.8}$$

where $\tilde{\varphi}_l$ is a prescribed value of the potential, and

$$\partial\mathcal{B}'^{\varphi}_\infty \cap \partial\mathcal{B}'^{\mathbf{b}}_\infty = \emptyset, \qquad \partial\mathcal{B}'^{\varphi}_\infty \cup \partial\mathcal{B}'^{\mathbf{b}}_\infty = \partial\mathcal{B}'_\infty. \tag{5.9}$$

Regarding the boundary conditions related to the mechanical part of the

problem, we impose:

$$\mathbf{u} = \mathbf{0} \quad \text{on } \partial\mathcal{B}_{\mathbf{u}}, \qquad \mathbf{TN} = \mathbf{t} \quad \text{on } \partial\mathcal{B}_{\mathbf{t}}, \tag{5.10}$$

where

$$\partial\mathcal{B}_{\mathbf{u}} \cap \partial\mathcal{B}_{\mathbf{t}} = \emptyset, \qquad \partial\mathcal{B}_{\mathbf{u}} \cup \partial\mathcal{B}_{\mathbf{t}} = \partial\mathcal{B}. \tag{5.11}$$

At this point we want to derive the weak formulation of the problem. Let us consider

$$\operatorname{Div} \mathbf{b}_l = 0 \quad \text{in } \mathcal{B} \cup \mathcal{B}'. \tag{5.12}$$

We multiply this equation by a *test function* $v \in \mathcal{V}$, where $\mathcal{V}$ is the functional space defined as

$$\mathcal{V} = \{v \in H^1(\mathcal{B} \cup \mathcal{B}') \,|\, v = 0 \text{ on } \partial\mathcal{B}'^{\varphi}_{\infty}\}, \tag{5.13}$$

$H^1(\mathcal{B} \cup \mathcal{B}')$ being the Hilbert space of square-integrable functions, equipped with square-integrable derivatives, defined on $\mathcal{B} \cup \mathcal{B}'$.

Integrating over the entire domain yields

$$\int_{\mathcal{B}\cup\mathcal{B}'} \operatorname{Div} \mathbf{b}_l \, v = 0. \tag{5.14}$$

Since $\mathbf{b}_l$ can be discontinuous across the separation surface between the inside and outside of the body (see Section 2.5), the internal and external contributions are separated:

$$\int_{\mathcal{B}} \operatorname{Div} \mathbf{b}_l \, v + \int_{\mathcal{B}'} \operatorname{Div} \mathbf{b}_l \, v = 0. \tag{5.15}$$

Applying integration by parts and the divergence theorem in each sub-domain, we obtain

$$\int_{\partial\mathcal{B}} \mathbf{b}_l \cdot \mathbf{N} \, v - \int_{\mathcal{B}} \mathbf{b}_l \cdot \operatorname{Grad} v + \int_{\partial\mathcal{B}'} \mathbf{b}_l \cdot \mathbf{N} \, v - \int_{\mathcal{B}'} \mathbf{b}_l \cdot \operatorname{Grad} v = 0. \tag{5.16}$$

Exploiting the continuity conditions (5.7) and the boundary conditions (5.8), the surface terms vanish and what remains is:

$$\int_{\mathcal{B}} \mathbf{b}_l \cdot \operatorname{Grad} v + \int_{\mathcal{B}'} \mathbf{b}_l \cdot \operatorname{Grad} v = 0. \tag{5.17}$$

In particular, in $\mathcal{B}$ and $\mathcal{B}'$ the magnetic fields are obtained from their respective

energy density functions:

$$\mathbf{b}_l = -\frac{\partial \Omega}{\partial \mathbf{h}_l} \quad \text{in } \mathcal{B}, \qquad \mathbf{b}_l = -\frac{\partial \Omega_{\text{out}}}{\partial \mathbf{h}_l} \quad \text{in } \mathcal{B}'. \tag{5.18}$$

These aspects will be discussed in detail in the following section; for a more in-depth exposition the reader is referred to [18].

At this point we analyze the mechanical equilibrium equation (5.5). We take a *test function* $\mathbf{w} \in \boldsymbol{\mathcal{W}}$, where

$$\boldsymbol{\mathcal{W}} = \left\{ \mathbf{w} \in [H^1(\mathcal{B})]^3 \;\middle|\; \mathbf{w} = \mathbf{0} \,\text{on}\, \partial\mathcal{B}_{\mathbf{u}} \right\}. \tag{5.19}$$

The space $[H^1(\mathcal{B})]^3$ is the *vector* Hilbert space consisting of displacement fields whose components are square-integrable functions, with square-integrable derivatives as well, equipped with the norm

$$\|\mathbf{w}\|^2_{[H^1(\mathcal{B})]^3} = \sum_{i=1}^{3} \left(\|w_i\|^2_{L^2(\mathcal{B})} + \|\nabla w_i\|^2_{L^2(\mathcal{B})} \right). \tag{5.20}$$

We multiply the equation scalarly by $\mathbf{w}$ and integrate over the body $\mathcal{B}$:

$$\int_{\mathcal{B}} \operatorname{Div} \mathbf{T} \cdot \mathbf{w} = 0. \tag{5.21}$$

Applying integration by parts and the divergence theorem, we obtain:

$$\int_{\mathcal{B}} \mathbf{T} \cdot \operatorname{Grad} \mathbf{w} = \int_{\partial\mathcal{B}} \mathbf{T}\mathbf{N} \cdot \mathbf{w}. \tag{5.22}$$

Exploiting the boundary conditions (7.10), the weak form of mechanical equilibrium is derived:

$$\int_{\mathcal{B}} \mathbf{T} \cdot \operatorname{Grad} \mathbf{w} = \int_{\partial\mathcal{B}_{\mathbf{t}}} \mathbf{t} \cdot \mathbf{w}. \tag{5.23}$$

Therefore, the weak formulation of the coupled magnetoelastic problem is the following: find $\varphi_l \in \varphi_l^* + \mathcal{V}$ and $\mathbf{u} \in \boldsymbol{\mathcal{W}}$ such that

$$\begin{aligned} &\int_{\mathcal{B}} \mathbf{b}_l \cdot \operatorname{Grad} v + \int_{\mathcal{B}'} \mathbf{b}_l \cdot \operatorname{Grad} v = 0, \\ &\int_{\mathcal{B}} \mathbf{T} \cdot \operatorname{Grad} \mathbf{w} = \int_{\partial\mathcal{B}_{\mathbf{t}}} \mathbf{t} \cdot \mathbf{w}, \end{aligned} \tag{5.24}$$

for every $v \in \mathcal{V}$ and $\mathbf{w} \in \boldsymbol{\mathcal{W}}$.

Here φ_l^* is the *boundary function*

$$\varphi_l^* = \begin{cases} \tilde{\varphi}_l, & \text{if } \mathbf{X} \in \partial\mathcal{B}_\infty'^{\varphi}, \\ 0, & \text{otherwise.} \end{cases} \tag{5.25}$$

5.2 Linearization and finite element approximation

Since the problem is highly nonlinear, we will adopt Newton's method for linearization. Here too, we rely on the text by Larson and Bengzon [16]. The method involves, initially, rewriting the unknown fields as follows:

$$\begin{aligned} \varphi_l &= \varphi_l^0 + \delta_\varphi, \\ \mathbf{u} &= \mathbf{u}^0 + \boldsymbol{\delta}_\mathbf{u}. \end{aligned} \tag{5.26}$$

where $\mathbf{u}^0$ and φ_l^0 are initial trial solutions, while $\boldsymbol{\delta}_\mathbf{u}$ and δ_φ are corrections. Using the Taylor series expansion truncated to the first order for $\mathbf{b}_l$ and $\mathbf{T}$ around the point $(\mathbf{u}^0, \varphi_l^0)$, we find:

$$\begin{aligned} \mathbf{b}_l(\operatorname{Grad}\mathbf{u}, \operatorname{Grad}\varphi_l) \approx\ & \mathbf{b}_l(\operatorname{Grad}\mathbf{u}^0, \operatorname{Grad}\varphi_l^0) + \mathbf{b}_\mathbf{u}'(\operatorname{Grad}\mathbf{u}^0, \operatorname{Grad}\varphi_l^0)\operatorname{Grad}\boldsymbol{\delta}_\mathbf{u} \\ & + \mathbf{b}_\varphi'(\operatorname{Grad}\mathbf{u}^0, \operatorname{Grad}\varphi_l^0)\operatorname{Grad}\delta_\varphi \end{aligned} \tag{5.27}$$

and

$$\begin{aligned} \mathbf{T}(\operatorname{Grad}\mathbf{u}, \operatorname{Grad}\varphi_l) \approx\ & \mathbf{T}(\operatorname{Grad}\mathbf{u}^0, \operatorname{Grad}\varphi_l^0) + \mathbf{T}_\mathbf{u}'(\operatorname{Grad}\mathbf{u}^0, \operatorname{Grad}\varphi_l^0)\operatorname{Grad}\boldsymbol{\delta}_\mathbf{u} \\ & + \mathbf{T}_\varphi'(\operatorname{Grad}\mathbf{u}^0, \operatorname{Grad}\varphi_l^0)\operatorname{Grad}\delta_\varphi \end{aligned} \tag{5.28}$$

where we have defined

$$\begin{aligned} \mathbf{b}_\mathbf{u}' &:= \frac{\partial\mathbf{b}_l}{\partial(\operatorname{Grad}\mathbf{u})}, & \mathbf{b}_\varphi' &:= \frac{\partial\mathbf{b}_l}{\partial(\operatorname{Grad}\varphi_l)}, \\ \mathbf{T}_\mathbf{u}' &:= \frac{\partial\mathbf{T}}{\partial(\operatorname{Grad}\mathbf{u})}, & \mathbf{T}_\varphi' &:= \frac{\partial\mathbf{T}}{\partial(\operatorname{Grad}\varphi_l)}. \end{aligned} \tag{5.29}$$

where $\mathbf{b}_\mathbf{u}'$ and $\mathbf{b}_\varphi'$ are third- and second-order tensors and $\mathbf{T}_\mathbf{u}'$ and $\mathbf{T}_\varphi'$ are fourth- and third-order tensors, respectively. To simplify the notation, we will use the superscript "0" if a given function is evaluated at the point $(\mathbf{u}^0, \varphi_l^0)$.

Neglecting higher-order terms, we arrive at the following linearized problem:

$$\int_{\mathcal{B}} \left(\mathbf{b}_{\mathbf{u}}'^{0} \operatorname{Grad} \boldsymbol{\delta}_{\mathbf{u}} + \mathbf{b}_{\varphi}'^{0} \operatorname{Grad} \delta_{\varphi}\right) \cdot \operatorname{Grad} v + \int_{\mathcal{B}'} \left(\mathbf{b}_{\mathbf{u}}'^{0} \operatorname{Grad} \boldsymbol{\delta}_{\mathbf{u}} + \mathbf{b}_{\varphi}'^{0} \operatorname{Grad} \delta_{\varphi}\right) \cdot \operatorname{Grad} v$$

$$= -\int_{\mathcal{B}} \mathbf{b}_{l}^{0} \cdot \operatorname{Grad} v - \int_{\mathcal{B}'} \mathbf{b}_{l}^{0} \cdot \operatorname{Grad} v \,, \tag{5.30}$$

$$\int_{\mathcal{B}} \left(\mathbf{T}_{\mathbf{u}}'^{0} \operatorname{Grad} \boldsymbol{\delta}_{\mathbf{u}} + \mathbf{T}_{\varphi}'^{0} \operatorname{Grad} \delta_{\varphi}\right) \cdot \operatorname{Grad} \mathbf{w}$$

$$= -\int_{\mathcal{B}} \mathbf{T}^{0} \cdot \operatorname{Grad} \mathbf{w} + \int_{\partial\mathcal{B}_{\mathbf{t}}} \mathbf{t} \cdot \mathbf{w}. \tag{5.31}$$

We define a partition $\mathscr{K}_1 = \{K_1\}$ of the domain $\mathcal{B}$ and a partition $\mathscr{K}_2 = \{K_2\}$ of the domain $\mathcal{B}'$ into tetrahedra K_1 and K_2. We choose to approximate the displacement field and the scalar potential using continuous piecewise linear functions (for each component of the displacement field). The resulting functional spaces will be:

$$\mathcal{V}_h = \{v \in S_{h1} \mid v = 0 \text{ on } \partial\mathcal{B}_{\infty}'^{\varphi}\} \tag{5.32}$$

$$\boldsymbol{\mathcal{W}}_h = \{\mathbf{w} \in [S_{h2}]^3 \mid \mathbf{w} = \mathbf{0} \text{ on } \partial\mathcal{B}_{\mathbf{u}}\} \tag{5.33}$$

where S_{h1} is the space of continuous piecewise linear functions on $\mathscr{K}_1 \cup \mathscr{K}_2$ and S_{h2} is the space of continuous piecewise linear functions on $\mathscr{K}_1$.

Therefore, the finite element approximation of the problem will be: find $\delta_{\varphi h} \in \mathcal{V}_h$ and $\boldsymbol{\delta}_{\mathbf{u}h} \in \boldsymbol{\mathcal{W}}_h$ such that

$$\int_{\mathcal{B}} \left(\mathbf{b}_{\mathbf{u}}'^{0} \operatorname{Grad} \boldsymbol{\delta}_{\mathbf{u}h} + \mathbf{b}_{\varphi}'^{0} \operatorname{Grad} \delta_{\varphi h}\right) \cdot \operatorname{Grad} v + \int_{\mathcal{B}'} \left(\mathbf{b}_{\mathbf{u}}'^{0} \operatorname{Grad} \boldsymbol{\delta}_{\mathbf{u}h} + \mathbf{b}_{\varphi}'^{0} \operatorname{Grad} \delta_{\varphi h}\right) \cdot \operatorname{Grad} v$$

$$= -\int_{\mathcal{B}} \mathbf{b}_{l}^{0} \cdot \operatorname{Grad} v - \int_{\mathcal{B}'} \mathbf{b}_{l}^{0} \cdot \operatorname{Grad} v \,, \tag{5.34}$$

$$\int_{\mathcal{B}} \left(\mathbf{T}_{\mathbf{u}}'^{0} \operatorname{Grad} \boldsymbol{\delta}_{\mathbf{u}h} + \mathbf{T}_{\varphi}'^{0} \operatorname{Grad} \delta_{\varphi h}\right) \cdot \operatorname{Grad} \mathbf{w}$$

$$= -\int_{\mathcal{B}} \mathbf{T}^{0} \cdot \operatorname{Grad} \mathbf{w} + \int_{\partial\mathcal{B}_{\mathbf{t}}} \mathbf{t} \cdot \mathbf{w}. \tag{5.35}$$

for every $v \in \mathcal{V}_h$ and $\mathbf{w} \in \boldsymbol{\mathcal{W}}_h$.

It is worth specifying that here the superscript "0" indicates that the function is

evaluated at the point $(\varphi^0_{lh}, \mathbf{u}^0_h)$.

We specify that for the approximate solutions it also holds true that

$$\begin{aligned} \varphi_{lh} &= \varphi^0_{lh} + \delta_{\varphi h}, \\ \mathbf{u}_h &= \mathbf{u}^0_h + \boldsymbol{\delta}_{\mathbf{u}h}. \end{aligned} \tag{5.36}$$

Writing the test functions as a linear combination of the *hat functions* $\{\phi_i\}$ and $\{\boldsymbol{\psi}_i\}$ of the respective spaces $\mathcal{V}_h$ and $\boldsymbol{\mathcal{W}}_h$, the expression is equivalent to

$$\begin{aligned} &\int_{\mathcal{B}} \left(\mathbf{b}'^0_{\mathbf{u}} \operatorname{Grad} \boldsymbol{\delta}_{\mathbf{u}h} + \mathbf{b}'^0_{\varphi} \operatorname{Grad} \delta_{\varphi h}\right) \cdot \operatorname{Grad} \phi_i + \int_{\mathcal{B}'} \left(\mathbf{b}'^0_{\mathbf{u}} \operatorname{Grad} \boldsymbol{\delta}_{\mathbf{u}h} + \mathbf{b}'^0_{\varphi} \operatorname{Grad} \delta_{\varphi h}\right) \cdot \operatorname{Grad} \phi_i \\ &\qquad = -\int_{\mathcal{B}} \mathbf{b}^0_l \cdot \operatorname{Grad} \phi_i - \int_{\mathcal{B}'} \mathbf{b}^0_l \cdot \operatorname{Grad} \phi_i \,, \end{aligned} \tag{5.37}$$

$$\begin{aligned} &\int_{\mathcal{B}} \left(\mathbf{T}'^0_{\mathbf{u}} \operatorname{Grad} \boldsymbol{\delta}_{\mathbf{u}h} + \mathbf{T}'^0_{\varphi} \operatorname{Grad} \delta_{\varphi h}\right) \cdot \operatorname{Grad} \boldsymbol{\psi}_i \\ &\qquad = -\int_{\mathcal{B}} \mathbf{T}^0 \cdot \operatorname{Grad} \boldsymbol{\psi}_i + \int_{\partial\mathcal{B}_{\mathbf{t}}} \mathbf{t} \cdot \boldsymbol{\psi}_i. \\ &\qquad\qquad i = 1, 2, \ldots, n_i \end{aligned} \tag{5.38}$$

where n_i is the number of internal nodes.

At this point, since $\delta_{\varphi h} \in \mathcal{V}_h$[1] and $\boldsymbol{\delta}_{\mathbf{u}h} \in \boldsymbol{\mathcal{W}}_h$, we can write

$$\begin{aligned} \delta_{\varphi h} &= \sum_{j=1}^{n_i} \xi_j \phi_j, \\ \boldsymbol{\delta}_{\mathbf{u}h} &= \sum_{j=1}^{n_i} \eta_j \boldsymbol{\psi}_j. \end{aligned} \tag{5.39}$$

therefore, we will obtain

$$\sum_{j=1}^{n_i} \xi_j \left(\int_{\mathcal{B}} \mathbf{b}'^0_{\varphi} \operatorname{Grad} \phi_j \cdot \operatorname{Grad} \phi_i + \int_{\mathcal{B}'} \mathbf{b}'^0_{\varphi} \operatorname{Grad} \phi_j \cdot \operatorname{Grad} \phi_i \right)$$

[1]We use a discrete lifting $\varphi^0_{lh} \in S_{h1}$ such that $\varphi^0_{lh}|_{\partial\mathcal{B}^{\varphi}_{\infty}} = \tilde{\varphi}_l$. The total field thus satisfies $\varphi_{lh} = \varphi^0_{lh} + \delta_{\varphi h}$ with $\delta_{\varphi h} \in \mathcal{V}_h$.

$$+\sum_{j=1}^{n_i}\eta_j\left(\int_{\mathcal{B}}\mathbf{b}_{\mathbf{u}}'^{0}\,\mathrm{Grad}\,\boldsymbol{\psi}_j\cdot\mathrm{Grad}\,\phi_i+\int_{\mathcal{B}'}\mathbf{b}_{\mathbf{u}}'^{0}\,\mathrm{Grad}\,\boldsymbol{\psi}_j\cdot\mathrm{Grad}\,\phi_i\right)$$

$$=-\int_{\mathcal{B}}\mathbf{b}_l^0\cdot\mathrm{Grad}\,\phi_i-\int_{\mathcal{B}'}\mathbf{b}_l^0\cdot\mathrm{Grad}\,\phi_i\,, \tag{5.40}$$

$$\sum_{j=1}^{n_i}\xi_j\int_{\mathcal{B}}\mathbf{T}_{\varphi}'^{0}\,\mathrm{Grad}\,\phi_j\cdot\mathrm{Grad}\,\boldsymbol{\psi}_i+\sum_{j=1}^{n_i}\eta_j\int_{\mathcal{B}}\mathbf{T}_{\mathbf{u}}'^{0}\,\mathrm{Grad}\,\boldsymbol{\psi}_j\cdot\mathrm{Grad}\,\boldsymbol{\psi}_i$$

$$=-\int_{\mathcal{B}}\mathbf{T}^0\cdot\mathrm{Grad}\,\boldsymbol{\psi}_i+\int_{\partial\mathcal{B}_{\mathbf{t}}}\mathbf{t}\cdot\boldsymbol{\psi}_i. \tag{5.41}$$

$$i=1,2,\ldots,n_i$$

Rewriting the two equations in matrix form, we obtain

$$\begin{aligned}\mathbf{J}^{\varphi\varphi}\boldsymbol{\xi}+\mathbf{J}^{u\varphi}\boldsymbol{\eta}&=-\mathbf{r}^{\varphi},\\ \mathbf{J}^{\varphi u}\boldsymbol{\xi}+\mathbf{J}^{uu}\boldsymbol{\eta}&=-\mathbf{r}^{u}.\end{aligned} \tag{5.42}$$

where

$$\begin{aligned}J_{ij}^{\varphi\varphi}&=\int_{\mathcal{B}}\mathbf{b}_{\varphi}'^{0}\,\mathrm{Grad}\,\phi_j\cdot\mathrm{Grad}\,\phi_i+\int_{\mathcal{B}'}\mathbf{b}_{\varphi}'^{0}\,\mathrm{Grad}\,\phi_j\cdot\mathrm{Grad}\,\phi_i,\\ J_{ij}^{u\varphi}&=\int_{\mathcal{B}}\mathbf{b}_{\mathbf{u}}'^{0}\,\mathrm{Grad}\,\boldsymbol{\psi}_j\cdot\mathrm{Grad}\,\phi_i+\int_{\mathcal{B}'}\mathbf{b}_{\mathbf{u}}'^{0}\,\mathrm{Grad}\,\boldsymbol{\psi}_j\cdot\mathrm{Grad}\,\phi_i,\\ J_{ij}^{\varphi u}&=\int_{\mathcal{B}}\mathbf{T}_{\varphi}'^{0}\,\mathrm{Grad}\,\phi_j\cdot\mathrm{Grad}\,\boldsymbol{\psi}_i,\\ J_{ij}^{uu}&=\int_{\mathcal{B}}\mathbf{T}_{\mathbf{u}}'^{0}\,\mathrm{Grad}\,\boldsymbol{\psi}_j\cdot\mathrm{Grad}\,\boldsymbol{\psi}_i,\\ r_i^{\varphi}&=\int_{\mathcal{B}}\mathbf{b}_l^0\cdot\mathrm{Grad}\,\phi_i+\int_{\mathcal{B}'}\mathbf{b}_l^0\cdot\mathrm{Grad}\,\phi_i,\\ r_i^{u}&=\int_{\mathcal{B}}\mathbf{T}^0\cdot\mathrm{Grad}\,\boldsymbol{\psi}_i-\int_{\partial\mathcal{B}_{\mathbf{t}}}\mathbf{t}\cdot\boldsymbol{\psi}_i.\end{aligned} \tag{5.43}$$

We can then assemble and write the following coupled linear system:

$$\mathbf{J}\Delta\mathbf{x}=-\mathbf{r} \tag{5.44}$$

where, in the component representation,

$$[\mathbf{J}] = \begin{pmatrix} [\mathbf{J}^{\varphi\varphi}] & [\mathbf{J}^{u\varphi}] \\ [\mathbf{J}^{\varphi u}] & [\mathbf{J}^{uu}] \end{pmatrix}, \quad \{\Delta\mathbf{x}\} = \begin{pmatrix} \{\xi\} \\ \{\eta\} \end{pmatrix}, \quad \{\mathbf{r}\} = \begin{pmatrix} \{\mathbf{r}^{\varphi}\} \\ \{\mathbf{r}^{u}\} \end{pmatrix} \tag{5.45}$$

One can then set up the following iterative algorithm to solve the problem:

Newton's algorithm for the magnetoelastic problem

1. Choose an initial solution $(\mathbf{u}_h^0, \varphi_{lh}^0)$, with $\mathbf{u}_h^0 \in \mathcal{W}_h$ and $\varphi_{lh}^0 \in \varphi_l^* + \mathcal{V}_h$, and a tolerance $\varepsilon > 0$.

2. **For** $k = 0, 1, 2, \ldots$, execute:

   (a) Assemble the matrix $\mathbf{J}^{(k)}$ and the residual $\mathbf{r}^{(k)}$, with:

$$
\begin{aligned}
J_{ij}^{\varphi\varphi(k)} &= \int_{\mathcal{B}} \mathbf{b}_{\varphi}'^{k} \operatorname{Grad} \phi_j \cdot \operatorname{Grad} \phi_i + \int_{\mathcal{B}'} \mathbf{b}_{\varphi}'^{k} \operatorname{Grad} \phi_j \cdot \operatorname{Grad} \phi_i \\
J_{ij}^{u\varphi(k)} &= \int_{\mathcal{B}} \mathbf{b}_{\mathbf{u}}'^{k} \operatorname{Grad} \boldsymbol{\psi}_j \cdot \operatorname{Grad} \phi_i + \int_{\mathcal{B}'} \mathbf{b}_{\mathbf{u}}'^{k} \operatorname{Grad} \boldsymbol{\psi}_j \cdot \operatorname{Grad} \phi_i \\
J_{ij}^{\varphi u(k)} &= \int_{\mathcal{B}} \mathbf{T}_{\varphi}'^{k} \operatorname{Grad} \phi_j \cdot \operatorname{Grad} \boldsymbol{\psi}_i, \\
J_{ij}^{uu(k)} &= \int_{\mathcal{B}} \mathbf{T}_{\mathbf{u}}'^{k} \operatorname{Grad} \boldsymbol{\psi}_j \cdot \operatorname{Grad} \boldsymbol{\psi}_i \\
r_i^{\varphi(k)} &= \int_{\mathcal{B}} \mathbf{b}_l^k \cdot \operatorname{Grad} \phi_i + \int_{\mathcal{B}'} \mathbf{b}_l^k \cdot \operatorname{Grad} \phi_i \\
r_i^{u(k)} &= \int_{\mathcal{B}} \mathbf{T}^k \cdot \operatorname{Grad} \boldsymbol{\psi}_i - \int_{\partial\mathcal{B}_{\mathbf{t}}} \mathbf{t} \cdot \boldsymbol{\psi}_i
\end{aligned}
$$

   (b) Solve the linear system:

$$
\mathbf{J}^{(k)} \Delta \mathbf{x}^{(k)} = -\mathbf{r}^{(k)}
$$

   (c) Update the solution:

$$
\mathbf{u}_h^{k+1} = \mathbf{u}_h^k + \boldsymbol{\delta}_{\mathbf{u}h}^k, \quad \varphi_{lh}^{k+1} = \varphi_{lh}^k + \delta_{\varphi h}^k.
$$

   (d) If $\left\|\boldsymbol{\delta}_h^k\right\| < \varepsilon$, terminate the iteration.

3. **End loop**

Where the increment vector is represented as:

$$
\{\boldsymbol{\delta}_h^k\} = \begin{pmatrix} \delta_{\varphi h}^k \\ \{\boldsymbol{\delta}_{\mathbf{u}h}^k\} \end{pmatrix}.
$$

5.3 Monolithic representation for the implementation

Following the discussion by Langtangen and Mardal [15], we adopt an alternative formulation of the weak form for systems of partial differential equations, which is the case under consideration.

Let there be a vector of unknown functions

$$\mathbf{U} = (U_1, \ldots, U_m), \tag{5.46}$$

and let us define the *product space*

$$\boldsymbol{\mathcal{V}} = \mathscr{V}_1 \times \ldots \times \mathscr{V}_m, \tag{5.47}$$

where, for each $i = 1, \ldots, m$,

$$\mathscr{V}_i = \{v \in H^1(\Omega) \,|\, v = 0 \text{ on } \partial\Omega_{U_i}\}. \tag{5.48}$$

It is further assumed that $U_i \in B_i + \mathscr{V}_i$, where B_i is a *boundary function* that accounts for any non-zero Dirichlet boundary conditions, and we define

$$\mathbf{B} = (B_1, \ldots, B_m). \tag{5.49}$$

Let us now consider the system of governing differential equations

$$\boldsymbol{\mathcal{L}}(\mathbf{U}) = \mathbf{0}, \tag{5.50}$$

where

$$\boldsymbol{\mathcal{L}}(\mathbf{U}) = \begin{pmatrix} \mathcal{L}_1(\mathbf{U}) \\ \vdots \\ \mathcal{L}_m(\mathbf{U}) \end{pmatrix} \tag{5.51}$$

is the vector of differential operators. By introducing a vector of test functions $\mathbf{v} \in \boldsymbol{\mathcal{V}}$,

$$\mathbf{v} = (v_1, \ldots, v_m), \tag{5.52}$$

the general weak formulation of the system is obtained by enforcing the inner product between the vector of equations and the vector of test functions: find $\mathbf{U} \in \mathbf{B} + \boldsymbol{\mathcal{V}}$ such that

$$\int_\Omega \boldsymbol{\mathcal{L}}(\mathbf{U}) \cdot \mathbf{v} = 0, \tag{5.53}$$

for all $\mathbf{v} \in \boldsymbol{\mathcal{V}}$.

5.3.1 Lagrangian scalar potential approach

In the specific case of the magneto-elastic problem presented in Section 5.1, we set

$$\mathbf{U} \equiv (\mathbf{u}, \varphi_l) \in \mathbf{B} + \boldsymbol{\mathcal{V}}, \tag{5.54}$$

with

$$\boldsymbol{\mathcal{V}} = \boldsymbol{\mathcal{W}} \times \mathcal{V}, \qquad \mathbf{B} = (\mathbf{0}, \varphi_l^*), \tag{5.55}$$

where $\boldsymbol{\mathcal{W}}$ and $\mathcal{V}$ are the functional spaces (5.19) and (5.13).

The system of governing differential equations takes the form

$$\begin{cases} \operatorname{Div} \mathbf{T} = \mathbf{0}, & \text{in } \mathcal{B}, \\ \operatorname{Div} \mathbf{b}_l = 0, & \text{in } \mathcal{B} \cup \mathcal{B}'. \end{cases} \tag{5.56}$$

Introducing the vector of test functions

$$\mathbf{v} = (\mathbf{w}, v) \in \boldsymbol{\mathcal{V}}, \tag{5.57}$$

the corresponding weak formulation is written as

$$\int_{\mathcal{B} \cup \mathcal{B}'} \operatorname{Div} \mathbf{T} \cdot \mathbf{w} + \operatorname{Div} \mathbf{b}_l \, v = 0. \tag{5.58}$$

Applying integration by parts and taking into account the boundary conditions, as shown in Section 5.1, we obtain the following *monolithic* weak formulation, equivalent to (7.24): find $\mathbf{U} \in \mathbf{B} + \boldsymbol{\mathcal{V}}$ such that

$$\int_{\mathcal{B}} \mathbf{T} \cdot \operatorname{Grad} \mathbf{w} - \int_{\partial \mathcal{B}_{\mathfrak{t}}} \mathbf{t} \cdot \mathbf{w} + \int_{\mathcal{B}} \mathbf{b}_l \cdot \operatorname{Grad} v + \int_{\mathcal{B}'} \mathbf{b}_l \cdot \operatorname{Grad} v = 0, \tag{5.59}$$

for all $\mathbf{v} = (\mathbf{w}, v) \in \boldsymbol{\mathcal{V}}$.

This is the weak form implemented in the FEM code. The linearization and solution via Newton's method are handled internally by the `FEniCSx` modules.

5.3.2 Lagrangian vector potential approach

To conclude the chapter, we introduce an alternative formulation of the magneto-elastic problem. In particular, we consider the representation in

$$\Omega^* = \Omega^*(\mathbf{F}, \mathbf{b}_l), \tag{5.60}$$

from which we obtain the constitutive relations

$$\mathbf{T} = \frac{\partial \Omega^*}{\partial \mathbf{F}}, \qquad \mathbf{h}_l = \frac{\partial \Omega^*}{\partial \mathbf{b}_l}. \tag{5.61}$$

In this formulation, which will be implemented in the last chapter, the analysis is conducted solely on the domain of the physical body $\mathcal{B}$, in the reference configuration, neglecting the vacuum region. A *vector* Lagrangian magnetic potential $\mathbf{a}_l$ is therefore introduced, such that

$$\mathbf{b}_l = \operatorname{Curl} \mathbf{a}_l, \tag{5.62}$$

so that the solenoidal condition $\operatorname{Div} \mathbf{b}_l = 0$ is automatically satisfied.

Starting from Maxwell's equation in the absence of surface currents,

$$\operatorname{Curl} \mathbf{h}_l = \mathbf{0} \quad \text{in } \mathcal{B}, \tag{5.63}$$

we derive the weak form of the magnetic problem. We multiply the equation by a test function $\mathbf{z} \in \mathcal{Z}$, with

$$\mathcal{Z} = \{\mathbf{v} \in H^3(\mathcal{B}) \,|\, \mathbf{v} = \mathbf{0} \text{ on } \partial\mathcal{B}_\mathbf{a}\}, \tag{5.64}$$

where $\partial\mathcal{B}_\mathbf{a} \subset \partial\mathcal{B}$, and integrate over $\mathcal{B}$:

$$\int_{\mathcal{B}} \operatorname{Curl} \mathbf{h}_l \cdot \mathbf{z} = 0. \tag{5.65}$$

Applying integration by parts and the divergence theorem yields:

$$\int_{\mathcal{B}} \mathbf{h}_l \cdot \operatorname{Curl} \mathbf{z} - \int_{\partial\mathcal{B}_\mathbf{h}} (\mathbf{h}_l \times \mathbf{N}) \cdot \mathbf{z} = 0, \tag{5.66}$$

where the boundary subsets satisfy $\partial\mathcal{B}_\mathbf{h} \cup \partial\mathcal{B}_\mathbf{a} = \partial\mathcal{B}$ and $\partial\mathcal{B}_\mathbf{h} \cap \partial\mathcal{B}_\mathbf{a} = \emptyset$.

Assuming the external magnetic field $\mathbf{h}_l^*$ is known, from (5.6) it follows that

$$\mathbf{h}_l \times \mathbf{N} = \mathbf{h}_l^* \times \mathbf{N}, \tag{5.67}$$

and therefore the previous expression becomes

$$\int_{\mathcal{B}} \mathbf{h}_l \cdot \operatorname{Curl} \mathbf{z} - \int_{\partial\mathcal{B}_{\mathbf{h}}} (\mathbf{h}_l^* \times \mathbf{N}) \cdot \mathbf{z} = 0. \tag{5.68}$$

Let us now introduce the product space

$$\mathcal{V}' = \mathcal{W} \times \mathcal{Z}, \tag{5.69}$$

where $\mathcal{W}$ is the functional space defined in (5.19). The weak formulation of the problem, in monolithic form, is therefore: find $\mathbf{U} \in \mathbf{B}' + \mathcal{V}'$ such that

$$\int_{\mathcal{B}} \mathbf{T} \cdot \operatorname{Grad} \mathbf{w} - \int_{\partial\mathcal{B}_{\mathbf{t}}} (\mathbf{t}_A + \mathbf{t}_M^*) \cdot \mathbf{w} + \int_{\mathcal{B}} \mathbf{h}_l \cdot \operatorname{Curl} \mathbf{z} - \int_{\partial\mathcal{B}_{\mathbf{h}}} (\mathbf{h}_l^* \times \mathbf{N}) \cdot \mathbf{z} = 0, \tag{5.70}$$

for all $\mathbf{v} = (\mathbf{w}, \mathbf{z}) \in \mathcal{V}'$.

The term $\mathbf{B}'$ represents the *boundary function* that accounts for the non-zero Dirichlet conditions imposed on the boundary. In this formulation, it is convenient to decompose the total surface traction $\mathbf{t}$ into its mechanical component $\mathbf{t}_A$ and magnetic component $\mathbf{t}_M^*$; this choice will prove useful in the analysis of the next chapter.

6 FEM Implementation

In this chapter, we implement the weak formulations of the magneto-elastic problem derived in the previous chapter. The entire numerical process is developed using the open-source software `FEniCSx`, a set of Python libraries designed for the finite element discretization of linear and nonlinear problems.

The chapter is divided into two main parts.

1. In the first part, we address a classical magneto-mechanical coupling problem: a magneto-elastic body immersed in a vacuum is described using the Lagrangian approach in $(\mathbf{u}, \varphi_l)$. The problem is similar to the one presented in [21]. The monolithic formulation of the problem is implemented, and we analyze its mechanical and magnetic response.

2. In the second part, we numerically reconstruct the *universal state* for isotropic magneto-elastic solids introduced in Section 3.1.4. We first solve an inverse problem, where the desired deformation and magnetic induction are prescribed; from the equilibrium conditions thus obtained, we extract the boundary conditions necessary to sustain this state. Finally, we use these conditions to solve the corresponding forward FEM problem in $(\mathbf{u}, \mathbf{a}_l)$, verifying the ability of the Lagrangian vector potential formulation to faithfully reproduce the universal state.

Recall that, in the previous chapter, the linearization and Newton's method were described explicitly for educational purposes. Here, these procedures are not implemented manually: instead, we make use of the built-in functionalities in `dolfinx`, specifically `NewtonSolver` and `NonlinearProblem`, which automate the resolution procedures illustrated in Chapter 5.

Regarding the constitutive model, we employ a *compressible neo-Hookean* energy. In the second part, to avoid the constraints associated with exact incompressibility, we adopt a sufficiently high bulk modulus value to guarantee a nearly incompressible behavior of the solid material.

Finally, we devote particular attention to the modeling of the vacuum. To obtain a numerically well-posed problem, it is necessary to assign it a (albeit minimal) elastic energy. The vacuum is therefore modeled as an extremely compressible material, characterized by a very low mechanical stiffness, while the magnetic

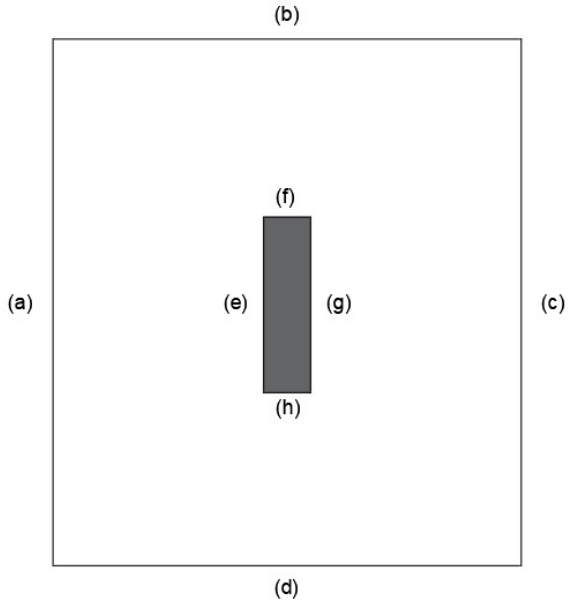

Figure 6.1: Geometry of the problem

energy is described using the vacuum permeability μ_0. The total absence of elastic energy in the vacuum would make the mechanical problem unsolvable, causing numerical instabilities.

The organization of the chapter follows the subdivision outlined above: Section 6.1 deals with the numerical formulation of the problem in $(\mathbf{u}, \varphi_l)$; Section 6.2 is dedicated to the reconstruction of the universal state and the corresponding forward FEM problem.

6.1 Magneto-elastic problem in $(\mathbf{u}, \varphi_l)$ with vacuum

The geometry of the problem is shown in Figure 6.1. The considered problem is entirely analogous to the one presented in [21]. The magneto-elastic body is represented by a rectangle of width $l_1 = 0.08$ m and height $l_2 = 0.30$ m.

Since it is not possible to numerically model a truly infinite vacuum domain, the dimensions of the external domain were chosen following the guidelines reported in [21]. In particular, a sufficiently extended vacuum domain was adopted to ensure that the fields calculated inside the body are practically indistinguishable from the ideal case of infinite vacuum. The dimensions used are $L_1 = 0.80$ m and $L_2 = 0.90$ m.

Before proceeding to the implementation, it is necessary to point out that, in accordance with what was discussed regarding the treatment of the vacuum, the weak form (5.59) undergoes a slight modification. In particular, the introduction of

an elastic energy in the vacuum implies that Ω_{out} also depends on the deformation gradient $\mathbf{F}$ and not only on $\mathbf{h}_l$. The weak form therefore becomes:

find $\mathbf{U} \in \mathbf{B} + \mathcal{V}$ such that

$$\int_{\mathcal{B}} \mathbf{T} \cdot \operatorname{Grad} \mathbf{w} + \int_{\mathcal{B}'} \mathbf{T} \cdot \operatorname{Grad} \mathbf{w} - \int_{\partial \mathcal{B}_{\mathfrak{t}}} \mathbf{t} \cdot \mathbf{w} + \int_{\mathcal{B}} \mathbf{b}_l \cdot \operatorname{Grad} v + \int_{\mathcal{B}'} \mathbf{b}_l \cdot \operatorname{Grad} v = 0, \quad (6.1)$$

for all $\mathbf{v} = (\mathbf{w}, v) \in \mathcal{V}$.

The expression for the stress tensor thus becomes

$$\mathbf{T} = \begin{cases} \dfrac{\partial \Omega}{\partial \mathbf{F}}, & \mathbf{X} \in \mathcal{B}, \\ \dfrac{\partial \Omega_{\text{out}}}{\partial \mathbf{F}}, & \mathbf{X} \in \mathcal{B}'. \end{cases} \quad (6.2)$$

Regarding the material's energy, we adopt — following [18] — the function

$$\Omega(\mathbf{F}, \mathbf{h}_l) = \frac{1}{2} G \left(\operatorname{tr} \mathbf{C} - 3 - 2 \ln \sqrt{\det \mathbf{C}} \right) + \frac{1}{2} \kappa \left(\sqrt{\det \mathbf{C}} - 1 \right)^2 - \frac{1}{2} \sqrt{\det \mathbf{C}}\, \mu_r \mu_0\, \mathbf{h}_l \cdot \mathbf{C}^{-1} \mathbf{h}_l, \quad (6.3)$$

and, analogously, for the vacuum:

$$\Omega_{\text{out}}(\mathbf{F}, \mathbf{h}_l) = \frac{1}{2} G_v \left(\operatorname{tr} \mathbf{C} - 3 - 2 \ln \sqrt{\det \mathbf{C}} \right) + \frac{1}{2} \kappa_v \left(\sqrt{\det \mathbf{C}} - 1 \right)^2 - \frac{1}{2} \sqrt{\det \mathbf{C}}\, \mu_0\, \mathbf{h}_l \cdot \mathbf{C}^{-1} \mathbf{h}_l. \quad (6.4)$$

For the mechanical part, the compressible neo-Hookean model [13] is used. The shear modulus G is chosen according to the values reported in [18], while the bulk modulus κ is assumed to be of the same order of magnitude.

The relative magnetic permeability of the material is fixed at $\mu_r = 1.5$ [18]. The vacuum permeability is denoted by μ_0.

Table 6.1: Material parameters of the magneto-elastic solid

Parameter	Value
Shear modulus G	1 MPa
Bulk modulus κ	1 MPa
Relative permeability μ_r	1.5
Vacuum permeability μ_0	$4\pi \times 10^{-7}$ H/m

The virtual mechanical parameters of the vacuum, G_v and κ_v, are chosen several

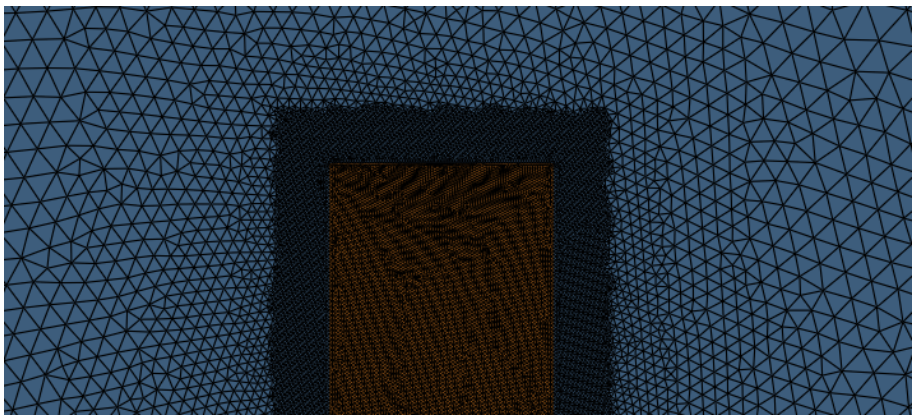

Figure 6.2: Various degrees of mesh refinement

orders of magnitude smaller than those of the solid, but still large enough to guarantee the convergence of the numerical solution.

The discretization of the domain was achieved using a non-uniform triangular mesh generated with `gmsh`, built starting from the geometry defined in Figure 6.1. The external domain $\mathcal{B}'$ is modeled as a rectangle of dimensions $L_1 \times L_2$, while the magneto-elastic body $\mathcal{B}$ is represented by a central rectangle with rounded corners of radius $r = 1$ mm. The introduction of a geometric fillet is not merely an aesthetic choice: as shown in [21], the presence of sharp corners can induce a high concentration of the magnetic field, with locally singular behavior of the solution, making the convergence of the nonlinear solver more difficult. The fillet thus allows physically regularizing the geometry and obtaining a more stable numerical behavior in the calculation of magnetic fields and derived quantities.

Particular attention was paid to the quality of the mesh in the region adjacent to the internal boundaries $\partial\mathcal{B}$. Indeed, a local refinement was applied around the inclusion, in order to obtain smaller elements in the areas where a greater variation of the magnetic and mechanical fields is expected, while a progressively coarser discretization is maintained in the external domain $\mathcal{B}'$.

This choice allows accurately describing the behavior of the fields near the interface, without resorting to a uniformly fine mesh over the entire domain, with considerable savings in terms of computational cost.

For the finite element implementation, the functional spaces for the problem unknowns are first defined. The domain is discretized using degree-one Lagrangian finite elements for both the displacement $\mathbf{u}$ and the magnetic potential φ_l. The

elements are then combined into a mixed space

$$W_h = \mathcal{W}_h \times \mathcal{V}_h, \tag{6.5}$$

necessary for the monolithic formulation of the coupled problem. The mixed space is built using the `mixed_element` constructor, while the overall solution function $\mathbf{U} = (\mathbf{u}, \varphi_l)$ is defined as a `Function(W)` object.

For convenience in applying the boundary conditions and for post-processing operations, the mixed spaces are subsequently "collapsed" into their components, obtaining the spaces $\mathcal{W}_h$ and $\mathcal{V}_h$ and the local indices of the associated degrees of freedom. The test functions $\mathbf{w}, v$ and the trial functions $\mathbf{u}_h, \varphi_{lh}$ are then introduced using `TestFunction` and `TrialFunction`, splitting the scalar and vector components[1].

Finally, the integration measures on the domain and boundaries are defined, specifying a quadrature degree of 8. The `dx` and `dS` measures are enriched with the marking information of the subdomains and facets, so as to be able to integrate distinctly over the magneto-elastic body and the vacuum.

```
from basix.ufl import element, mixed_element

# Functional spaces
cellname = domain.ufl_cell().cellname()
Ve_u = element("Lagrange", cellname, 1, shape=(gdim,))
Ve_m = element("Lagrange", cellname, 1)
W = fem.functionspace(domain, mixed_element([Ve_u, Ve_m]))

# Collapsed spaces for constraints and post-processing
V_u_c, dofs_u = W.sub(0).collapse()
V_phi_c, dofs_phi = W.sub(1).collapse()
interp_pts_vec = V_u_c.element.interpolation_points()

# Unknowns / test / trial
U = fem.Function(W)
dU = ufl.TrialFunction(W)
```

[1]Unlike what was shown in the previous chapter, where the linearization and Newton's method were applied to the continuous formulation prior to discretization, in `FEniCSx` the procedure is reversed: the weak formulation is first discretized using finite elements, and only subsequently does the solver automatically apply Newton's method to the discrete version of the problem. Conceptually, the difference is marginal, since both procedures lead to the same result.

```
Z = ufl.TestFunction(W)

u_, phi_ = ufl.split(U)
w_, v_ = ufl.split(Z)
du_, dphi_ = ufl.split(dU)

# Measures
metadata = {"quadrature_degree": 8}
dx = ufl.Measure("dx", domain=domain, subdomain_data=cell_tag,
metadata=metadata)
dS = ufl.Measure("dS", domain=domain, subdomain_data=facet_tag,
metadata=metadata)
```

The variables $\mathbf{T}$ and $\mathbf{b}_l$ are obtained via symbolic differentiation of the energies Ω and Ω_{out}, using the `ufl.diff` function provided by `FEniCSx`. In particular:

```
Pin   = ufl.diff(Omega, F)
blin  = -ufl.diff(Omega, hl)
Pout  = ufl.diff(OmegaOut, F)
blout = -ufl.diff(OmegaOut, hl)
```

The boundary conditions, referring to the edges (a)–(h) shown in Figure 6.1, are as follows:

- edges (a), (c): natural magnetic condition

$$\mathbf{b}_l \cdot \mathbf{N} = 0,$$

- edges (b), (d): fixed magnetic potential

$$\varphi_l = \mp\tilde{\varphi}_l,$$

  respectively;

- edge (f): application of a vertical traction

$$\mathbf{t} = t\,\mathbf{e}_2,$$

- edges (e), (g): traction-free conditions;

- edge (h): condition on the displacement

$$u_2 = 0.$$

The imposition of the potential values on edges (b) and (d) is aimed at modeling an external magnetic field that, far from the body, is approximately uniform and directed along $\mathbf{e}_2$. In particular, we consider as a reference field

$$\tilde{\mathbf{h}}_l = h_0\, \mathbf{e}_2. \tag{6.6}$$

Since in our approach we know that $\mathbf{h}_l = -\operatorname{Grad} \varphi_l$, if the field were exactly uniform throughout the external domain we would have:

$$\tilde{h}_{l1} = -\frac{\partial \tilde{\varphi}_l}{\partial X_1} = 0, \qquad \tilde{h}_{l2} = -\frac{\partial \tilde{\varphi}_l}{\partial X_2} = h_0. \tag{6.7}$$

Integrating along the height of the external domain L_2, the corresponding potential difference is

$$\Delta \tilde{\varphi}_l = -\int_0^{L_2} h_0\, dX_2 = -h_0\, L_2. \tag{6.8}$$

To construct a boundary condition consistent with this reference field, we impose two constant and symmetric values on the upper and lower edges:

$$\mp \tilde{\varphi}_l = \mp \frac{\Delta \tilde{\varphi}_l}{2} = \mp \frac{1}{2} h_0\, L_2. \tag{6.9}$$

In the absence of the magneto-elastic body, this choice would produce a uniform field throughout the external domain; in the present configuration, it provides a good approximation of a uniform applied field far from the body, allowing the actual field to adapt locally to the presence of the solid.

Alongside the continuous functional spaces used for the problem unknowns, we also introduce the discrete space of type DG0, defined as

$$\mathrm{DG}_0 = \left\{ q_h \in L^2(\mathcal{B} \cup \mathcal{B}') \;\middle|\; q_h|_K \text{ is constant for each cell } K \in \mathscr{K} \right\}, \tag{6.10}$$

where $L^2(\mathcal{B} \cup \mathcal{B}')$ denotes the space of square-integrable functions on the physical domain $\mathcal{B} \cup \mathcal{B}'$. The functions belonging to DG_0 are therefore piecewise constant and do not impose any continuity between adjacent cells.

In this space, we construct the discrete characteristic function

$$\chi_h \in \mathrm{DG}_0, \qquad \chi_h(\mathbf{X}) = \begin{cases} 1, & \mathbf{X} \in \mathcal{B}, \\ 0, & \mathbf{X} \in \mathcal{B}', \end{cases} \tag{6.11}$$

which allows for precisely distinguishing the magneto-elastic body from the surrounding vacuum directly at the discretization level. The discontinuous nature of DG_0 makes it possible to correctly select the boundary contributions in the integrations with traces $(\cdot)^+$ and $(\cdot)^-$, as in the case of applying external tractions on the upper side of the domain.

The discrete form of the residual, constructed in accordance with the variational formulation presented above, is given below in the `ufl` syntax of `FEniCSx`:

```
F_u = (
      ufl.inner(Pin,   ufl.grad(w_)) * dx(1)
    + ufl.inner(Pout,  ufl.grad(w_)) * dx(2)
    - (chi('-')*ufl.inner(t_surface, w_('+'))
       + chi('+')*ufl.inner(t_surface, w_('-'))) * dS_top
)

F_phi = (
      ufl.inner(blout, ufl.grad(v_)) * dx(2)
    + ufl.inner(blin,  ufl.grad(v_)) * dx(1)
)

F_tot = F_u + F_phi
```

The numerical solution of the nonlinear problem is entrusted to the Newton solver integrated into `FEniCSx`. The finite element residual is provided to the `NonlinearProblem` constructor, and the corresponding Jacobian matrix is computed automatically.

The `NewtonSolver` uses a globalized Newton method via backtracking *line search*, which reduces the update step[2] if the Newton increment does not produce an adequate decrease in the residual. The absolute and relative tolerances are set to `atol` $= 10^{-8}$ and `rtol` $= 10^{-8}$ respectively, while the maximum number of

[2]The backtracking strategy implements a form of *damped Newton method*, where the Newton increment is scaled by a relaxation parameter $\alpha \in (0, 1]$. The update therefore takes the form $\mathbf{U}^{(k+1)} = \mathbf{U}^{(k)} + \alpha\, \boldsymbol{\delta}_{\mathbf{U}}^{(k)}$, with α chosen automatically to ensure a sufficient reduction of the nonlinear residual.

iterations is limited to `max_it` = 50. At the end of the iteration, the convergence of the residual is verified before updating the total field of the unknowns.

6.1.1 Results

In this first configuration, we consider an applied magnetic field with an intensity of $h_0 = 10^6$ A/m and zero surface traction on the body's edges. The distributions of the mechanical and magnetic fields are shown in Figures 6.3–6.5.

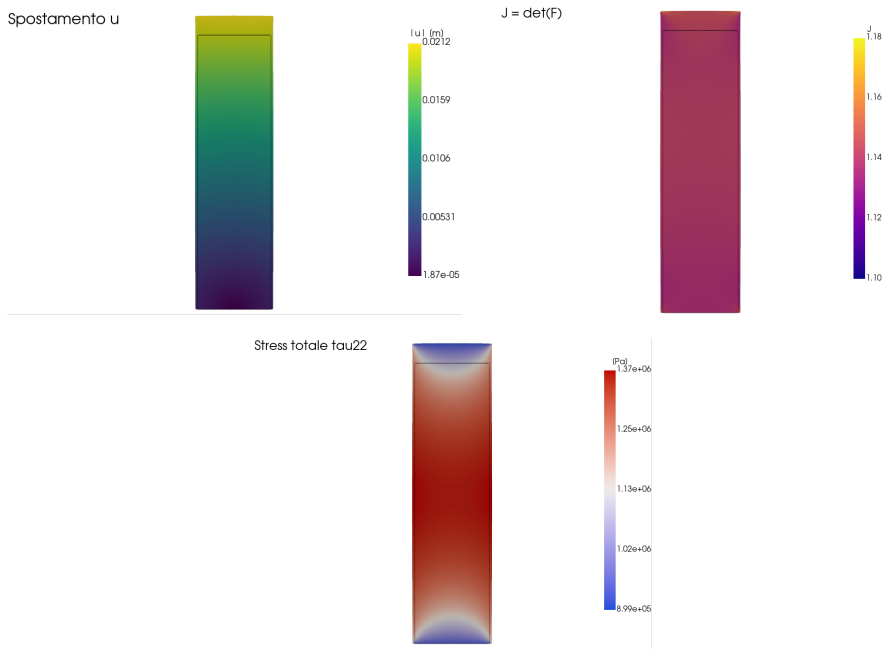

Figure 6.3: Displacement, Jacobian $J = \det \mathbf{F}$, and total stress component τ_{22} in the configuration $h_0 = 10^6$ A/m without surface traction.

The resulting deformation is significant (the scale is unitary), as expected from a soft material subjected to such intense magnetic fields. The Jacobian J shows limited variations within the body, with maximums located near the inclusion's vertices, where the magnetic fields are most intense.

The total stress τ_{22} is almost uniform in the central part of the body and shows more pronounced variations only near the upper and lower edges. The range between the maximum and minimum values is relatively small (about 0.4 MPa).

In cases where the magnetic stress is comparable to or greater than the mechanical one (see the present test), the presence of a gradient in τ_{22} is natural and follows the distribution of the magnetic field component h_2. Conversely, when the mechanical response dominates the magnetic one (a case not reported here, but numerically verified), the stress state tends to become almost uniaxial, and the τ_{22} component is nearly uniform throughout the domain.

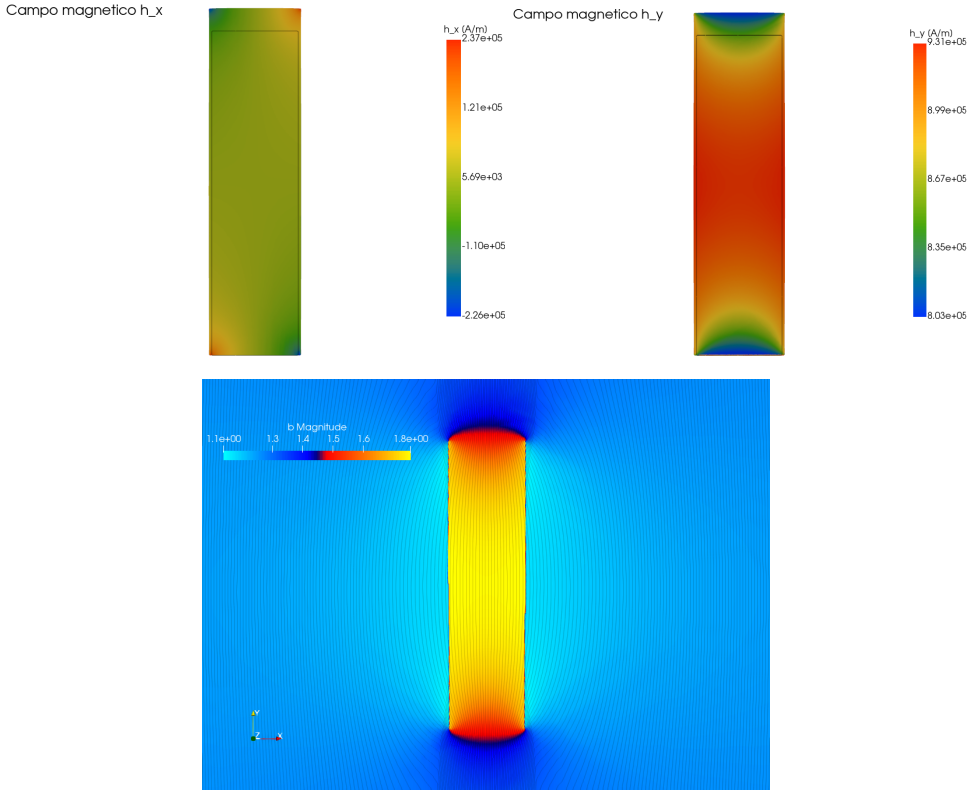

Figure 6.4: Components h_1, h_2 of the magnetic field and streamlines of the magnetic induction $\mathbf{b}$.

The horizontal component h_1 presents local peaks at the vertices of the inclusion, where the flow lines curve more sharply, as highlighted by the streamlines of $\mathbf{b}$.

The vertical component h_2, on the other hand, reaches maximum values in the central part of the body and decreases steadily toward the upper and lower edges. The structure of the field is consistent with the imposed boundary conditions and the flux concentration effects introduced by the inclusion.

Figure 6.5 shows the profiles of the Lagrangian magnetic components along vertical lines intersecting the inclusion. It can be observed that:

- b_{l2} (normal component of the magnetic induction) is continuous at the two interfaces, in agreement with the physical condition

$$[\![\mathbf{b}_l]\!] \cdot \mathbf{N} = 0.$$

- h_{l1} (tangential component of the magnetic field) is also continuous, as required by the condition

$$[\![\mathbf{h}_l]\!] \times \mathbf{N} = \mathbf{0}.$$

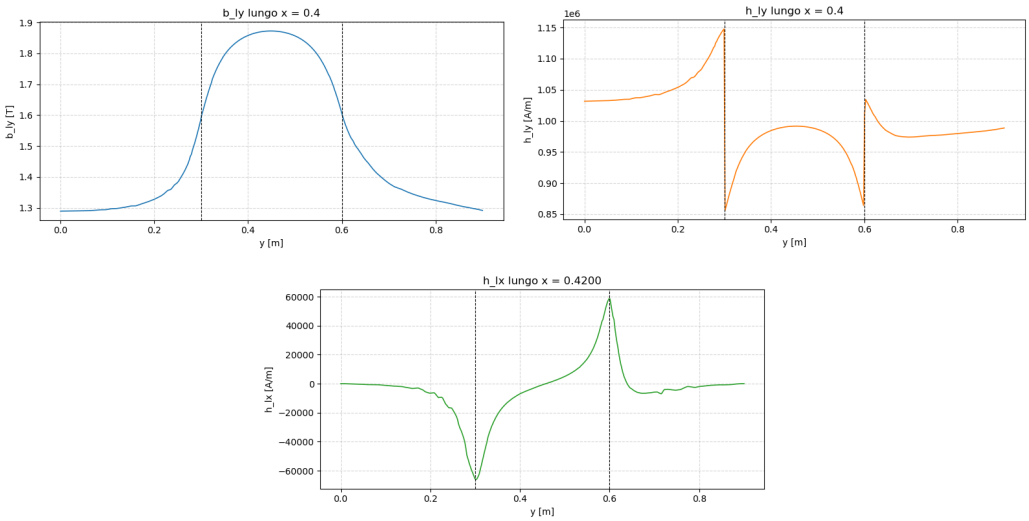

Figure 6.5: Profile of b_{l2}, h_{l2}, and h_{l1} along vertical lines crossing the inclusion. The dashed lines indicate the two vacuum-solid interfaces.

- h_{l2} instead presents a jump, consistent with the different magnetic permeabilities of the vacuum and the solid.

Let us now move on to the second case, where we apply a vertical surface traction $\mathbf{t} = t\,\mathbf{e}_2$ with $t = 200$ kPa. The intensity of the applied magnetic field is halved compared to the previous case, setting $h_0 = 5 \times 10^5$ A/m. The results are shown in Figure 6.6.

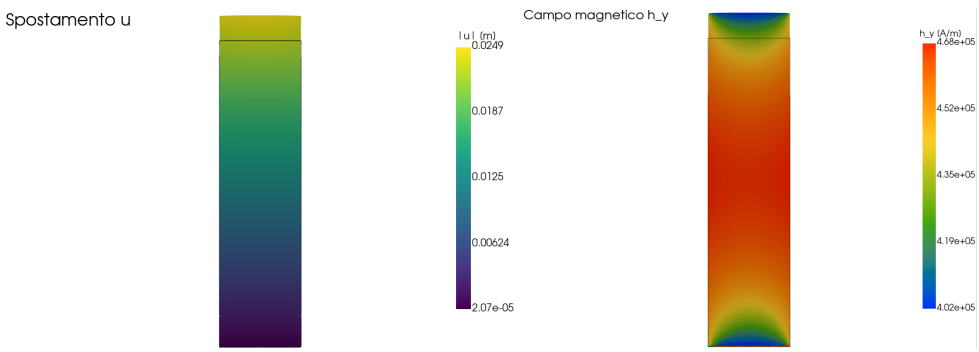

Figure 6.6: Distribution of the displacement $\mathbf{u}$ and the Eulerian magnetic component h_2 in the case with surface traction $t = 200$ kPa and magnetic field $h_0 = 5 \times 10^5$ A/m.

First, it can be observed that the magnitude of the displacements is of the same order as in the previous case: approximately 21 mm under the magnetic action alone and about 25 mm when surface traction is also present. However, the overall mechanical behavior is different: in the first case, the body tends to elongate vertically *and* widen laterally; in the second case, the application of the traction $t\,\mathbf{e}_2$ instead induces an elongation accompanied by a lateral contraction.

The distribution of the vertical component of the magnetic field h_2 is qualitatively very similar to that of the case without traction. The reduction in the applied field only results in a decrease in intensity, without altering the pattern of the lines of force.

6.2 Numerical reconstruction of a controllable state

6.2.1 Inverse problem

The objective of this section is to consider a controllable state case belonging to the family of homogeneous deformations with a uniform magnetic quantity introduced in Section 3.1.4, and to calculate the surface tractions and external fields required to maintain this state. In particular, for the infinite plate of Section 3.1.4, we impose a simple shear deformation:

$$x_1 = X_1 + \gamma X_3, \quad x_2 = X_2, \quad x_3 = X_3. \tag{6.12}$$

The plate is bounded along the X_3 direction by the surfaces $X_3 = 0$ and $X_3 = 0.12$ m. We want the plate to deform with a shear angle of 3°; therefore,

$$\gamma = \tan 3^\circ \approx 0.0524. \tag{6.13}$$

The deformation gradient $\mathbf{F}$, uniform over the entire body, is

$$[\mathbf{F}] = \begin{pmatrix} 1 & 0 & 0.0524 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{6.14}$$

We also impose an Eulerian magnetic induction $\{\mathbf{b}\} = (0, 0, b_0)^{\mathrm{T}}$ with $b_0 = 0.2$ T.

For this problem, we adopt the formulation in $\Omega(\mathbf{F}, \mathbf{b}_l)$, with the corresponding constitutive relations

$$\mathbf{T} = \frac{\partial \Omega}{\partial \mathbf{F}}, \qquad \mathbf{h}_l = \frac{\partial \Omega}{\partial \mathbf{b}_l}. \tag{6.15}$$

We assume that the material is an isotropic magneto-elastic solid and choose an energy function of the type

$$\Omega(\mathbf{F}, \mathbf{b}_l) = \frac{1}{2} G \left(\operatorname{tr} \mathbf{C} - 3 - 2 \ln \sqrt{\det \mathbf{C}}\right) + \frac{1}{2} \kappa \left(\sqrt{\det \mathbf{C}} - 1\right)^2 - \frac{1}{2} \sqrt{\det \mathbf{C}}\, \mu_r \mu_0\, \mathbf{b}_l \cdot \mathbf{C} \mathbf{b}_l. \tag{6.16}$$

The values of the material parameters are summarized in Table 6.2.

Table 6.2: Material parameters of the magneto-elastic solid

Parameter	Value
Shear modulus G	4 MPa
Bulk modulus κ	800 MPa
Relative permeability μ_r	1.5
Vacuum permeability μ_0	$4\pi \times 10^{-7}$ H/m

The value of κ was chosen to be particularly high to simulate the incompressibility of the material without imposing the exact constraint[3].

The calculations whose results we report were performed using the Python library `jax`; the procedure implemented in the code follows the one described in Section 3.1.4.

It should be noted that, for the purpose of comparison with the forward FEM problem that will be presented later, the surface tractions and external fields are reported in Lagrangian form, since in the forward problem it is not possible to work directly with Eulerian variables.

Therefore, it turns out that to maintain the uniform deformation state and magnetic induction described above, the following values are required on the boundaries $X_3 = 0$ and $X_3 = 0.12$ m with normals $\{\mathbf{N}\} = (0, 0, \mp 1)^{\mathrm{T}}$:

- *mechanical* surface traction $\mathbf{t}_A$:

$$t_{A1} \approx \pm 210 \text{ kPa}, \qquad t_{A3} = 0,$$

  with the positive sign on the upper edge ($X_3 = 0.12$ m) and the negative sign on the lower edge ($X_3 = 0$);

- Lagrangian external magnetic induction $\mathbf{b}_l^*$:

$$b_{l1}^* = -0.0105 \text{ T}, \qquad b_{l3}^* = 0.2 \text{ T},$$

  identical on both sides;

- Lagrangian external magnetic field $\mathbf{h}_l^*$:

$$h_{l1}^* = 0, \qquad h_{l3}^* = 1.59 \times 10^5 \text{ A/m},$$

[3]In the inverse problem presented here, the value of κ is nonetheless not decisive: the simple shear deformation is isochoric, meaning $J = \det \mathbf{F} = 1$.

also equal on both edges.

These values constitute the boundary conditions that must be imposed in the forward finite element problem to numerically reconstruct the universal state of simple shear with uniform magnetic induction.

6.2.2 Forward problem

The first aspect to address in the FEM implementation is the modeling of a domain that, in the theoretical model, represents an infinite plate in the X_1 direction. Since an infinite discretization is not feasible, we chose to truncate the plate in the X_1X_3 plane at the values

$$X_1 = 0, \qquad X_1 = 0.80 \text{ m}.$$

The length was chosen large enough to minimize edge effects and obtain a central region in which the solution approximates that of the infinite plate.

Mechanical and magnetic tractions derived from the inverse problem are applied to the upper edge; the lower edge is clamped by imposing $u_1 = u_3 = 0$. The prescribed external fields are imposed on both edges.

The FEM problem is solved using the Lagrangian formulation in $(\mathbf{u}, \mathbf{a}_l)$ described in Section 5.3.2. It is therefore necessary to impose a Dirichlet condition on the Lagrangian vector potential on the boundaries such that

$$\operatorname{Curl} \mathbf{a}_l = \mathbf{b}_l^* \qquad \text{on } \partial\mathcal{B}.$$

A compatible choice satisfying this constraint is:

$$a_{l1} = \frac{1}{2} b_{l2}^* X_3, \qquad a_{l2} = b_{l3}^* X_1 - b_{l1}^* X_3, \qquad a_{l3} = -\frac{1}{2} b_{l2}^* X_1.$$

The results are shown in Figure 6.7. It is observed that in the central region the body is subjected to a simple shear deformation and the magnetic induction is almost uniform, while near the edges the effects of the unrealized infinite nature of the domain manifest.

To more precisely quantify the correspondence with the ideal state, we report some statistics in the central portion of the domain $X_1 \in (0.24 \text{ m}, 0.56 \text{ m})$. Table 6.3 compares the mean values of the considered components and the RMS error relative to the target values.

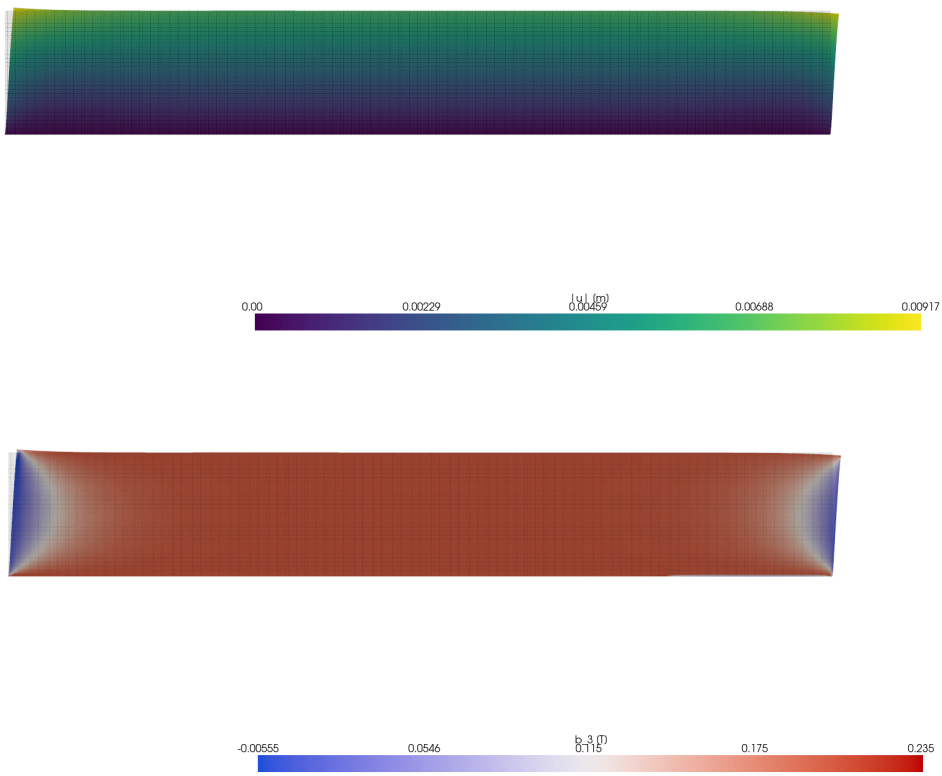

Figure 6.7: Resulting displacement field and component b_3 of the magnetic induction.

Table 6.3: Statistics of the magnetic components and deformation gradient in the region $X_1 \in (0.24\text{ m}, 0.56\text{ m})$.

Component	Mean value	Target	RMS Error
b_1 [T]	0.00021	0	9.8×10^{-4}
b_3 [T]	0.19992	0.2	1.7×10^{-4}
F_{11}	1.00002	1	3.4×10^{-4}
F_{13}	0.05384	0.0524	15.4×10^{-4}
F_{31}	−0.00010	0	1.8×10^{-4}
F_{33}	0.99978	1	3.4×10^{-4}

The resulting mean shear angle is

$$\arctan(0.05384) \approx 3.08^\circ.$$

By further increasing the length in the X_1 direction, the approximation improves. Considering a domain bounded by $X_1 = 0$ and $X_1 = 1.60$ m, the following values are obtained:

Table 6.4: Statistics in the central region upon doubling the domain size.

Component	Mean value	Target	RMS Error
b_1 [T]	0.00013	0	1.7×10^{-4}
b_3 [T]	0.20001	0.2	0.7×10^{-4}
F_{11}	0.99972	1	3.6×10^{-4}
F_{13}	0.05314	0.0524	8.7×10^{-4}
F_{31}	−0.00004	0	1.0×10^{-4}
F_{33}	1.00029	1	3.6×10^{-4}

The mean angle is now

$$\arctan(0.05314) \approx 3.04^\circ.$$

We can therefore conclude that, despite working on a finite domain, it is possible to achieve the ideal controllable state with good accuracy in the region far from the edges: as the length of the domain increases, the approximation systematically improves.

7 Conclusions

In this thesis, we have explored various theoretical and numerical aspects related to anisotropic elastic materials, specifically magneto-elastic ones, and their corresponding universal relations, as well as the numerical simulation of the behavior of such materials using the finite element method.

- **Universal relations in anisotropic elastic materials:** We developed a new method to rapidly derive universal relations in anisotropic elastic materials. Our proposal is distinguished by the adoption of an integral condition of isotropy on the internal energy, rather than the locally used traditional condition. In addition, we provided an interpretation of these relations, highlighting the orthogonality between the skew-symmetric part of the Eshelby tensor and the referential direction of anisotropy. With reference to the literature, we discussed how the symmetric part of the Eshelby tensor, defined as the Eshelbian couple, performs work on the possible remodeling of the referential fibers. Finally, we illustrated an application example in which the power per unit of reference volume expended by the Eshelbian couple in an anisotropic elastic solid, where fibers are allowed to reorganize, was calculated.

- **Numerical part and implementation of the nonlinear coupled magneto-elastic problem:** We derived the weak form of the balance problem starting from the differential equations in local (strong) form, and developed an algorithm based on Newton's method to solve the system. Subsequently, we implemented a FEM code using Python/`FEniCSx` for the numerical resolution of the problem. The obtained results were in agreement with those reported in the literature, confirming the validity of our numerical approach. Furthermore, we numerically reconstructed a universal state for a magneto-elastic solid and verified the results through an additional FEM code, confirming the accuracy of our simulations.

The considerations regarding universal relations require further investigation. In particular, it would be interesting to explore a generalization of them, extending them to the case in which the fiber distribution is not fixed but admits a remodeling, possibly dissipative. It is known, in fact, that in the case of passive remodeling, i.e., when there is no external action on the fibers, the fibers evolve towards a

configuration that makes the material's response isotropic [8]. An in-depth study of this phenomenon could lead to original and extremely interesting conclusions.

Furthermore, regarding the numerical implementation, it would be appropriate to explore more effective architectures for the solver. In our approach, in fact, we used a relatively simple energy function, while previous studies, such as that of [21], have employed more complex energy models.

Overall, the conclusions of this thesis are to be considered more than satisfactory, especially in light of the initial objectives, and offer a good starting point for further theoretical and applicative developments.

Appendix

Isotropic functions

This section collects some fundamental definitions and results, accompanied by relative proofs, concerning *isotropic functions*. The discussion closely follows that proposed by Gurtin [12]. For the sake of brevity, the proofs of some theorems are not reported here in full; for a complete exposition, please refer to the cited text.

Definition 1. Let $\mathscr{V}$ denote the vector space associated with the three-dimensional Euclidean space.

Definition 2. Given a set $\{\mathbf{u}, \mathbf{v}, \ldots, \mathbf{w}\}$ of vectors (or tensors), its *span* is defined as the subspace of $\mathscr{V}$ consisting of all possible linear combinations of such vectors:

$$\operatorname{span}\{\mathbf{u}, \mathbf{v}, \ldots, \mathbf{w}\} = \{\alpha\mathbf{u} + \beta\mathbf{v} + \ldots + \gamma\mathbf{w} \mid \alpha, \beta, \ldots, \gamma \in \mathbb{R}\}.$$

This notation will also be used for spaces other than $\mathscr{V}$.

Definition 3. Given a vector $\mathbf{v}$, we define

$$\{\mathbf{v}\}^{\perp} = \{\mathbf{u} \mid \mathbf{u} \cdot \mathbf{v} = 0\},$$

which is the subspace of $\mathscr{V}$ consisting of all vectors orthogonal to $\mathbf{v}$.

Spectral theorem. Let $\mathbf{S} \in \mathrm{Sym}$. Then there exists an orthonormal basis of $\mathscr{V}$ consisting entirely of eigenvectors of $\mathbf{S}$. Furthermore, for any such basis $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$, the corresponding eigenvalues $\omega_1, \omega_2, \omega_3$, when ordered, constitute the entire spectrum of $\mathbf{S}$, and the following holds:

$$\mathbf{S} = \sum_i \omega_i \, \mathbf{e}_i \otimes \mathbf{e}_i.$$

Conversely, if $\mathbf{S}$ has the form (8.1) with $\{\mathbf{e}_i\}$ orthonormal, then $\omega_1, \omega_2, \omega_3$ are eigenvalues of $\mathbf{S}$ with corresponding eigenvectors $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$. Furthermore:

(a) $\mathbf{S}$ has exactly three distinct eigenvalues if and only if the characteristic spaces of $\mathbf{S}$ are three mutually perpendicular lines passing through the origin.

(b) **S** has exactly two distinct eigenvalues if and only if it admits the representation

$$\mathbf{S} = \omega_1\, \mathbf{e} \otimes \mathbf{e} + \omega_2\, (\mathbf{I} - \mathbf{e} \otimes \mathbf{e})\,, \quad |\mathbf{e}| = 1, \quad \omega_1 \neq \omega_2.$$

In this case, ω_1 and ω_2 are the two distinct eigenvalues, and the corresponding characteristic spaces are span$\{\mathbf{e}\}$ and $\{\mathbf{e}\}^\perp$, respectively. Conversely, if span$\{\mathbf{e}\}$ (with $|\mathbf{e}| = 1$) and $\{\mathbf{e}\}^\perp$ are the characteristic spaces of **S**, then **S** has the form (8.2).

(c) **S** has exactly one eigenvalue if and only if

$$\mathbf{S} = \omega\, \mathbf{I}.$$

In this case, ω is the eigenvalue and $\mathscr{V}$ is the corresponding characteristic space. Conversely, if $\mathscr{V}$ is a characteristic space for **S**, then **S** has the form (8.3).

Note. Relation (8.1) is called the *spectral decomposition* of **S**. Note that, by (8.1), the matrix of **S** with respect to a basis $\{\mathbf{e}_i\}$ of eigenvectors is diagonal:

$$[\mathbf{S}] = \begin{pmatrix} \omega_1 & 0 & 0 \\ 0 & \omega_2 & 0 \\ 0 & 0 & \omega_3 \end{pmatrix}. \tag{8.4}$$

Furthermore, by virtue of points (a)–(c) of the spectral theorem, the characteristic spaces of a symmetric tensor **S** are three mutually perpendicular lines passing through the origin; or a line l passing through the origin and the plane through the origin perpendicular to l; or, again, **S** possesses a single characteristic space, coinciding with $\mathscr{V}$ itself. Let $\mathscr{U}_\alpha$ ($\alpha = 1, \ldots, n \leq 3$) be the set of characteristic spaces of **S**; then every vector **v** can be written in the form

$$\mathbf{v} = \sum_\alpha \mathbf{v}_\alpha, \qquad \mathbf{v}_\alpha \in \mathscr{U}_\alpha. \tag{8.5}$$

Commutation theorem. Let **S** and **T** be two tensors that commute. Then **T** leaves each characteristic space of **S** invariant; that is, if **v** belongs to a characteristic space of **S**, then **Tv** belongs to the same characteristic space. Conversely, if **T** leaves each characteristic space of a symmetric tensor **S** invariant, then **S** and **T** commute.

Proof. Let $\mathbf{S}$ and $\mathbf{T}$ be two tensors that commute. Suppose that $\mathbf{Sv} = \omega\mathbf{v}$. Then

$$\mathbf{S}(\mathbf{Tv}) = \mathbf{T}(\mathbf{Sv}) = \omega(\mathbf{Tv}), \tag{8.6}$$

and therefore $\mathbf{Tv}$ belongs to the same characteristic space as $\mathbf{v}$.

To prove the converse statement, let $\mathbf{v}$ be a vector and decompose $\mathbf{v}$ as in (8.5). If $\mathbf{T}$ leaves each characteristic space $\mathscr{U}_\alpha$ of $\mathbf{S}$ invariant, then $\mathbf{Tv}_\alpha \in \mathscr{U}_\alpha$ and

$$\mathbf{S}(\mathbf{Tv}_\alpha) = \omega_\alpha(\mathbf{Tv}_\alpha) = \mathbf{T}(\omega_\alpha\mathbf{v}_\alpha) = \mathbf{T}(\mathbf{Sv}_\alpha), \tag{8.7}$$

where ω_α is the eigenvalue corresponding to $\mathscr{U}_\alpha$. It follows, using the previous decomposition, that

$$\mathbf{STv} = \sum_\alpha \mathbf{STv}_\alpha = \sum_\alpha \mathbf{TSv}_\alpha = \mathbf{TSv}. \tag{8.8}$$

Since $\mathbf{v}$ was chosen arbitrarily, we thus have

$$\mathbf{ST} = \mathbf{TS}. \tag{8.9}$$

Definition 4. Let $\mathcal{G} \subset \mathrm{Orth}$. A set $\mathcal{A} \subset \mathrm{Lin}$ is said to be *invariant with respect to* $\mathcal{G}$ if

$$\mathbf{QAQ}^{\mathrm{T}} \in \mathcal{A}, \quad \forall\, \mathbf{A} \in \mathcal{A},\ \mathbf{Q} \in \mathcal{G}. \tag{8.10}$$

Proposition 1. The following sets are invariant with respect to Orth:

$$\mathrm{Lin}, \quad \mathrm{Lin}^+, \quad \mathrm{Orth}, \quad \mathrm{Orth}^+, \quad \mathrm{Sym}, \quad \mathrm{Skw}, \quad \mathrm{Psym}.$$

Definition 5. Let $\mathcal{A} \subset \mathrm{Lin}$. A *scalar function*

$$\varphi : \mathcal{A} \to \mathbb{R} \tag{8.11}$$

is said to be *invariant with respect to* $\mathcal{G}$ if

$$\varphi(\mathbf{A}) = \varphi(\mathbf{QAQ}^{\mathrm{T}}), \quad \forall\, \mathbf{A} \in \mathcal{A},\ \mathbf{Q} \in \mathcal{G}. \tag{8.12}$$

Similarly, a *tensor function*

$$\mathbf{G} : \mathcal{A} \to \mathrm{Lin} \tag{8.13}$$

is said to be *invariant with respect to* $\mathcal{G}$ if

$$\mathbf{Q}\mathbf{G}(\mathbf{A})\mathbf{Q}^{\mathrm{T}} = \mathbf{G}(\mathbf{Q}\mathbf{A}\mathbf{Q}^{\mathrm{T}}), \quad \forall\, \mathbf{A} \in \mathcal{A},\ \mathbf{Q} \in \mathcal{G}. \tag{8.14}$$

Definition (Isotropic function). An *isotropic function* is a function invariant with respect to Orth.

Proposition 2. Let Φ be a scalar or tensor function defined on a domain in Lin. Then Φ is *isotropic* if and only if it is invariant with respect to Orth^{+}.

Theorem 2.

(a) The functions det and tr, considered on Lin, are isotropic.

(b) The set of principal invariants

$$\mathscr{I}_{\mathbf{A}} = \big(I_1(\mathbf{A}),\, I_2(\mathbf{A}),\, I_3(\mathbf{A})\big) \tag{8.15}$$

where

$$\begin{aligned} I_1(\mathbf{A}) &= \operatorname{tr}\mathbf{A}, \\ I_2(\mathbf{A}) &= \tfrac{1}{2}\Big[\big(\operatorname{tr}\mathbf{A}\big)^2 - \operatorname{tr}\big(\mathbf{A}^2\big)\Big], \\ I_3(\mathbf{A}) &= \det\mathbf{A}, \end{aligned} \tag{8.16}$$

is isotropic, that is

$$\mathscr{I}_{\mathbf{A}} = \mathscr{I}_{\mathbf{Q}\mathbf{A}\mathbf{Q}^{\mathrm{T}}}, \quad \forall\, \mathbf{A} \in \mathrm{Lin},\ \mathbf{Q} \in \mathrm{Orth}. \tag{8.17}$$

Note. For brevity, we introduce the notation

$$\mathscr{I}(\mathcal{A}) = \{\mathscr{I}_{\mathbf{A}} \mid \mathbf{A} \in \mathcal{A}\} \tag{8.18}$$

for the set of all triplets of principal invariants associated with the elements of $\mathcal{A}$. Thus $\mathscr{I}(\mathcal{A}) \subset \mathbb{R}^3$.

In what follows, we consider some important representation theorems for functions defined on subsets of Sym. It is assumed, from now on, that $\mathcal{A}$ is a set invariant with respect to Orth.

Representation theorem for isotropic scalar functions. A function

$$\varphi : \mathcal{A} \to \mathbb{R}, \quad \mathcal{A} \subset \mathrm{Sym}, \tag{8.19}$$

is isotropic if and only if there exists a function

$$\tilde{\varphi} : \mathscr{I}(\mathcal{A}) \to \mathbb{R}, \tag{8.20}$$

such that

$$\varphi(\mathbf{A}) = \tilde{\varphi}(\mathscr{I}_{\mathbf{A}}), \quad \forall\, \mathbf{A} \in \mathcal{A}. \tag{8.21}$$

Proof. Suppose that φ is isotropic. To establish the representation (8.15), it is sufficient to show that

$$\varphi(\mathbf{A}) = \varphi(\mathbf{B}) \tag{8.22}$$

whenever

$$\mathscr{I}_{\mathbf{A}} = \mathscr{I}_{\mathbf{B}}. \tag{8.23}$$

Let therefore $\mathbf{A}, \mathbf{B} \in \mathrm{Sym}$ and suppose that (8.17) holds. By the latter and by the following proposition, the proof of which will not be reported, the desired result is obtained.

Proposition 3. Let $\mathbf{S}, \mathbf{T} \in \mathrm{Sym}$. If

$$\mathscr{I}_{\mathbf{S}} = \mathscr{I}_{\mathbf{T}}, \tag{8.24}$$

then $\mathbf{S}$ and $\mathbf{T}$ have the same spectrum.

It therefore follows that $\mathbf{A}$ and $\mathbf{B}$ have the same spectrum. By the spectral theorem, there exist orthonormal bases $\{\mathbf{e}_i\}$ and $\{\mathbf{f}_i\}$ such that

$$\mathbf{A} = \sum_i \omega_i\, \mathbf{e}_i \otimes \mathbf{e}_i, \qquad \mathbf{B} = \sum_i \omega_i\, \mathbf{f}_i \otimes \mathbf{f}_i. \tag{8.25}$$

Let $\mathbf{Q}$ be the orthogonal tensor that maps the basis $\{\mathbf{f}_i\}$ to the basis $\{\mathbf{e}_i\}$, that is

$$\mathbf{Q}\mathbf{f}_i = \mathbf{e}_i. \tag{8.26}$$

Then, since

$$\mathbf{Q}(\mathbf{f}_i \otimes \mathbf{f}_i)\mathbf{Q}^{\mathrm{T}} = (\mathbf{Q}\mathbf{f}_i) \otimes (\mathbf{Q}\mathbf{f}_i), \tag{8.27}$$

it follows that

$$\mathbf{Q}\mathbf{B}\mathbf{Q}^{\mathrm{T}} = \mathbf{A}. \tag{8.28}$$

Since φ is isotropic, we have

$$\varphi(\mathbf{QBQ}^{\mathrm{T}}) = \varphi(\mathbf{B}), \tag{8.29}$$

from which, by (8.22), the equality (8.16) is verified. It therefore follows that φ admits the representation (8.15).

The converse statement, namely that (8.15) defines an isotropic function, follows immediately from (8.11) and the definition of isotropy.

Transfer theorem. Let

$$\mathbf{G} : \mathcal{A} \to \mathrm{Lin}, \qquad (\mathcal{A} \subset \mathrm{Sym}) \tag{8.30}$$

be an isotropic function. Then every eigenvector of $\mathbf{A} \in \mathcal{A}$ is also an eigenvector of $\mathbf{G}(\mathbf{A})$.

Proof. Let $\mathbf{e}$ be an eigenvector of $\mathbf{A} \in \mathcal{A}$, and let $\mathbf{Q} \in \mathrm{Orth}$ be the reflection with respect to the plane perpendicular to $\mathbf{e}$; that is

$$\mathbf{Qe} = -\mathbf{e}, \qquad \mathbf{Qf} = \mathbf{f} \quad \text{if } \mathbf{f} \cdot \mathbf{e} = 0. \tag{8.31}$$

By the spectral theorem, $\mathbf{Q}$ leaves the characteristic spaces of $\mathbf{A}$ invariant; therefore, based on the commutation theorem, we have

$$\mathbf{QAQ}^{\mathrm{T}} = \mathbf{A}. \tag{8.32}$$

Since $\mathbf{G}$ is isotropic, it follows that

$$\mathbf{QG}(\mathbf{A})\mathbf{Q}^{\mathrm{T}} = \mathbf{G}(\mathbf{QAQ}^{\mathrm{T}}) = \mathbf{G}(\mathbf{A}), \tag{8.33}$$

and thus $\mathbf{Q}$ commutes with $\mathbf{G}(\mathbf{A})$. Therefore,

$$\mathbf{QG}(\mathbf{A})\mathbf{e} = \mathbf{G}(\mathbf{A})\mathbf{Qe} = -\,\mathbf{G}(\mathbf{A})\mathbf{e}, \tag{8.34}$$

that is, $\mathbf{Q}$ maps $\mathbf{G}(\mathbf{A})\mathbf{e}$ to its opposite. But, by relation (8.31), this can only happen if $\mathbf{G}(\mathbf{A})\mathbf{e}$ is parallel to $\mathbf{e}$; hence

$$\mathbf{G}(\mathbf{A})\mathbf{e} = \omega\mathbf{e}. \tag{8.35}$$

It follows that $\mathbf{e}$ is an eigenvector of $\mathbf{G}(\mathbf{A})$. □

Wang's Lemma. Let $\mathbf{A} \in \mathrm{Sym}$.

(a) Consider the spectral decomposition

$$\mathbf{A} = \sum_i \omega_i \, \mathbf{e}_i \otimes \mathbf{e}_i. \tag{8.36}$$

If the eigenvalues ω_i are distinct, then the set $\{\mathbf{I}, \mathbf{A}, \mathbf{A}^2\}$ is linearly independent and

$$\mathrm{span}\{\mathbf{I}, \mathbf{A}, \mathbf{A}^2\} = \mathrm{span}\{\mathbf{e}_1 \otimes \mathbf{e}_1,\ \mathbf{e}_2 \otimes \mathbf{e}_2,\ \mathbf{e}_3 \otimes \mathbf{e}_3\}. \tag{8.37}$$

(b) Suppose that $\mathbf{A}$ has exactly two distinct eigenvalues, so that

$$\mathbf{A} = \omega_1 \, \mathbf{e} \otimes \mathbf{e} + \omega_2 \left(\mathbf{I} - \mathbf{e} \otimes \mathbf{e}\right), \qquad |\mathbf{e}| = 1. \tag{8.38}$$

Then $\{\mathbf{I}, \mathbf{A}\}$ is linearly independent and

$$\mathrm{span}\{\mathbf{I}, \mathbf{A}\} = \mathrm{span}\{\mathbf{e} \otimes \mathbf{e},\ \mathbf{I} - \mathbf{e} \otimes \mathbf{e}\}. \tag{8.39}$$

First representation theorem for isotropic tensor functions. Let

$$\mathbf{G} : \mathcal{A} \to \mathrm{Sym}, \qquad (\mathcal{A} \subset \mathrm{Sym}). \tag{8.40}$$

The function $\mathbf{G}$ is isotropic if and only if there exist scalar functions

$$\varphi_0,\ \varphi_1,\ \varphi_2 : \mathscr{I}(\mathcal{A}) \to \mathbb{R} \tag{8.41}$$

such that

$$\mathbf{G}(\mathbf{A}) = \varphi_0(\mathscr{I}_{\mathbf{A}})\mathbf{I} + \varphi_1(\mathscr{I}_{\mathbf{A}})\mathbf{A} + \varphi_2(\mathscr{I}_{\mathbf{A}})\mathbf{A}^2, \quad \forall\, \mathbf{A} \in \mathcal{A}. \tag{8.42}$$

Proof. Suppose that $\mathbf{G}$ admits the representation (8.42). Let $\mathbf{A} \in \mathcal{A}$ and $\mathbf{Q} \in \mathrm{Orth}$. Then, by (8.17),

$$\begin{aligned}
\mathbf{G}(\mathbf{Q}\mathbf{A}\mathbf{Q}^{\mathrm{T}}) &= \varphi_0(\mathscr{I}_{\mathbf{Q}\mathbf{A}\mathbf{Q}^{\mathrm{T}}})\mathbf{I} + \varphi_1(\mathscr{I}_{\mathbf{Q}\mathbf{A}\mathbf{Q}^{\mathrm{T}}})\mathbf{Q}\mathbf{A}\mathbf{Q}^{\mathrm{T}} + \varphi_2(\mathscr{I}_{\mathbf{Q}\mathbf{A}\mathbf{Q}^{\mathrm{T}}})(\mathbf{Q}\mathbf{A}\mathbf{Q}^{\mathrm{T}})^2 \\
&= \varphi_0(\mathscr{I}_{\mathbf{A}})\mathbf{I} + \varphi_1(\mathscr{I}_{\mathbf{A}})\mathbf{Q}\mathbf{A}\mathbf{Q}^{\mathrm{T}} + \varphi_2(\mathscr{I}_{\mathbf{A}})\mathbf{Q}\mathbf{A}^2\mathbf{Q}^{\mathrm{T}} \\
&= \mathbf{Q}\big[\varphi_0(\mathscr{I}_{\mathbf{A}})\mathbf{I} + \varphi_1(\mathscr{I}_{\mathbf{A}})\mathbf{A} + \varphi_2(\mathscr{I}_{\mathbf{A}})\mathbf{A}^2\big]\mathbf{Q}^{\mathrm{T}} = \mathbf{Q}\mathbf{G}(\mathbf{A})\mathbf{Q}^{\mathrm{T}}.
\end{aligned} \tag{8.43}$$

To prove the converse implication, we assume that $\mathbf{G}$ is isotropic and consider a generic tensor $\mathbf{A} \in \mathcal{A}$.

Case 1. $\mathbf{A}$ has exactly three distinct eigenvalues. Let (8.36) be the spectral decomposition of $\mathbf{A}$. Then, from the transfer theorem,

$$\mathbf{G}(\mathbf{A}) = \sum_i \beta_i \, \mathbf{e}_i \otimes \mathbf{e}_i. \tag{8.44}$$

From (8.37) it follows that there exist scalar functions $\alpha_0(\mathbf{A})$, $\alpha_1(\mathbf{A})$, $\alpha_2(\mathbf{A})$ such that

$$\mathbf{G}(\mathbf{A}) = \alpha_0(\mathbf{A})\mathbf{I} + \alpha_1(\mathbf{A})\mathbf{A} + \alpha_2(\mathbf{A})\mathbf{A}^2. \tag{8.45}$$

Case 2. $\mathbf{A}$ has exactly two distinct eigenvalues. Then $\mathbf{A}$ admits the representation (8.38), and its characteristic spaces are $\mathrm{span}\{\mathbf{e}\}$ and $\{\mathbf{e}\}^\perp$. From the transfer theorem, each of these two subspaces is contained in a characteristic space of $\mathbf{G}(\mathbf{A})$. This is possible only if:

(i) $\mathbf{G}(\mathbf{A})$ has characteristic spaces $\mathrm{span}\{\mathbf{e}\}$ and $\{\mathbf{e}\}^\perp$; or

(ii) $\mathscr{V}$ is the only characteristic space for $\mathbf{G}(\mathbf{A})$.

From the spectral theorem we have, in both cases,

$$\mathbf{G}(\mathbf{A}) = \beta_1 \, \mathbf{e} \otimes \mathbf{e} + \beta_2 \, (\mathbf{I} - \mathbf{e} \otimes \mathbf{e}), \tag{8.46}$$

with $\beta_1 \neq \beta_2$ in case (i) and $\beta_1 = \beta_2$ in case (ii). By virtue of (8.39), (8.46) can be written in the form (8.45) with

$$\alpha_2(\mathbf{A}) = 0. \tag{8.47}$$

Case 3. $\mathbf{A}$ has only one eigenvalue. By the spectral theorem and the transfer theorem, $\mathscr{V}$ is the common characteristic space of $\mathbf{A}$ and $\mathbf{G}(\mathbf{A})$, therefore

$$\mathbf{G}(\mathbf{A}) = \beta \, \mathbf{I}. \tag{8.48}$$

It follows that $\mathbf{G}(\mathbf{A})$ admits the representation (8.45) with

$$\alpha_0(\mathbf{A}) = \beta, \qquad \alpha_1(\mathbf{A}) = \alpha_2(\mathbf{A}) = 0. \tag{8.49}$$

Since these three cases exhaust all possibilities, we conclude that every isotropic tensor function $\mathbf{G}$ admits the representation (8.45). It remains to be shown that the functions $\alpha_0, \alpha_1, \alpha_2$ are also isotropic, that is

$$\alpha_k(\mathbf{A}) = \alpha_k(\mathbf{Q}\mathbf{A}\mathbf{Q}^{\mathrm{T}}), \quad k = 0, 1, 2, \tag{8.50}$$

for every $\mathbf{A} \in \mathcal{A}$ and $\mathbf{Q} \in \mathrm{Orth}$. Thus, we choose $\mathbf{A} \in \mathcal{A}$, $\mathbf{Q} \in \mathrm{Orth}$. Since $\mathbf{G}$ is isotropic, we have

$$\mathbf{Q}\mathbf{G}(\mathbf{A})\mathbf{Q}^{\mathrm{T}} = \mathbf{G}(\mathbf{Q}\mathbf{A}\mathbf{Q}^{\mathrm{T}}), \tag{8.51}$$

or, in an equivalent form,

$$\mathbf{G}(\mathbf{A}) - \mathbf{Q}^{\mathrm{T}}\mathbf{G}(\mathbf{Q}\mathbf{A}\mathbf{Q}^{\mathrm{T}})\mathbf{Q} = \mathbf{0}. \tag{8.52}$$

Since furthermore

$$\mathbf{Q}^{\mathrm{T}}(\mathbf{Q}\mathbf{A}\mathbf{Q}^{\mathrm{T}})^2\mathbf{Q} = \mathbf{Q}^{\mathrm{T}}\mathbf{Q}\mathbf{A}\mathbf{Q}^{\mathrm{T}}\mathbf{Q}\mathbf{A}\mathbf{Q}^{\mathrm{T}}\mathbf{Q} = \mathbf{A}^2, \tag{8.53}$$

we obtain

$$[\alpha_0(\mathbf{A}) - \alpha_0(\mathbf{Q}\mathbf{A}\mathbf{Q}^{\mathrm{T}})]\mathbf{I} + [\alpha_1(\mathbf{A}) - \alpha_1(\mathbf{Q}\mathbf{A}\mathbf{Q}^{\mathrm{T}})]\mathbf{A} + [\alpha_2(\mathbf{A}) - \alpha_2(\mathbf{Q}\mathbf{A}\mathbf{Q}^{\mathrm{T}})]\mathbf{A}^2 = \mathbf{0}. \tag{8.54}$$

Note. Let us reconsider the three cases studied previously.

Case 1. By point (a) of Wang's Lemma, $\{\mathbf{I}, \mathbf{A}, \mathbf{A}^2\}$ is linearly independent, so (8.54) implies (8.50).

Case 2. By (8.17) and Proposition 3, $\mathbf{A}$ and $\mathbf{Q}\mathbf{A}\mathbf{Q}^{\mathrm{T}}$ have the same spectrum. Therefore, both have exactly two distinct eigenvalues and (8.47) implies $\alpha_2(\mathbf{A}) = \alpha_2(\mathbf{Q}\mathbf{A}\mathbf{Q}^{\mathrm{T}}) = 0$. Furthermore, by point (b) of Wang's Lemma, the set $\{\mathbf{I}, \mathbf{A}\}$ is linearly independent and (8.54) implies (8.50).

Case 3. Here $\mathbf{A} = \lambda\mathbf{I}$, so that $\mathbf{A} = \mathbf{Q}\mathbf{A}\mathbf{Q}^{\mathrm{T}}$ and (8.50) holds trivially.

By the Cayley-Hamilton theorem (4.21),

$$\mathbf{A}^2 = I_1(\mathbf{A})\,\mathbf{A} - I_2(\mathbf{A})\,\mathbf{I} + I_3(\mathbf{A})\,\mathbf{A}^{-1}, \tag{8.55}$$

provided, of course, that $\mathbf{A}$ is invertible. Therefore, the preceding theorem has the following corollary.

Second representation theorem for isotropic tensor functions. Let $\mathcal{A}$ be a subset of the set of all invertible symmetric tensors. Then

$$\mathbf{G} : \mathcal{A} \to \mathrm{Sym} \tag{8.56}$$

is isotropic if and only if there exist scalar functions $\beta_0, \beta_1, \beta_2 : \mathscr{I}(\mathcal{A}) \to \mathbb{R}$ such

that

$$\mathbf{G}(\mathbf{A}) = \beta_0(\mathscr{I}_{\mathbf{A}})\,\mathbf{I} + \beta_1(\mathscr{I}_{\mathbf{A}})\,\mathbf{A} + \beta_2(\mathscr{I}_{\mathbf{A}})\,\mathbf{A}^{-1}, \quad \forall\,\mathbf{A} \in \mathscr{A}. \tag{8.57}$$

Simultaneous diagonalization of commuting tensors

We have seen, through the commutation theorem, that if two tensors **S** and **T** commute, then every eigenspace of **S** is invariant under the action of **T**. In this section, we show that, assuming also the diagonalizability of the two tensors, this property allows us to obtain a stronger result: *simultaneous diagonalization*. The following proof is based on the one provided by [23].

Theorem. Let **S** and **T** be two diagonalizable tensors. Then they are simultaneously diagonalizable if and only if they commute.

Proof.

(⇒) If **S** and **T** are simultaneously diagonalizable, there exists a basis $\{\mathbf{v}_i\}$ such that

$$\mathbf{S}\mathbf{v}_i = s_i\,\mathbf{v}_i, \qquad \mathbf{T}\mathbf{v}_i = t_i\,\mathbf{v}_i. \tag{8.58}$$

In this basis both tensors are diagonal. Applying them in succession to any eigenvector $\mathbf{v}_i$ we obtain

$$\mathbf{S}\mathbf{T}\mathbf{v}_i = \mathbf{S}(t_i\mathbf{v}_i) = t_i s_i\,\mathbf{v}_i = s_i t_i\,\mathbf{v}_i = \mathbf{T}(s_i\mathbf{v}_i) = \mathbf{T}\mathbf{S}\mathbf{v}_i. \tag{8.59}$$

Since the basis is complete, it follows that $\mathbf{ST} = \mathbf{TS}$. Thus simultaneous diagonalization implies commutativity.

(⇐) Let us now suppose that **S** and **T** commute. Since **S** is diagonalizable, there exists a basis of eigenvectors $\{\mathbf{v}_i\}$.

We also assume that these eigenvectors have been ordered so that those associated with the same eigenvalue of **S** are contiguous.

For each i we have

$$\mathbf{S}\mathbf{v}_i = s_i\,\mathbf{v}_i. \tag{8.60}$$

Since **S** and **T** commute, the commutation theorem guarantees that every eigenspace of **S** is invariant under the action of **T**; therefore,

$$\mathbf{T}\mathbf{v}_i \in \operatorname{span}\{\mathbf{v}_j : s_j = s_i\}. \tag{8.61}$$

In the basis $\{\mathbf{v}_i\}$ the representative matrix of $\mathbf{T}$ thus assumes a block structure:

$$[\mathbf{T}] = \begin{pmatrix} [\mathbf{T}_1] & & \\ & [\mathbf{T}_2] & \\ & & \ddots \end{pmatrix}, \tag{8.62}$$

where each block $[\mathbf{T}_k]$ acts on an eigenspace of $\mathbf{S}$ and has a dimension equal to the multiplicity of the corresponding eigenvalue.

Since $\mathbf{T}$ is diagonalizable, so is each block $[\mathbf{T}_k]$. We can therefore diagonalize each block separately through a change of basis *internal* to the corresponding eigenspace. Such a change does not alter the diagonality of $[\mathbf{S}]$.

After having diagonalized all the blocks, both matrices $[\mathbf{S}]$ and $[\mathbf{T}]$ are diagonal in the same basis.

In conclusion, there exists a common basis of eigenvectors of $\mathbf{S}$ and $\mathbf{T}$, and therefore the two tensors are simultaneously diagonalizable.

Ericksen's theorem

Casey's proof of Ericksen's theorem [7] is reported here. Let $\mathcal{B}$ be a homogeneous and elastic three-dimensional body. Let X_A $(A = 1, 2, 3)$ denote the rectangular Cartesian coordinates of a point $X \in \mathcal{B}$ in the reference configuration $\mathcal{B}$, and $x_i = \chi_i(X_A)$ $(i = 1, 2, 3)$ the coordinates of the same point in the deformed configuration. Then

$$\mathbf{F} = \operatorname{Grad} \boldsymbol{\chi} \equiv \frac{\partial \chi_i}{\partial X_A} =: x_{i,A}, \qquad \mathbf{B} = \mathbf{F}\mathbf{F}^{\mathrm{T}} \equiv x_{i,A} x_{i,B}, \qquad I_1 = \operatorname{tr} \mathbf{B} = x_{k,K} x_{k,K}. \tag{8.63}$$

Let us now consider two homogeneous, isotropic, and hyperelastic materials, denoted by $m^{(1)}$ and $m^{(2)}$, whose free energy functions are respectively

$$W^{(1)} = I_1, \qquad W^{(2)} = I_1^2. \tag{8.64}$$

Theorem 1. If a deformation can be maintained solely by surface forces for both material $m^{(1)}$ and material $m^{(2)}$, then such a deformation must necessarily be homogeneous.

Proof. Suppose there exists a deformation that can be maintained solely by surface forces for both materials $m^{(1)}$ and $m^{(2)}$. In this case, the equilibrium

equation

$$\operatorname{Div} \mathbf{P} = \mathbf{0} \tag{8.65}$$

must be satisfied for both free energy functions, when substituted into the constitutive relation

$$\mathbf{P} = \frac{\partial W}{\partial \mathbf{F}}. \tag{8.66}$$

For material $m^{(1)}$ we have

$$\mathbf{P}^{(1)} = \frac{\partial W^{(1)}}{\partial \mathbf{F}} = 2\mathbf{F}, \tag{8.67}$$

from which, imposing equilibrium,

$$\operatorname{Div} \mathbf{F} = \mathbf{0}. \tag{8.68}$$

For material $m^{(2)}$, instead, we have

$$\mathbf{P}^{(2)} = \frac{\partial W^{(2)}}{\partial \mathbf{F}} = 4I_1\mathbf{F}. \tag{8.69}$$

Therefore, the equilibrium equation becomes

$$\mathbf{F} \operatorname{Grad} I_1 + I_1 \operatorname{Div} \mathbf{F} = \mathbf{0}, \tag{8.70}$$

from which it follows, thanks to the previous result,

$$\mathbf{F} \operatorname{Grad} I_1 = \mathbf{0}. \tag{8.71}$$

Premultiplying by $\mathbf{F}^{-1}$, we obtain

$$\operatorname{Grad} I_1 = \mathbf{0}. \tag{8.72}$$

Substituting the expression for I_1, we obtain

$$x_{k,KA}x_{k,K} = 0. \tag{8.73}$$

Differentiating again with respect to the material coordinate, it follows

$$x_{k,KAA}x_{k,K} + x_{k,KA}x_{k,KA} = 0. \tag{8.74}$$

Since we know that

$$\operatorname{Div} \mathbf{F} \equiv x_{i,AA} = 0, \tag{8.75}$$

we have

$$x_{k,KAA} = x_{k,AAK} = 0. \tag{8.76}$$

It therefore remains

$$x_{k,KA} x_{k,KA} = 0, \tag{8.77}$$

which implies

$$x_{k,KA} = 0. \tag{8.78}$$

It follows that

$$x_{k,K} = \text{constant}, \tag{8.79}$$

that is, the deformation is necessarily homogeneous and of the form

$$x_i = A_{iA} X_A + a_i, \tag{8.80}$$

with A_{iA} and a_i constant.

Ericksen's theorem. If a deformation can be maintained solely by surface forces in every member of the set $\mathscr{I}$ of homogeneous, isotropic, and hyperelastic materials, then such a deformation must be homogeneous.

Proof. If a deformation can be maintained solely by surface forces in every member of $\mathscr{I}$, then, in particular, this also applies to materials $m^{(1)}$ and $m^{(2)}$, which belong to $\mathscr{I}$. From the previous theorem, the deformation must therefore be homogeneous.

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